

Shape Deformation

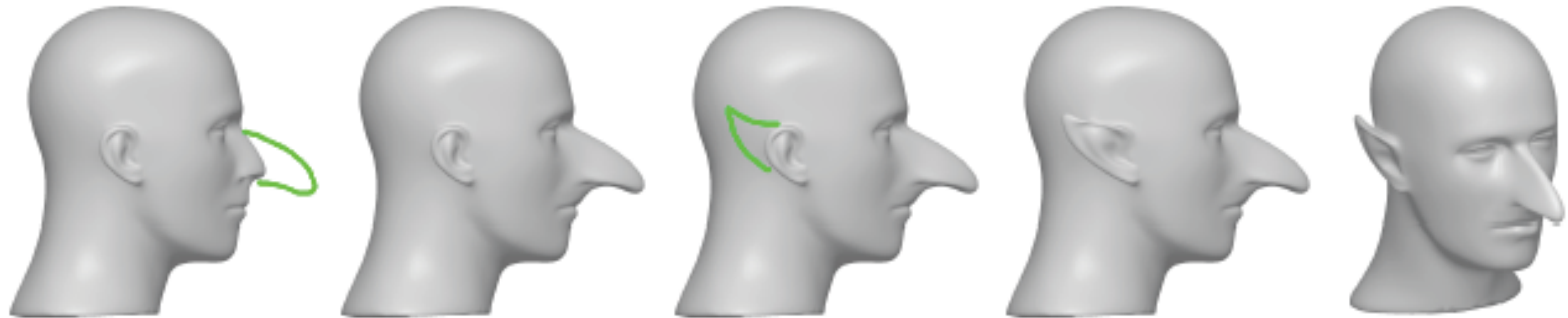
Why Shape Deformation?

Animation



Why Shape Deformation?

Mesh Editing



Why Shape Deformation?

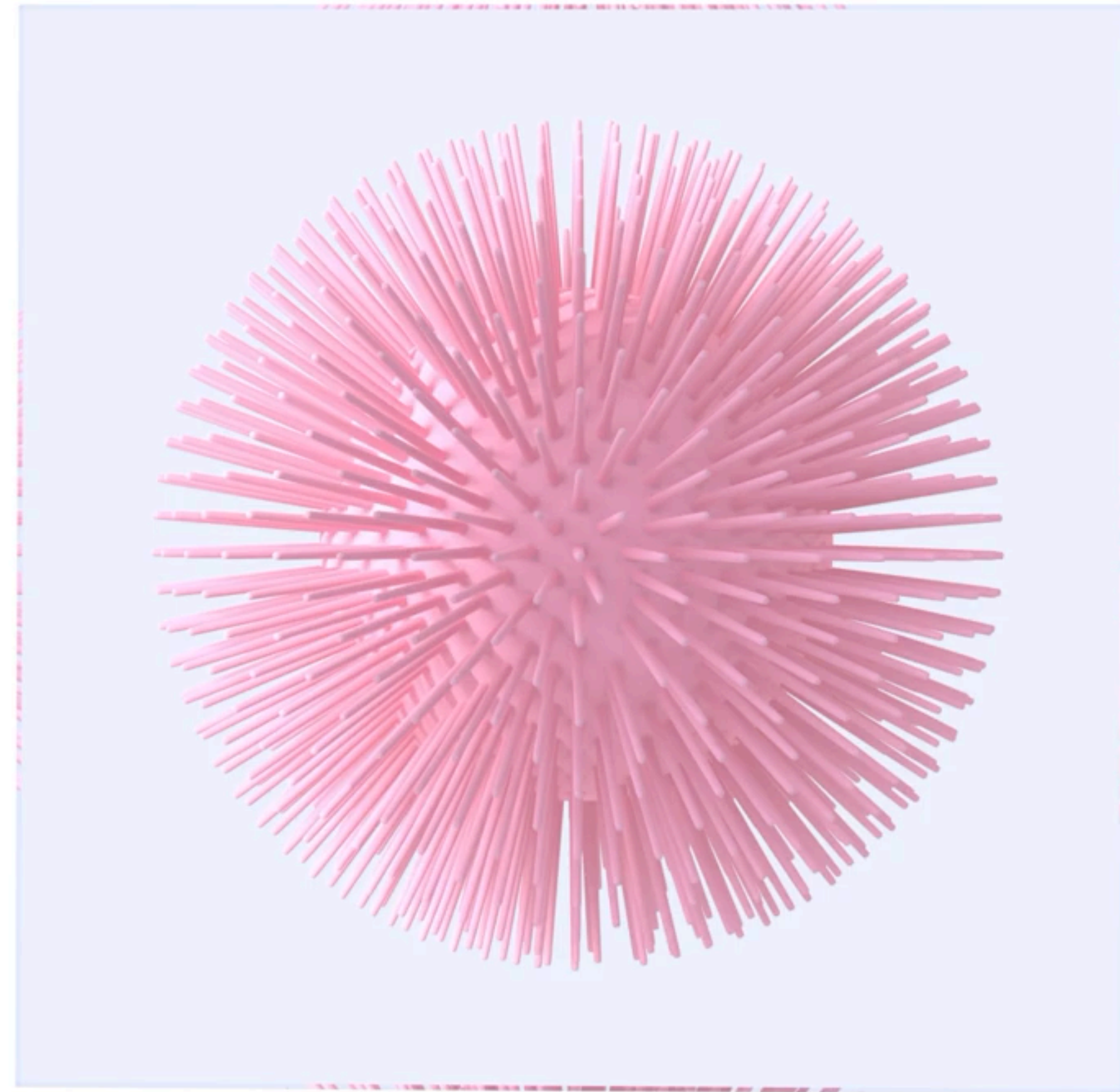
Simulation of Physical Phenomena

IPC

7e-3X



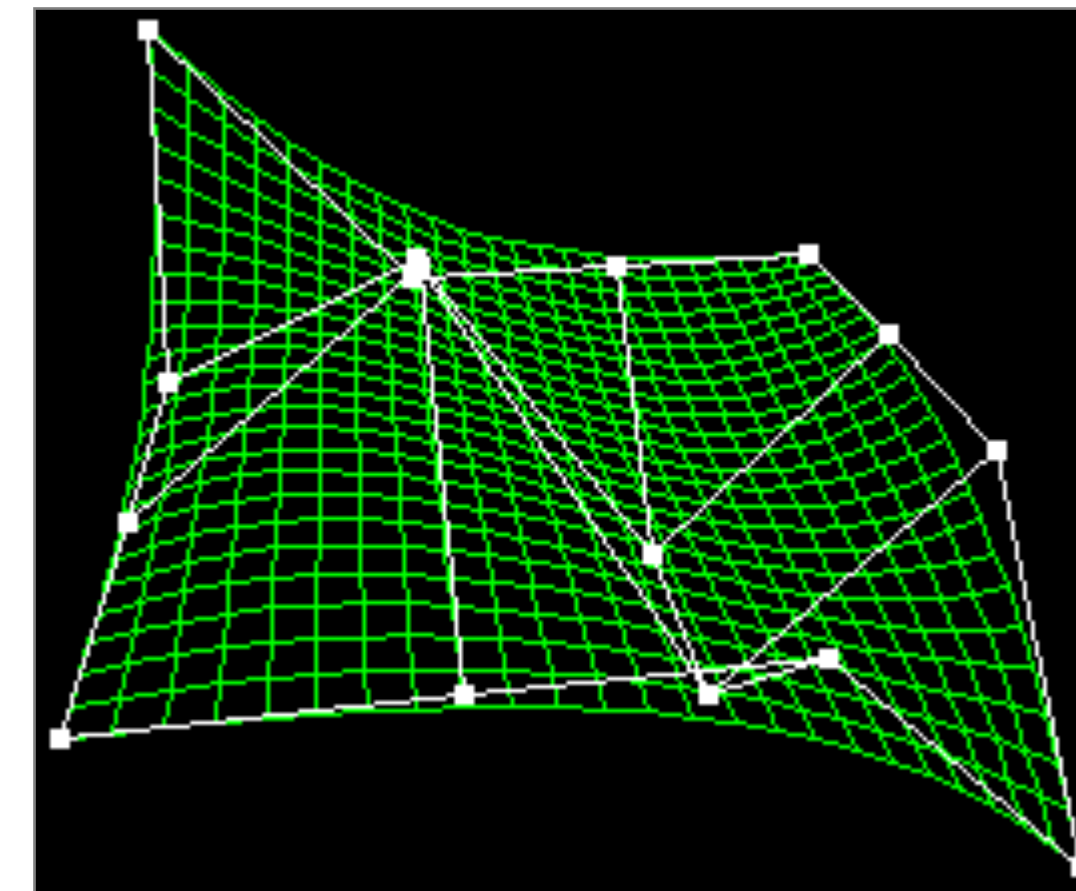
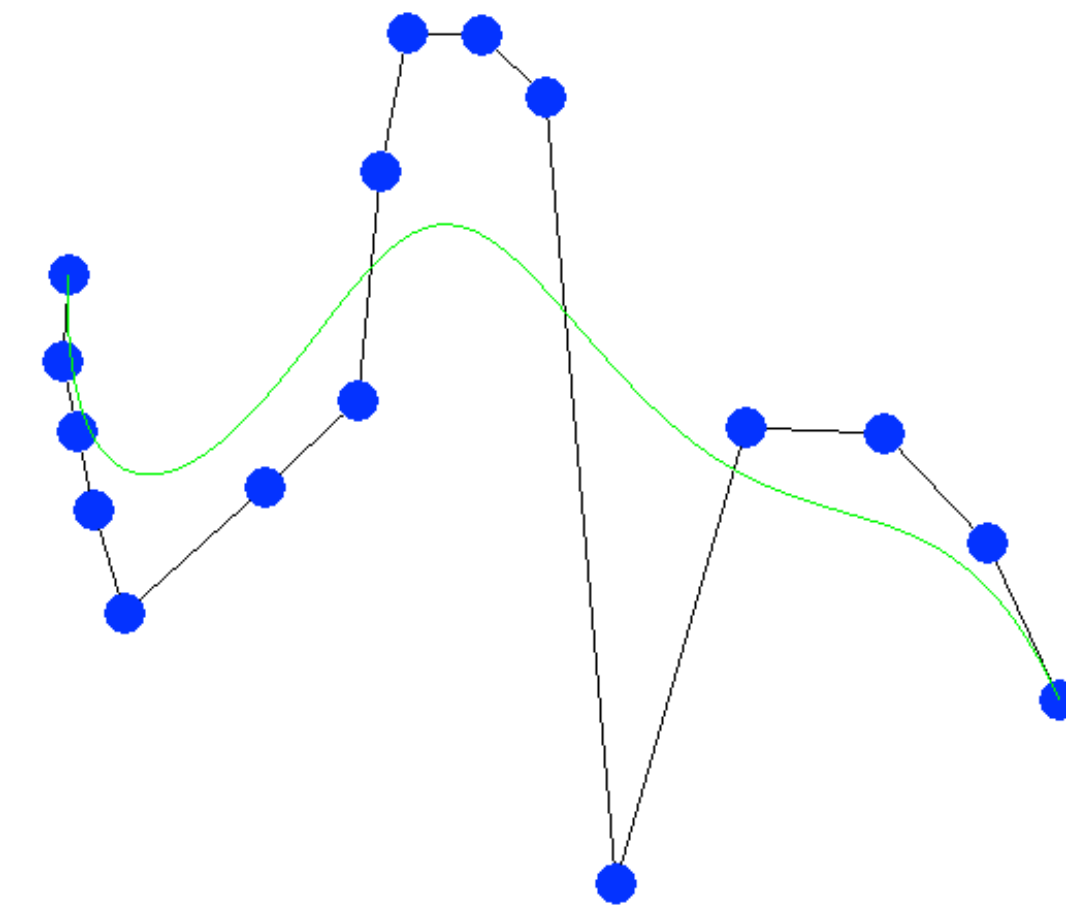
tetrahedra: 2314K
contacts per step (max): 105K
dt: 0.001
 μ : 0



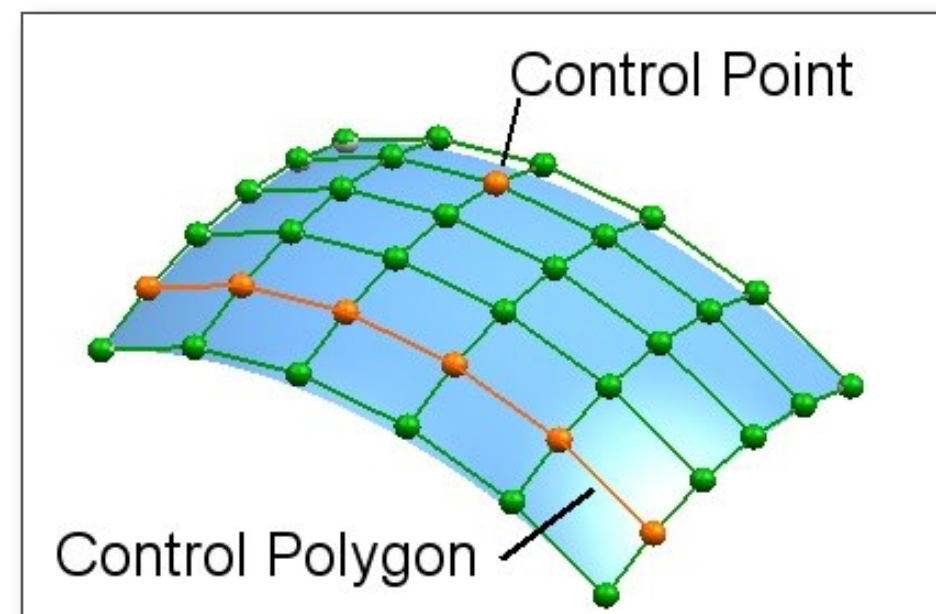
<https://ipc-sim.github.io>

Parametric Curves and Surfaces

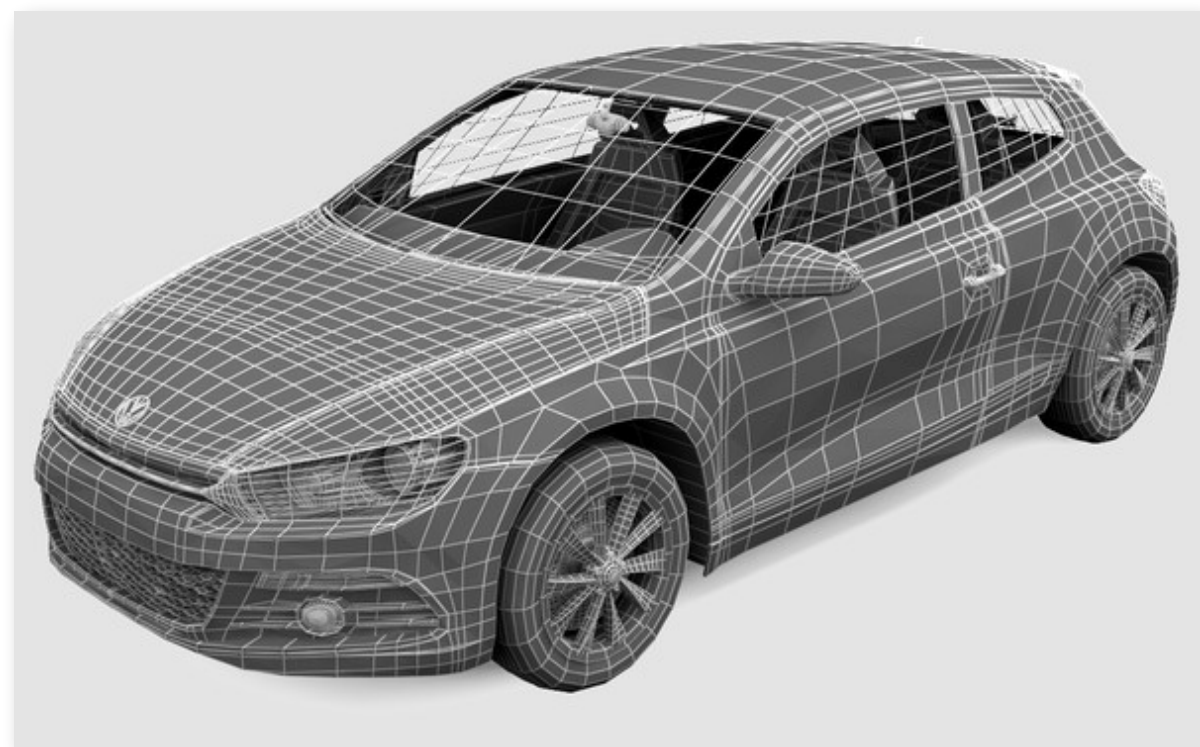
- Deformation by control point manipulation
- Built-in deformation mechanism
- Control structure is pre-set (you cannot pull on arbitrary points)



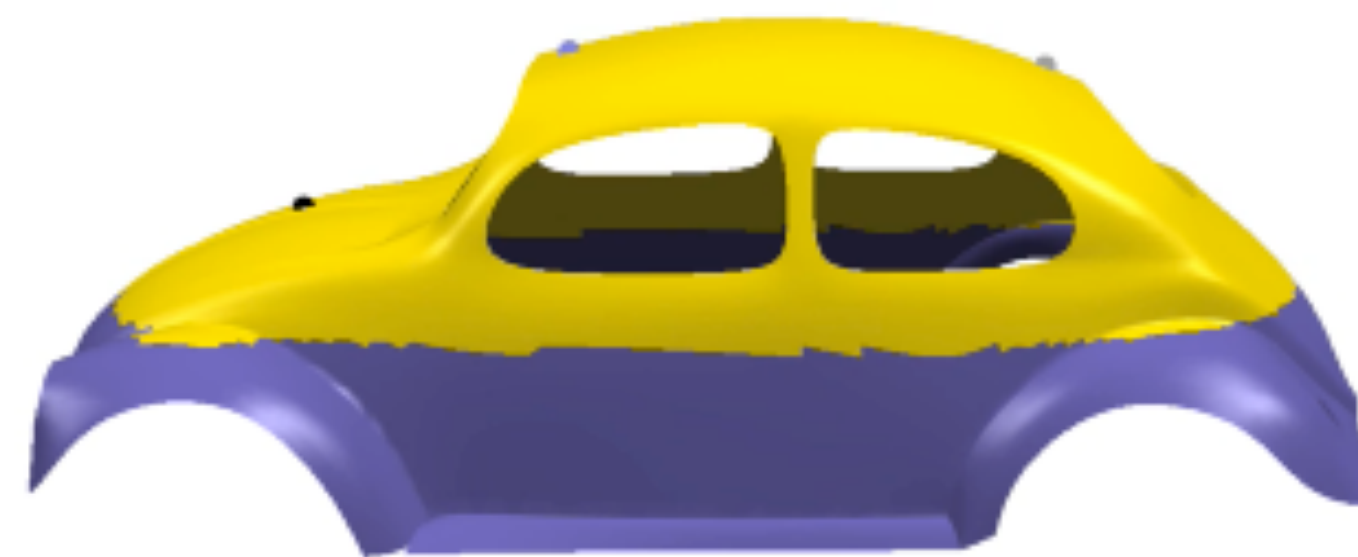
Traditional CAD vs Unstructured Meshes



$$s(u, v) = \sum_{i,j} \mathbf{p}_{i,j} B_i(u) B_j(v)$$

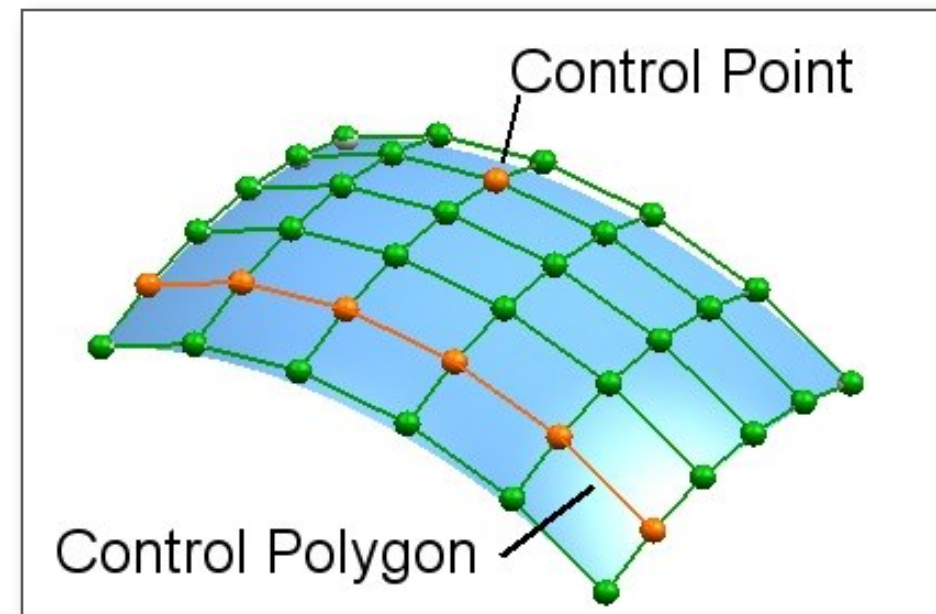


$$\min_{\mathbf{x}'} \int_{\mathcal{S}} \|\Delta_{\mathcal{S}} \mathbf{x}' - \delta_0\|^2$$
$$s.t. \mathbf{x}'|_{\mathcal{C}} = \mathbf{x}_{\text{fixed}}$$

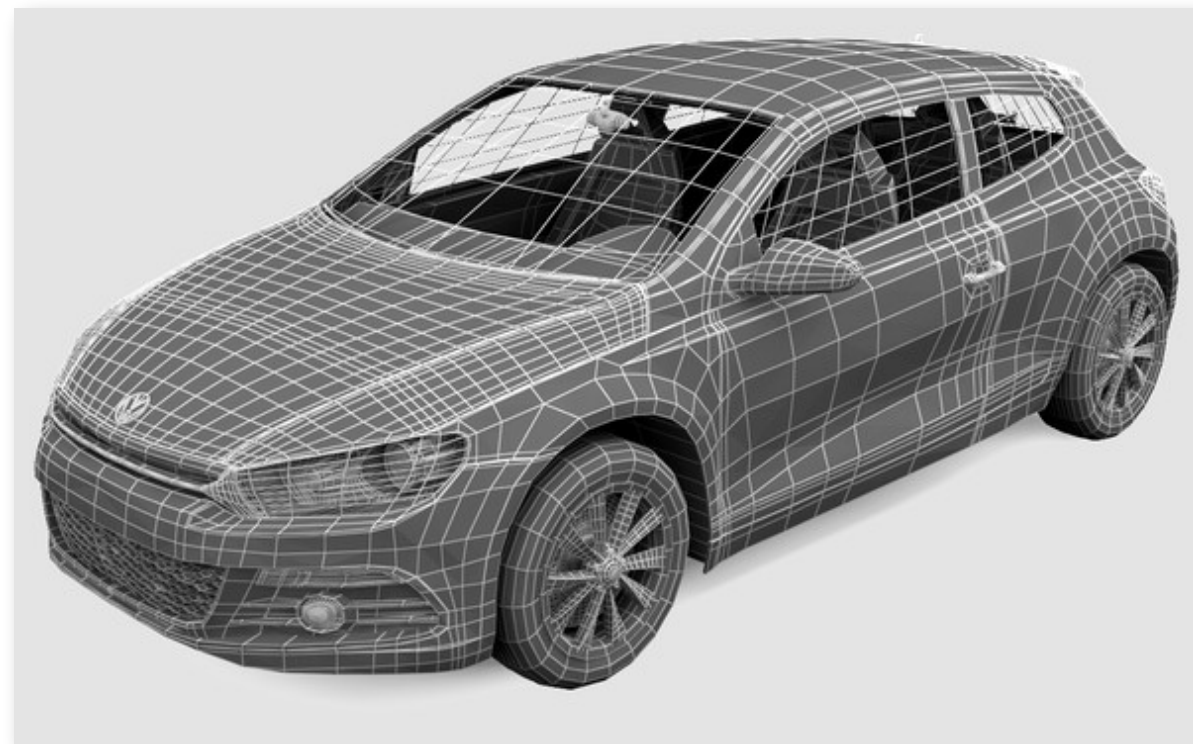


images from Jacobson et al., SGP 2010

Traditional CAD vs Unstructured Meshes



$$s(u, v) = \sum_{i,j} \mathbf{p}_{i,j} B_i(u) B_j(v)$$

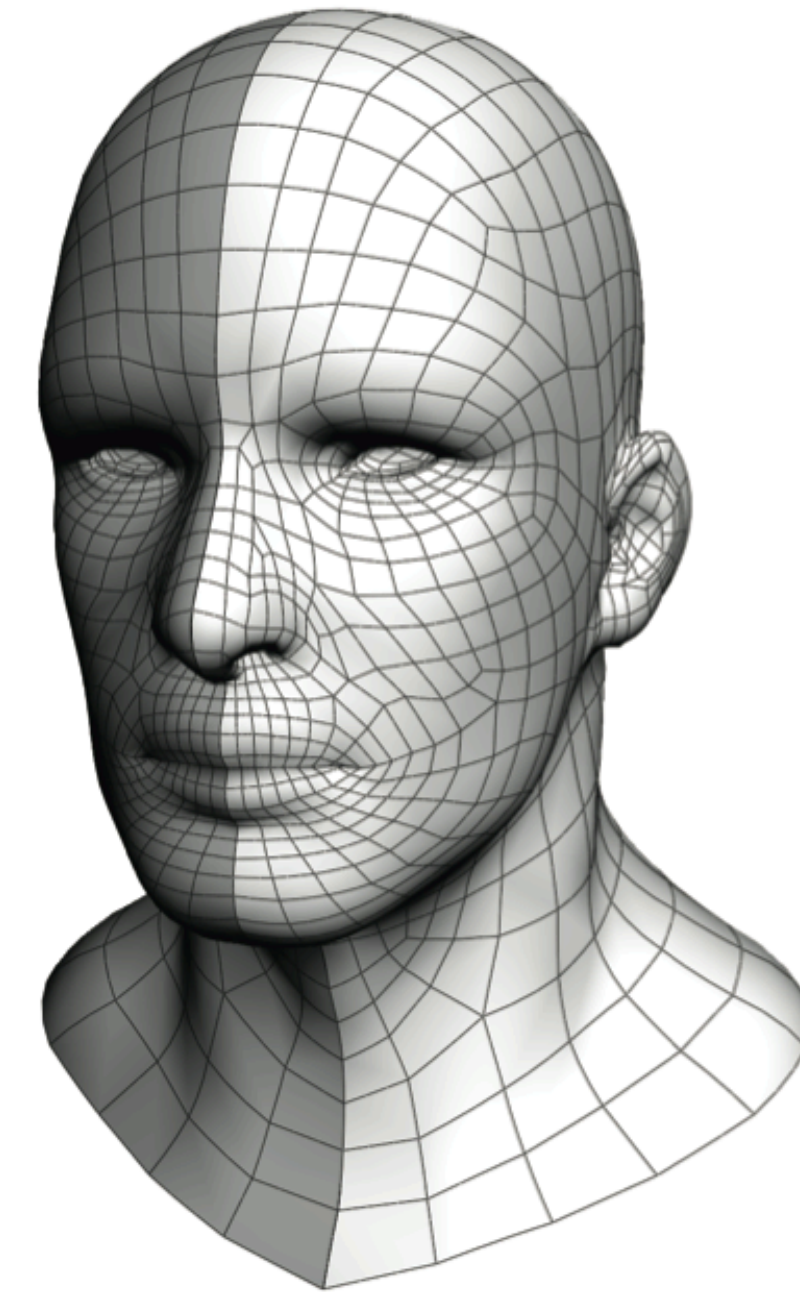
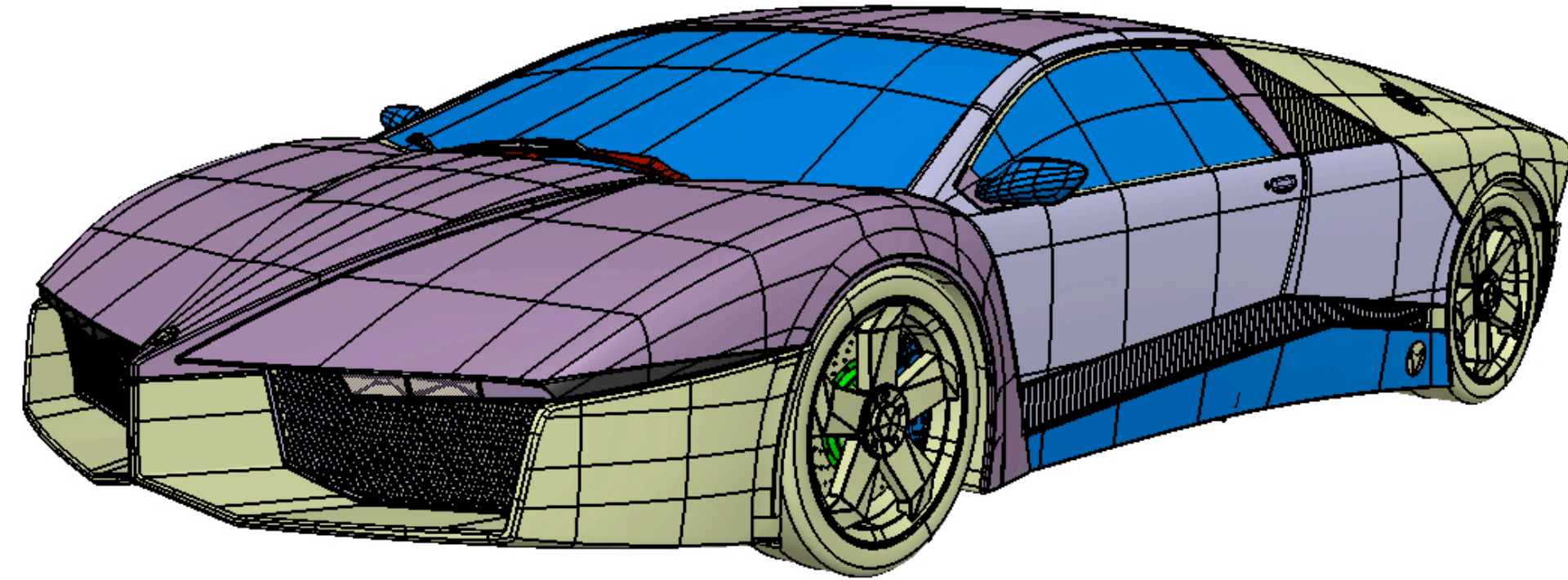


$$\min_{\mathbf{x}'} \int_{\mathcal{S}} \|\Delta_{\mathcal{S}} \mathbf{x}' - \delta_0\|^2$$
$$s.t. \mathbf{x}'|_{\mathcal{C}} = \mathbf{x}_{\text{fixed}}$$



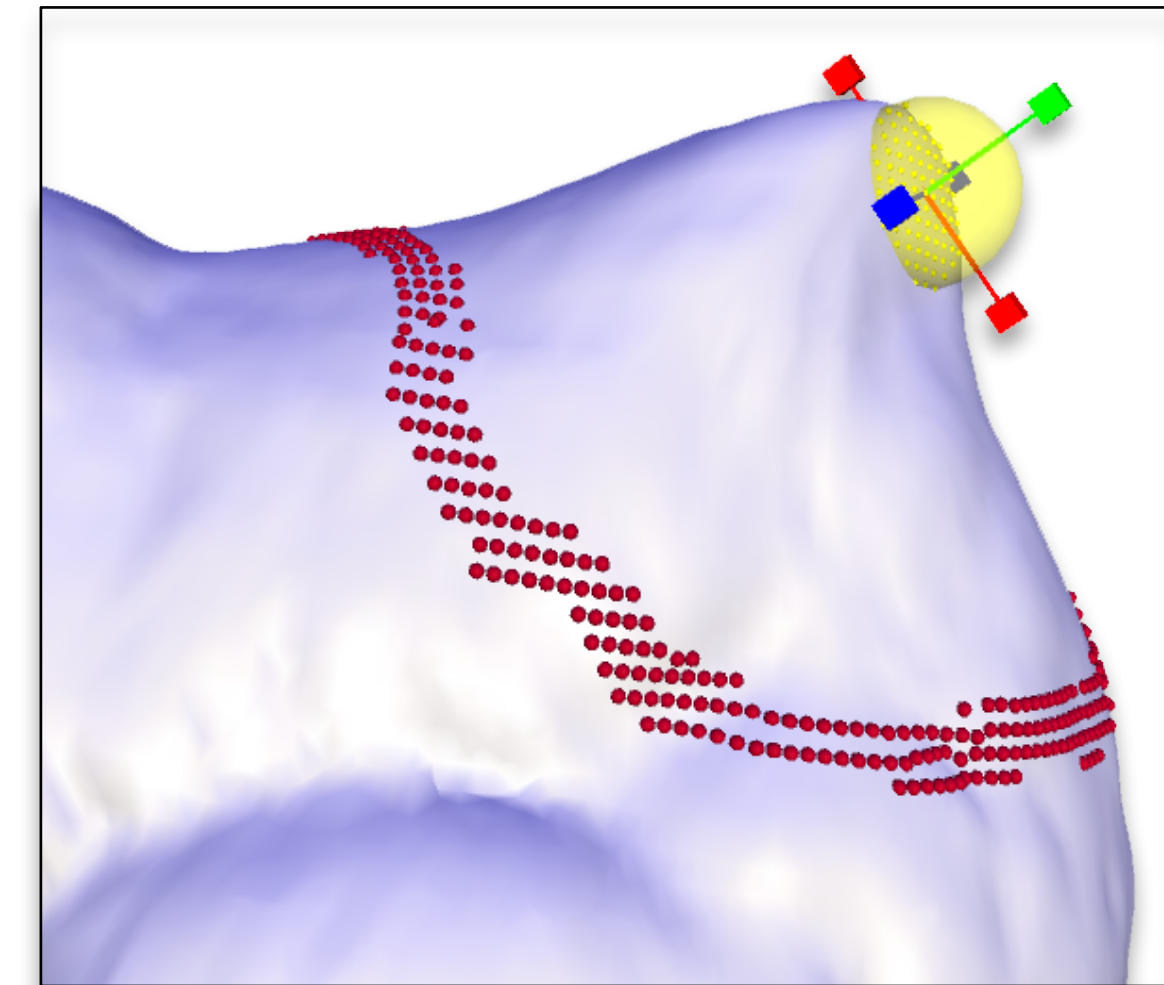
Parametric Curves and Surfaces

- Hard to change / adapt control structure to user needs
- Hard to experiment, need a precise idea of what will be modeled.



Mesh Deformation

- Naïve method: dragging single vertices
- Smarter:
 - Introduce a small set of deformation handles
 - Makes deformation/editing easier
 - Introduces a trade-off between degrees of freedom and simplicity of the deformation task
 - Create a small set of control parameters
 - Affine transformations



Common Paradigms

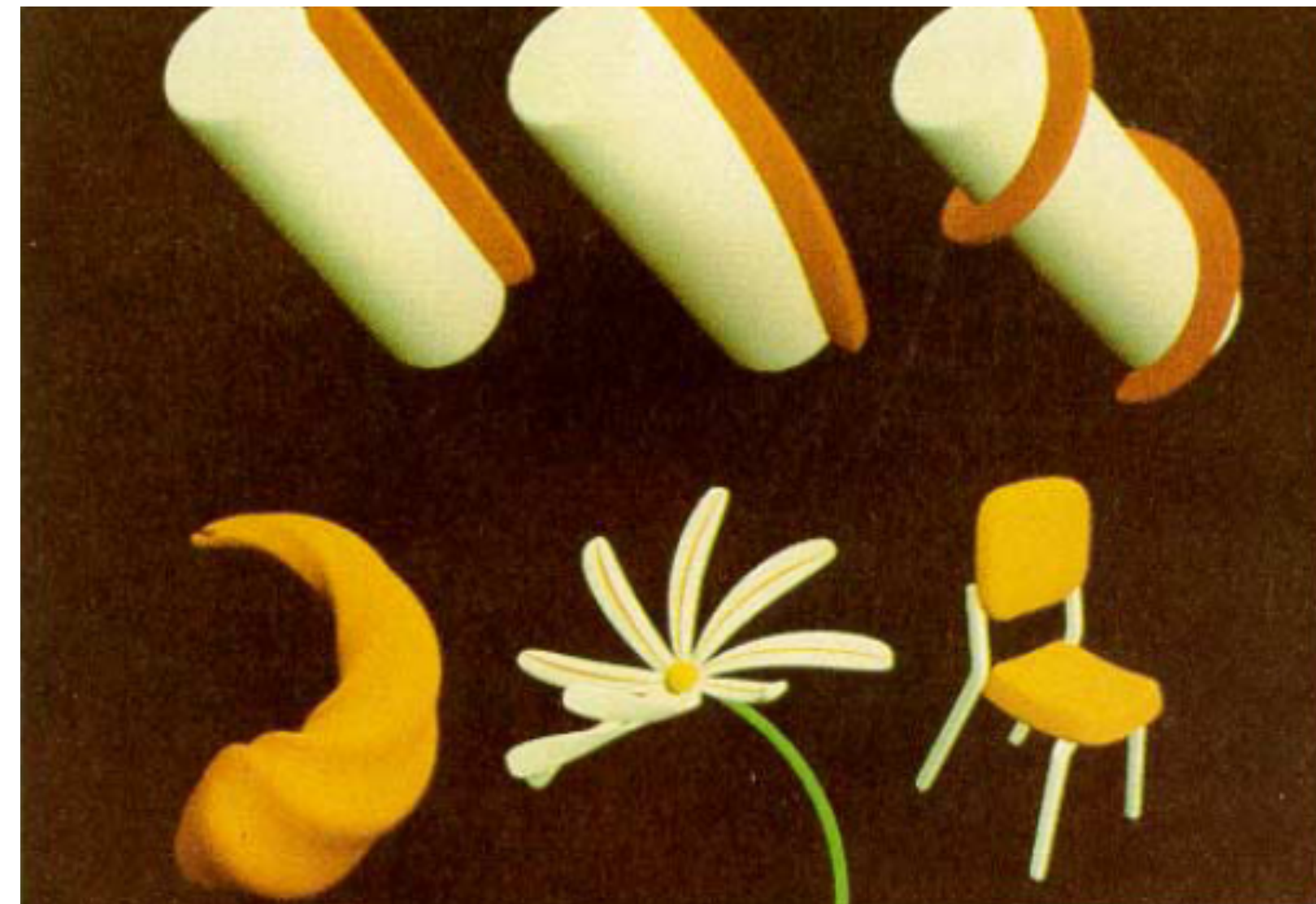
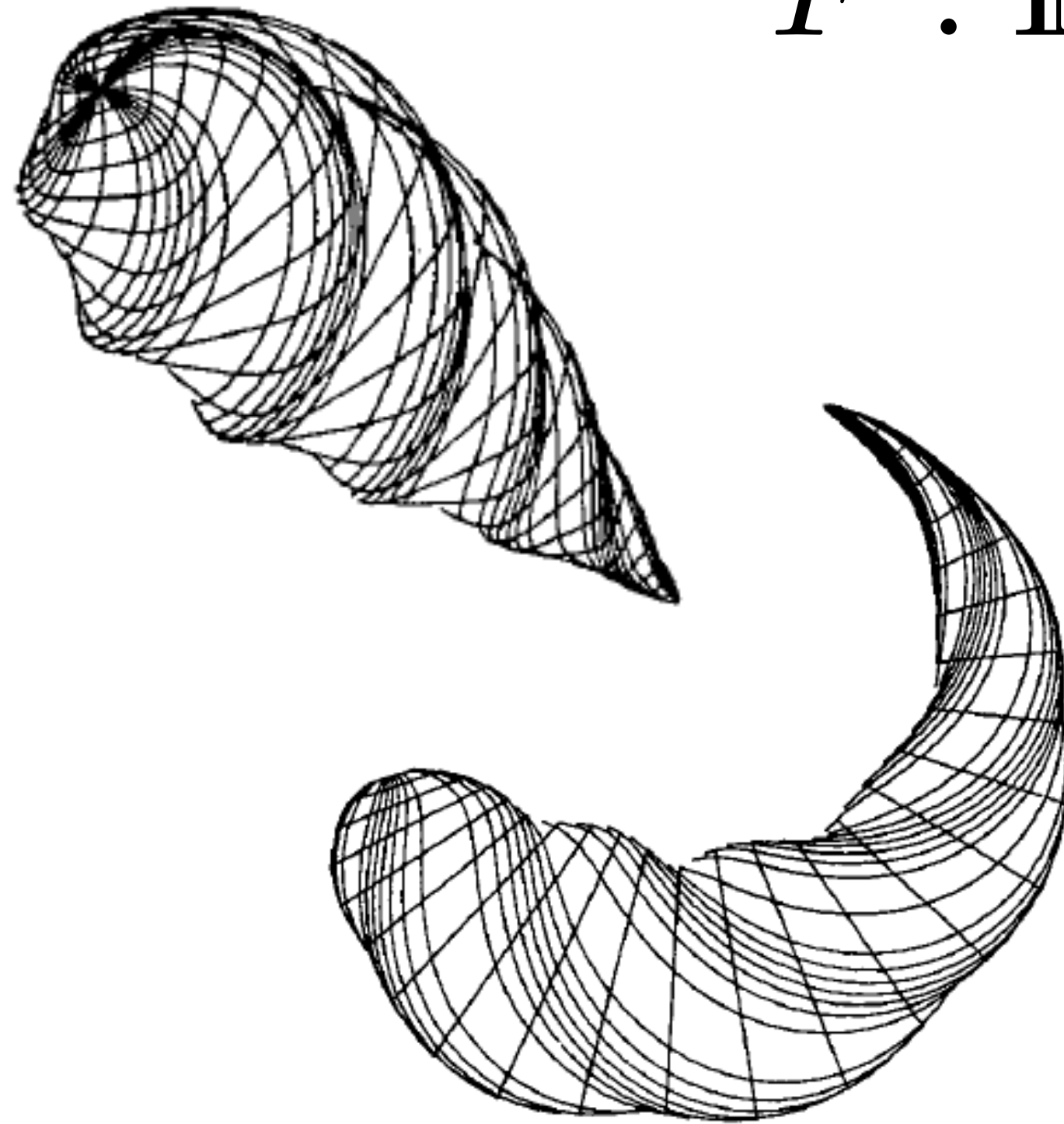
- Space deformations
 - Deforms some 2D/3D space using a *cage*
 - Deformation propagation to all points in the space
 - Independent of shape representation
- Surface based deformations
 - Optimization on the surface
 - Physically motivated: variants of elastic energy minimization

Space Deformations

Early seminal works in computer graphics

- Global and local deformation of solids [Barr 1984] <http://dl.acm.org/citation.cfm?id=808573>

$$F : \mathbb{R}^3 \rightarrow \mathbb{R}^3$$



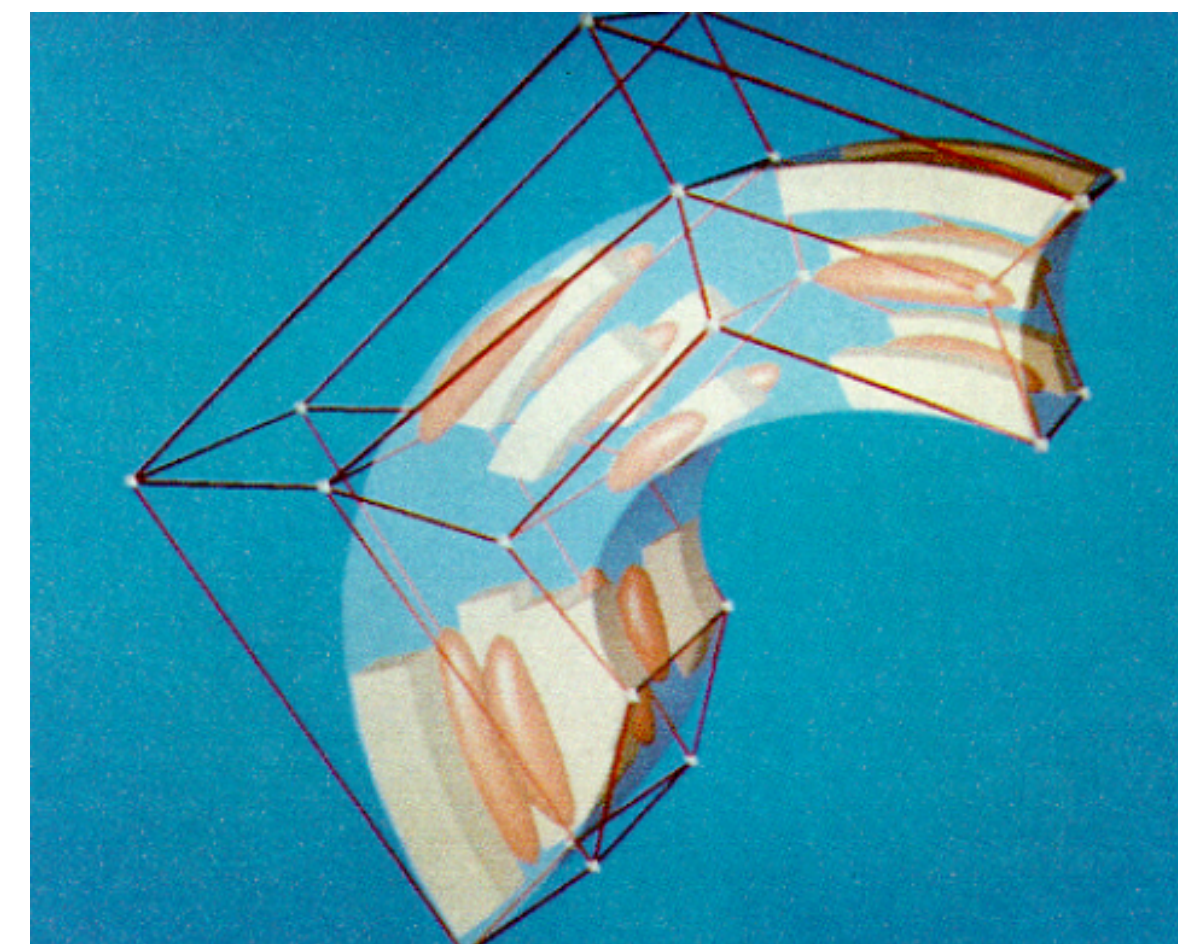
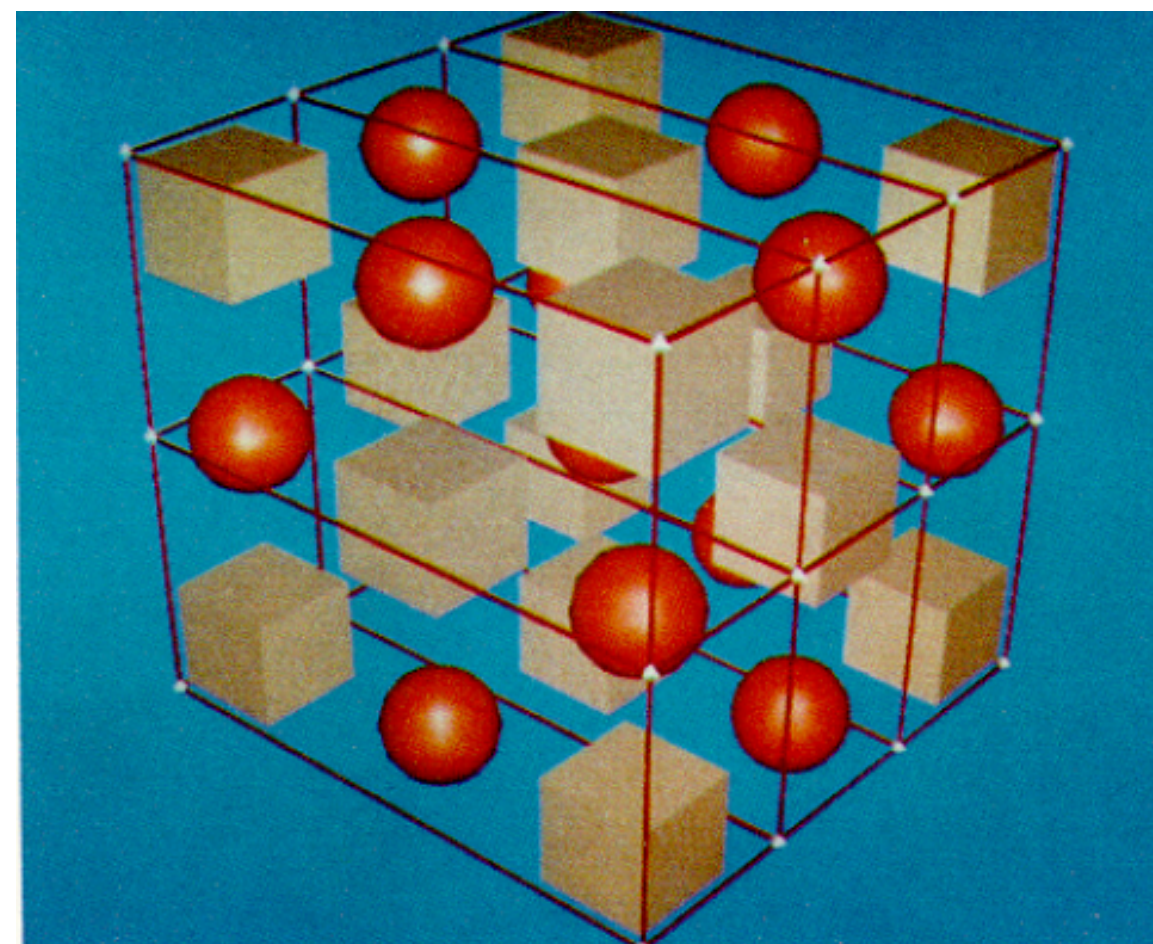
Early seminal works in computer graphics

- Free form deformations

[Sederberg and Parry 1986] <http://dl.acm.org/citation.cfm?id=15903>

- Uses trivariate tensor product polynomial basis

$$f : \mathbb{R}^3 \rightarrow \mathbb{R}^3$$



Early seminal works in computer graphics

- Can be designed to be volume preserving



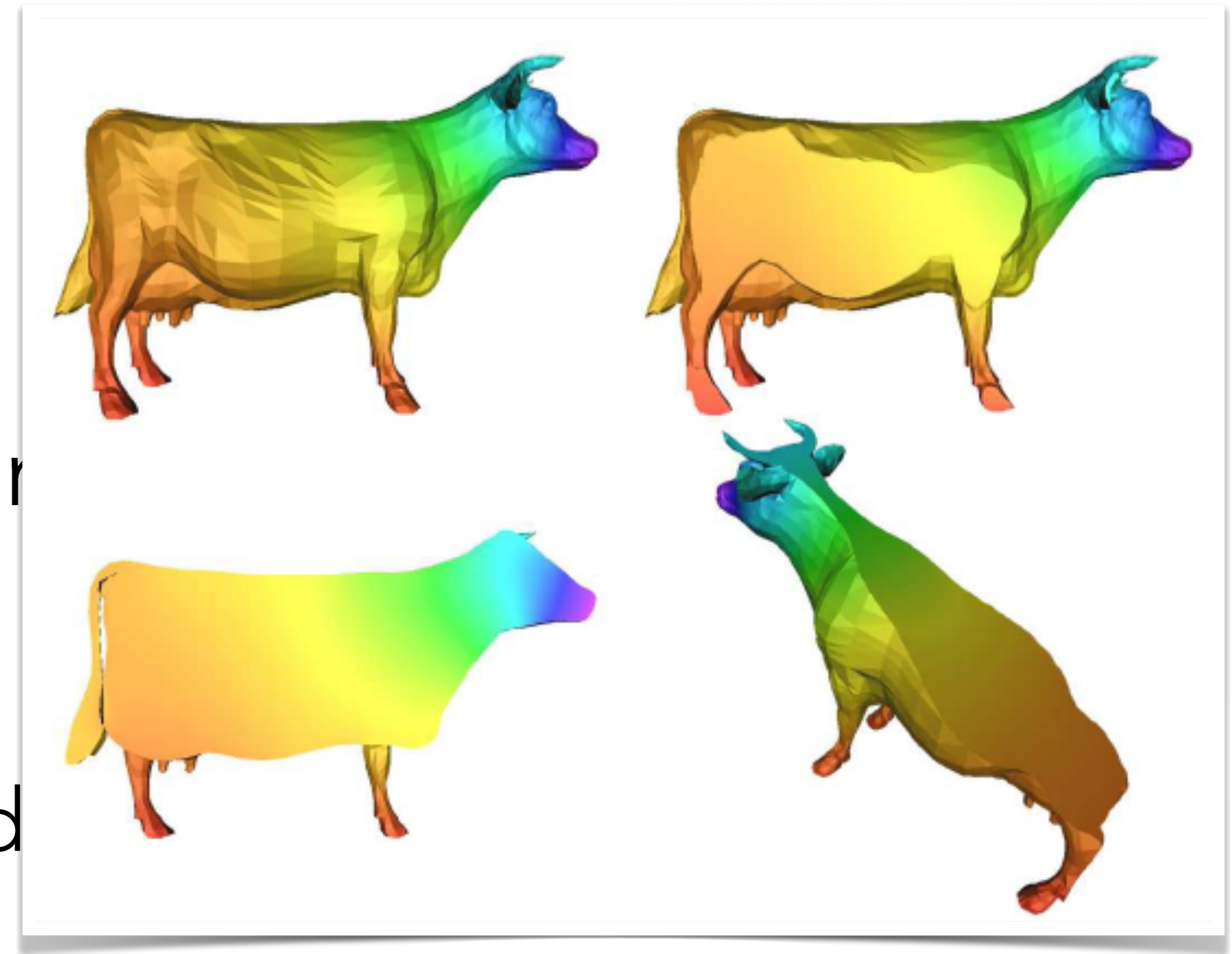
$$f : \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$\mathbf{F}(x,y,z) = (F(x,y,z), G(x,y,z), H(x,y,z))$$

then the Jacobian is the determinant

$$Jac(\mathbf{F}) = \begin{vmatrix} \frac{\partial F}{\partial x} & \frac{\partial F}{\partial y} & \frac{\partial F}{\partial z} \\ \frac{\partial G}{\partial x} & \frac{\partial G}{\partial y} & \frac{\partial G}{\partial z} \\ \frac{\partial H}{\partial x} & \frac{\partial H}{\partial y} & \frac{\partial H}{\partial z} \end{vmatrix}$$

Basic Idea



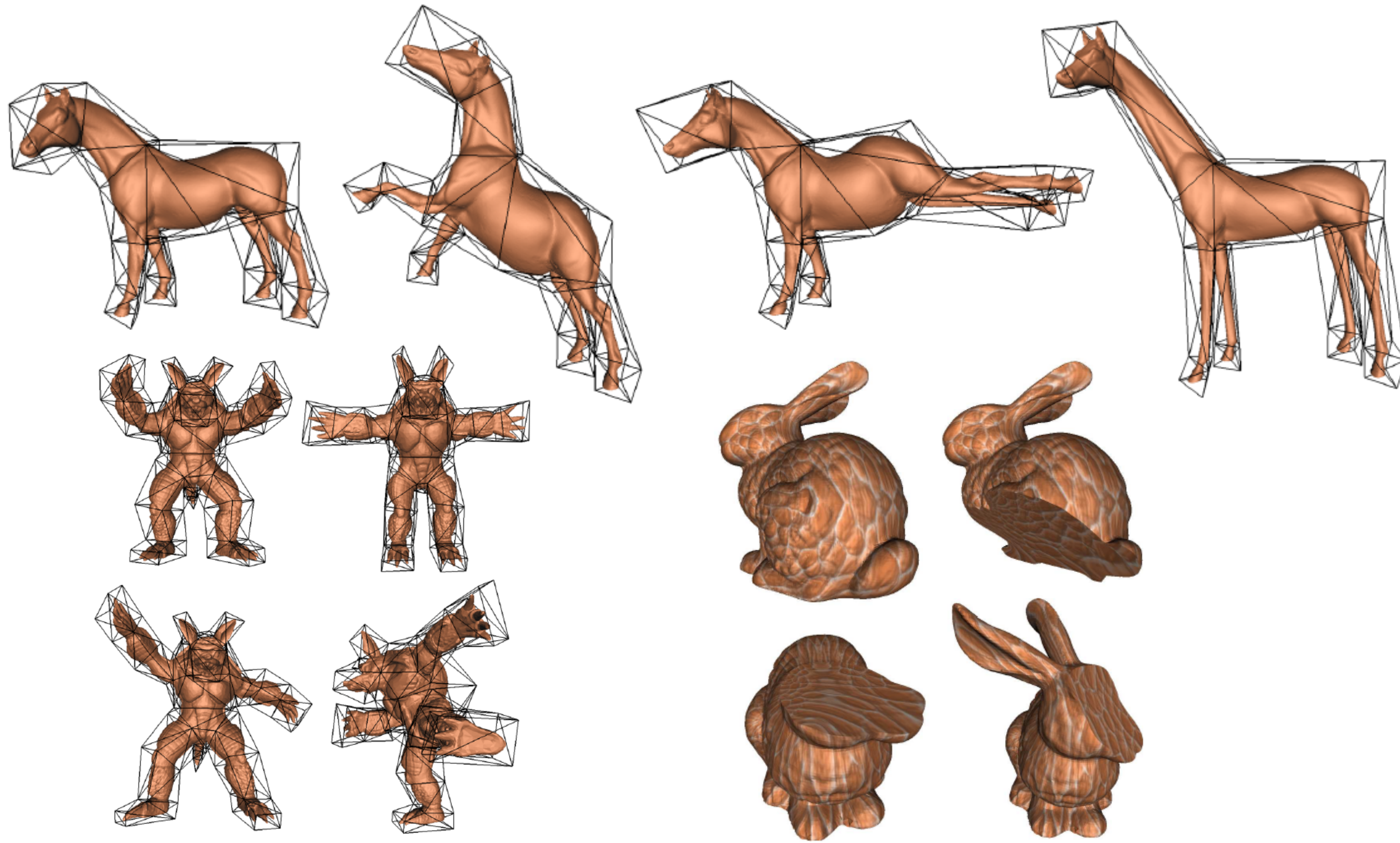
- Design a set of coordinates for all points in the cage w.r.t. the "cage" vertices
- Each point $\mathbf{x}$ can be represented as a weighted sum of the cage vertices

$$\mathbf{x} = \sum_{i=1}^k w_i(\mathbf{x}) \mathbf{p}_i$$

- When the cage changes, the coords stay the same, substitute the new cage geometry:

$$\mathbf{x}' = \sum_{i=1}^k w_i(\mathbf{x}) \mathbf{p}'_i$$

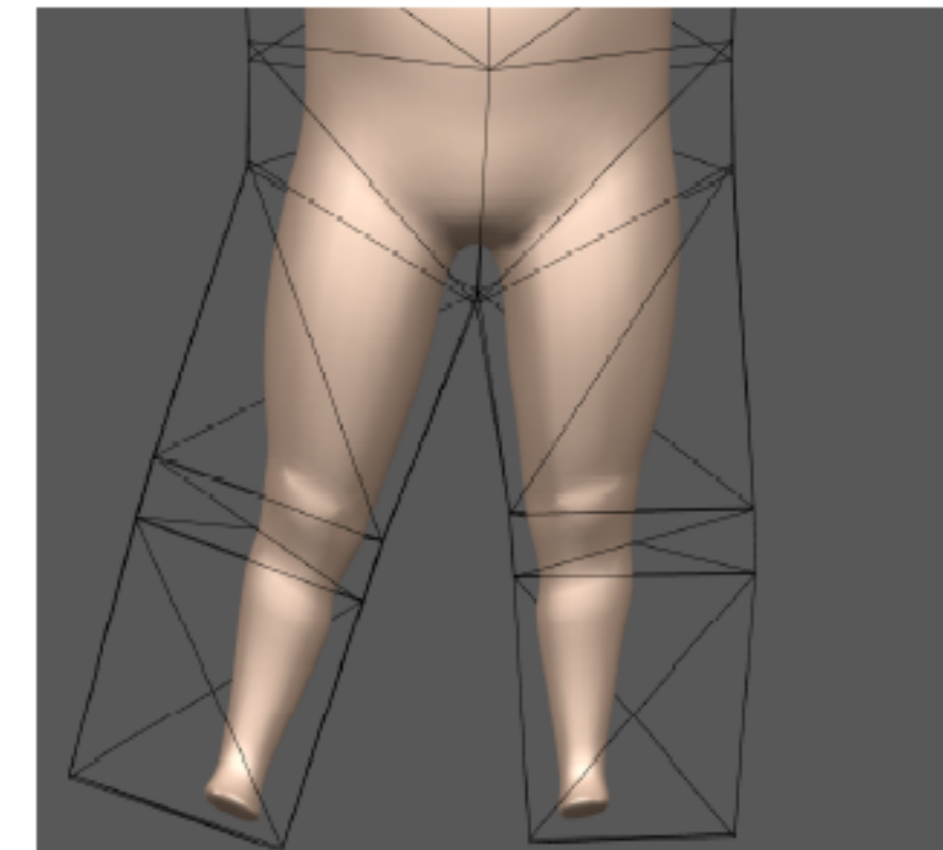
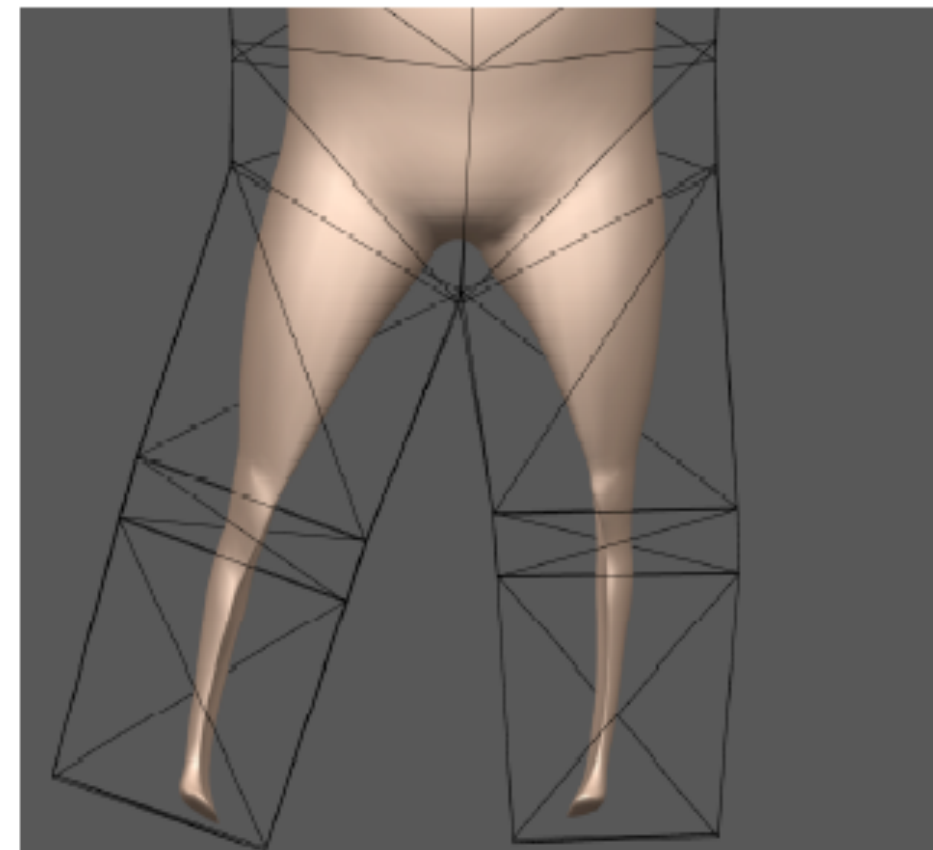
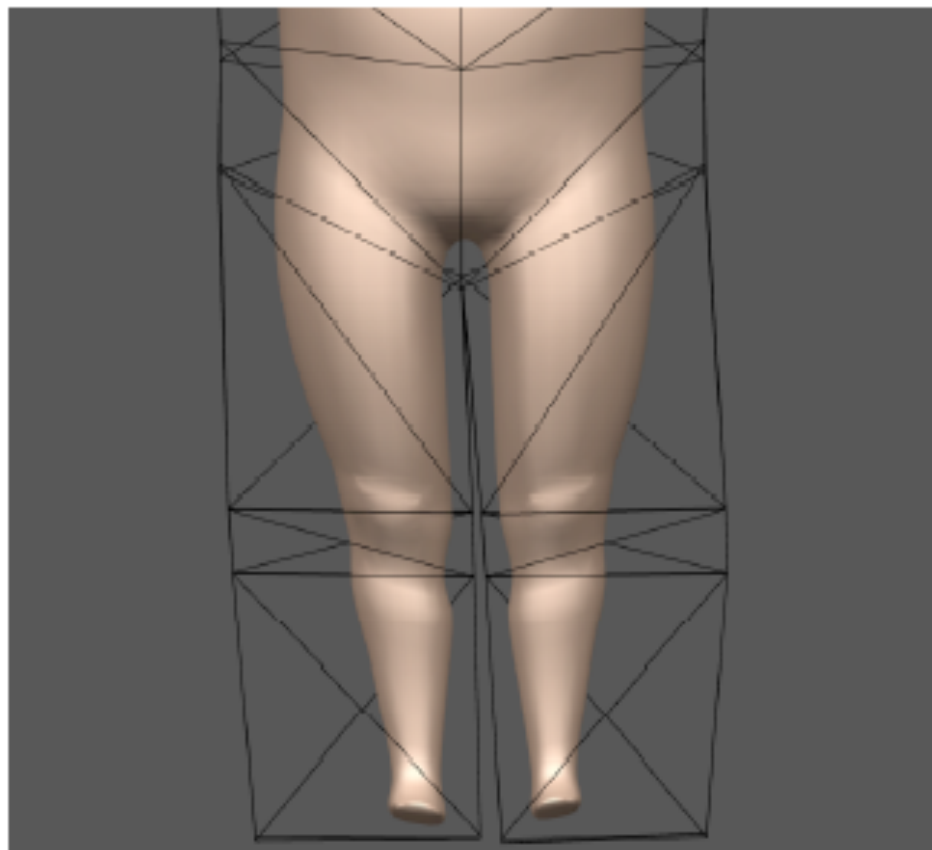
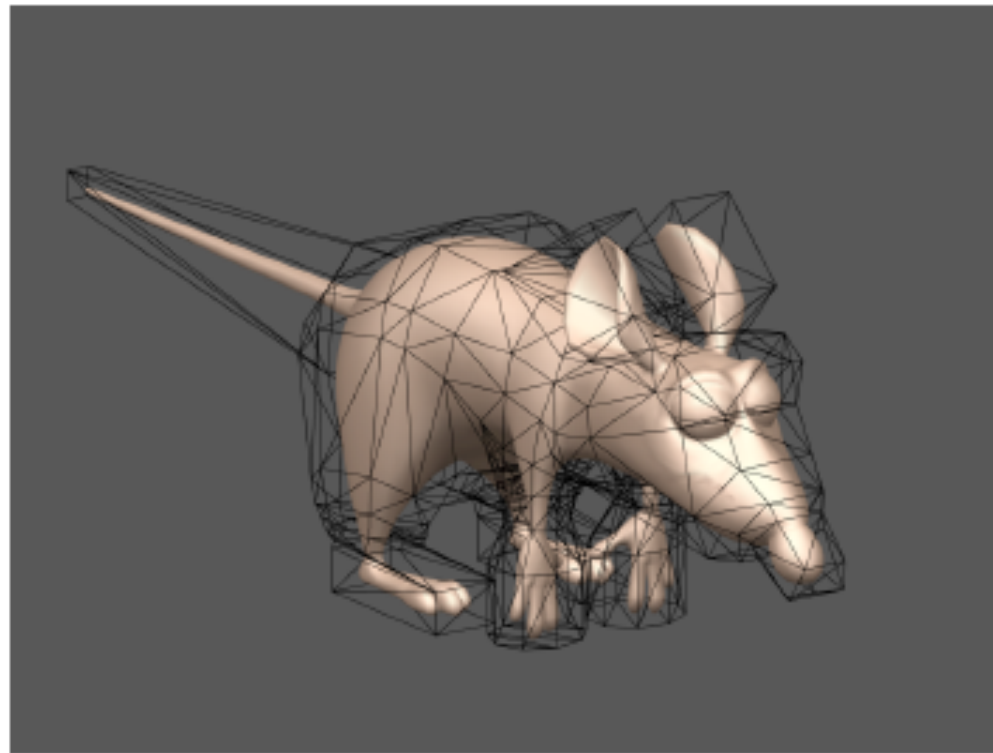
Mean Value Coordinates



[Ju et al. 2005]

<http://dl.acm.org/citation.cfm?id=1186822.1073229>

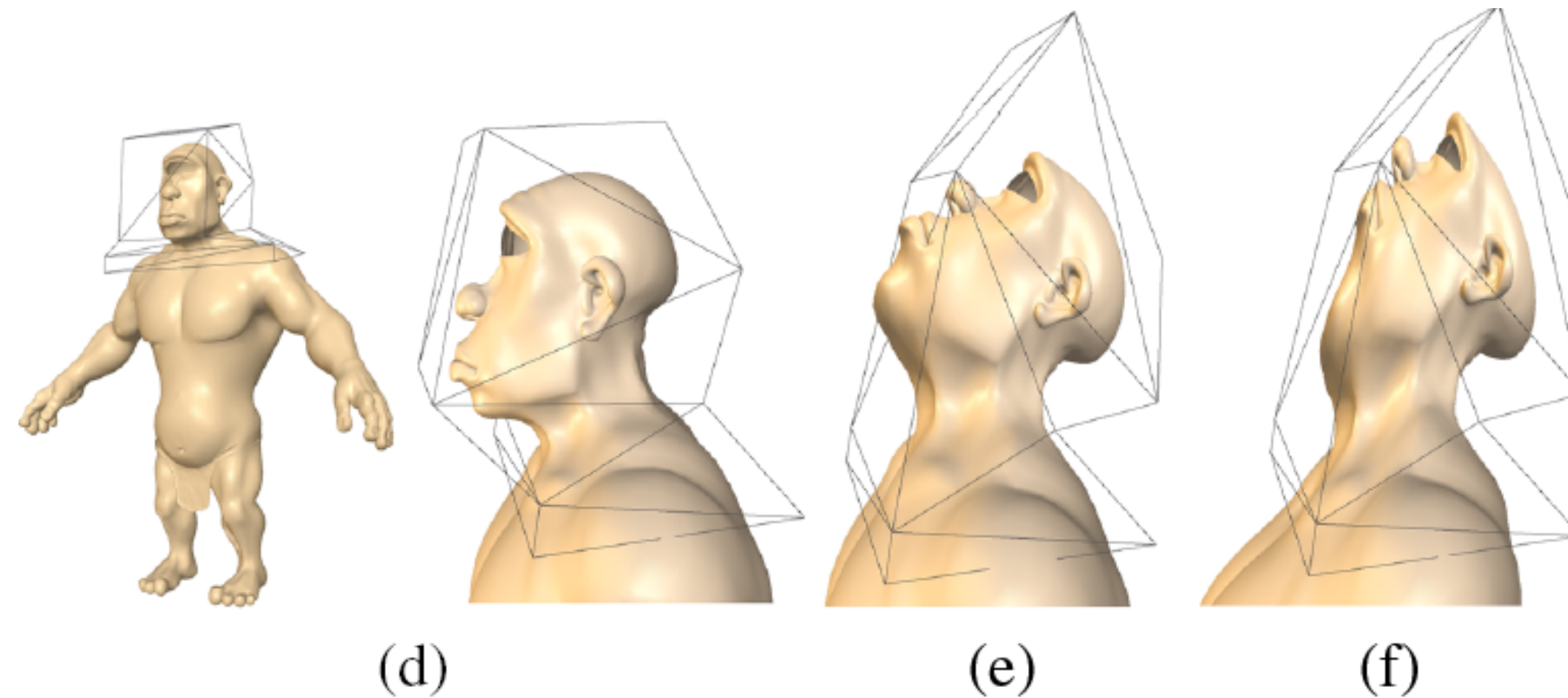
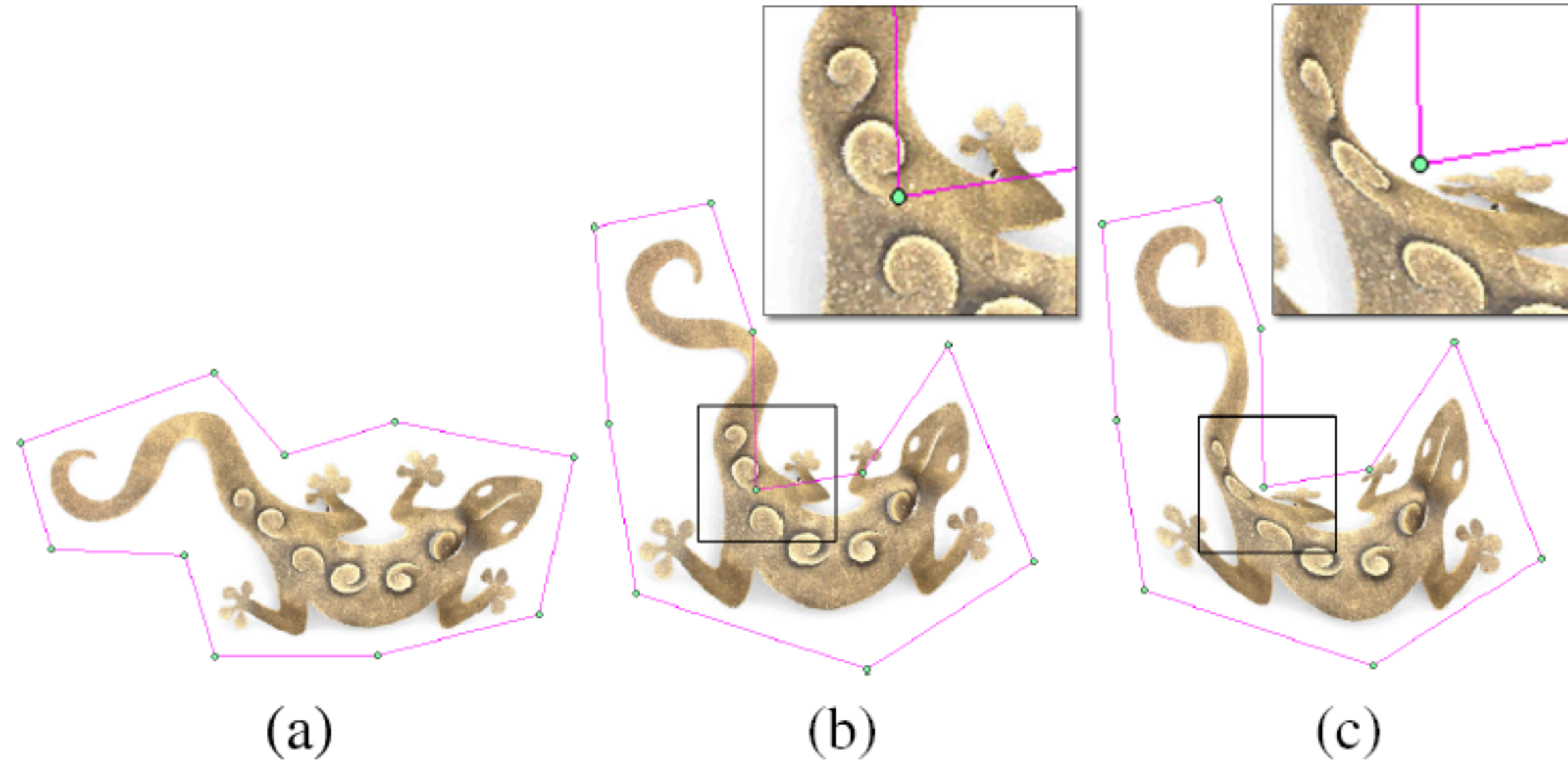
Harmonic Coordinates



[Joshi et al. 2007]

<http://dl.acm.org/citation.cfm?id=1276466>

Green Coordinates



<http://dl.acm.org/citation.cfm?id=1360677>



**University
of Victoria**

Computer Science

Space Deformations

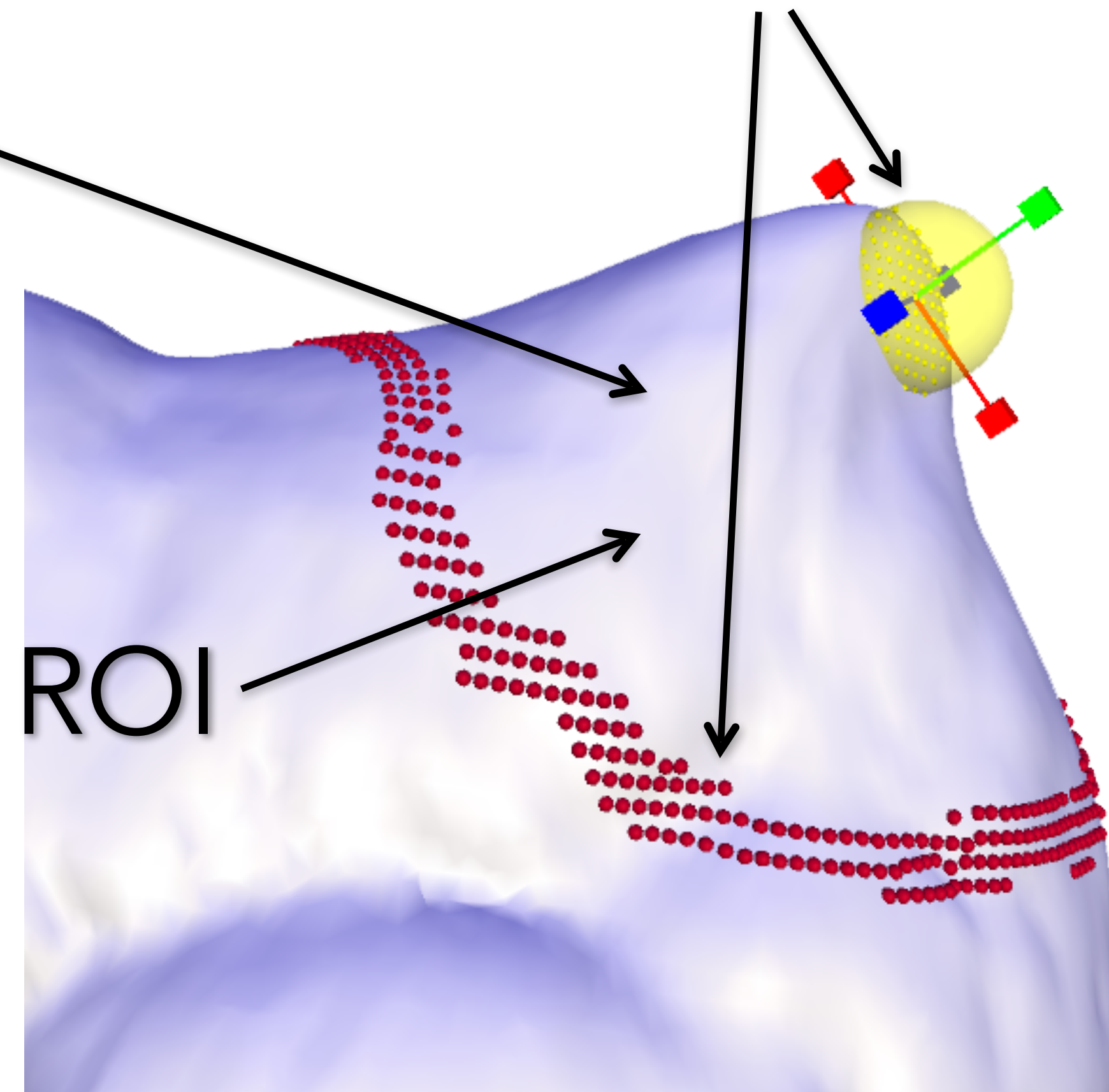
- Complexity depends mainly on the cage; linear in the number of mesh elements
 - Parallel execution with GPU accelerators
- Can handle disconnected components or even just point sets
- Harder to control the surface properties since the whole space is being warped

Surface-Based Deformations

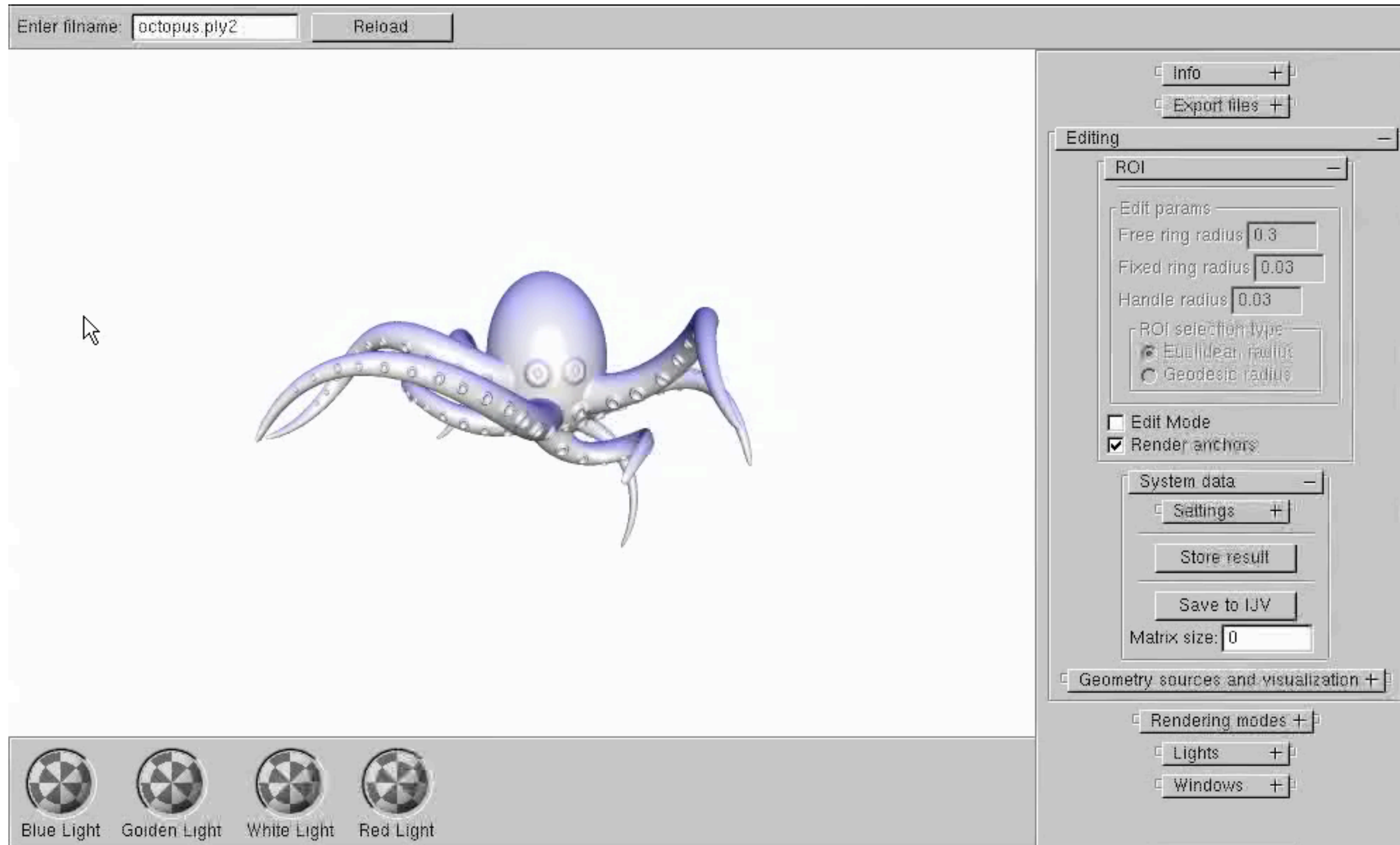
Surface-based Deformation: ROI-Handle Editing Metaphor

$$\mathbf{x}_{\text{def}} = \underset{\mathbf{x}'}{\operatorname{argmin}} E(\mathbf{x}') \quad s.t. \quad \mathbf{x}'_i = \mathbf{c}_i$$

- ROI is bounded by a belt (static anchors)
- Manipulation through handle(s) – affine transformations

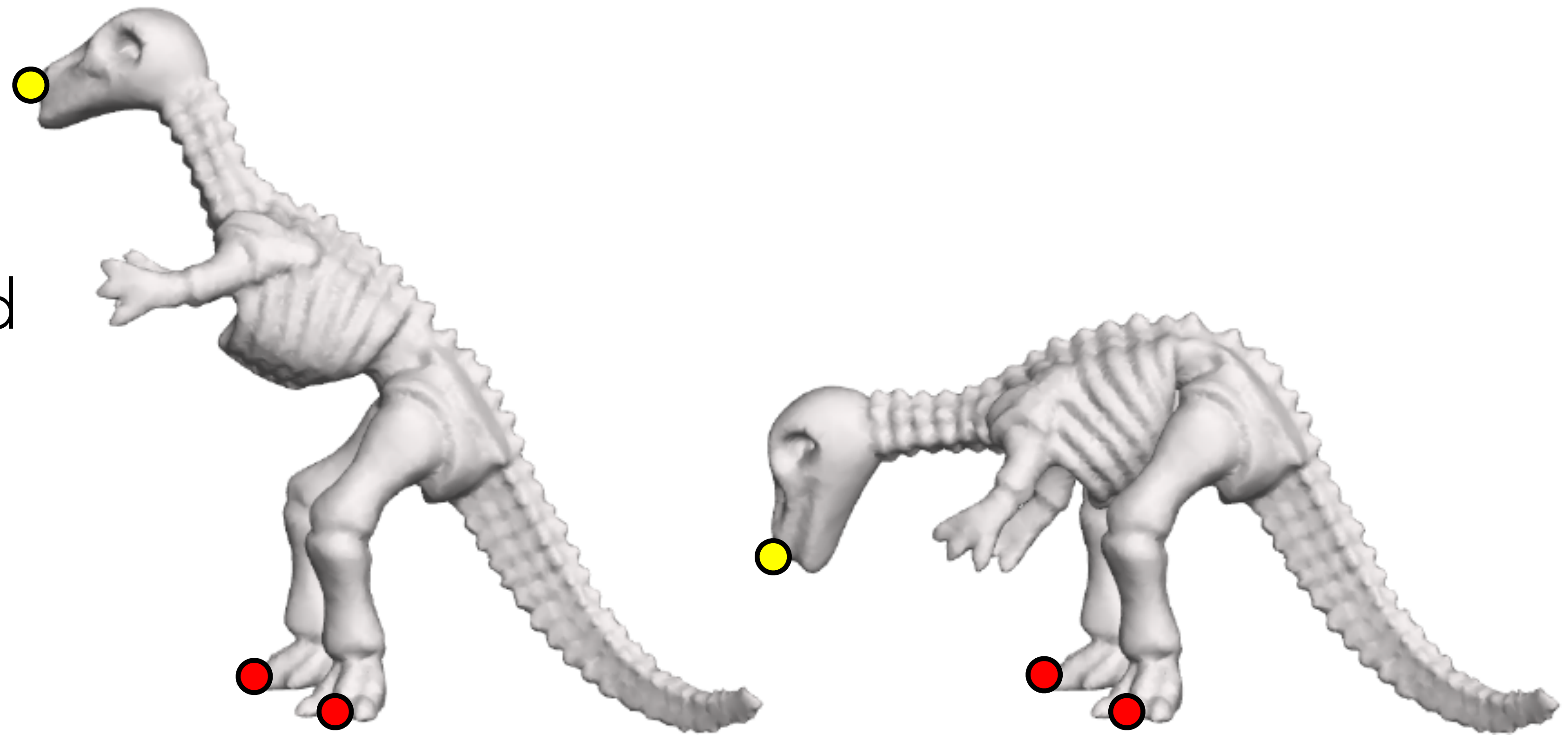


Surface-based Deformation: ROI-Handle Editing Metaphor



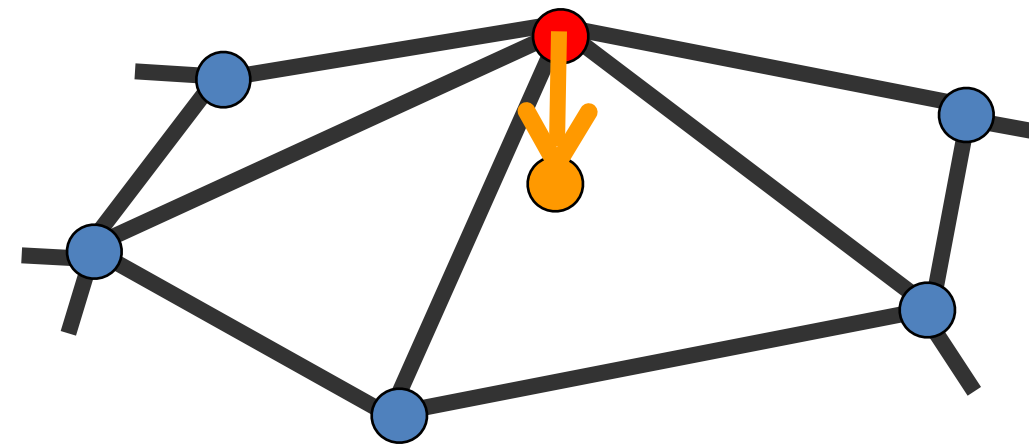
How to Define $E(\mathbf{x}')$?

- Intuitive deformations:
 - Smooth deformation on the global scale
 - Preserve local details (curvatures)
- Invariants: $E(\mathbf{x}')$ should be zero if $\mathbf{x}'$ is a rigid $\mathbf{x}$ transformation of original geometry



Recap: Differential Coordinates

- Detail = *smooth*(surface) – surface
- Smoothing = averaging



$$\delta_i = \frac{1}{W_i} \sum_{j \in \mathcal{N}(i)} w_{ij} (\mathbf{x}_j - \mathbf{x}_i) \approx -2H_i \mathbf{n}_i$$

Recap: Differential Coordinates

- Represent ***local detail*** at each surface point
 - More descriptive of the shape than just xyz
- Linear transition from xyz to δ
- Useful for operations on surfaces where surface details are important



Simple Laplacian Editing

- Preserve mean curvature normal [$\approx$ differential coordinates] at every point in the ROI [$\approx$ every vertex of the ROI]

continuous: $E(\mathcal{S}') = \int_{\mathcal{S}'} \|\Delta \mathbf{x}' - \delta\|^2 d\mathbf{x}'$

discrete: $E(\mathbf{x}') = \sum_{i=1}^n A_i \|\Delta(\mathbf{x}'_i) - \delta_i\|^2$

Simplifying the Laplacian Energy

$$\begin{aligned} E(\mathbf{x}') &= \sum_{i=1}^n A_i \|\Delta(\mathbf{x}'_i) - \delta_i\|^2 = \sum_{i=1}^n A_i (\Delta(\mathbf{x}'_i)^T \Delta(\mathbf{x}'_i) - 2\Delta(\mathbf{x}'_i)^T \delta_i + \delta_i^T \delta_i) = \\ &= \mathbf{x}'^T \underline{L^T M L} \mathbf{x}' - 2\mathbf{x}'^T \underline{L^T M} \delta + \text{const} \end{aligned}$$

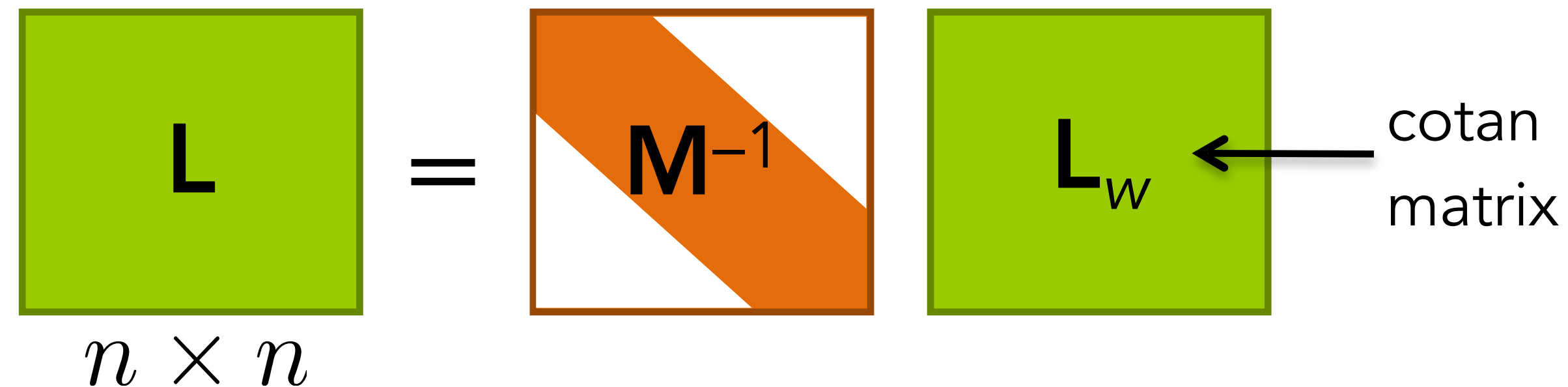
The diagram illustrates the decomposition of the Laplacian matrix L into the product of two matrices: M^{-1} and L_w . On the left, a green square labeled L is shown with the dimension $n \times n$ below it. This is followed by an equals sign. To the right of the equals sign is an orange square labeled M^{-1} , which has a diagonal line from the top-left to the bottom-right. To the right of this is another green square labeled L_w . An arrow points from the text "cotan matrix" to the L_w matrix.

$$\begin{matrix} \text{Green box } L \\ n \times n \end{matrix} = \begin{matrix} \text{Orange box } M^{-1} \end{matrix} \begin{matrix} \text{Green box } L_w \\ \leftarrow \text{cotan matrix} \end{matrix}$$

Simplifying the Laplacian Energy

$$E(\mathbf{x}') = \sum_{i=1}^n A_i \|\Delta(\mathbf{x}'_i) - \delta_i\|^2 = \sum_{i=1}^n A_i (\Delta(\mathbf{x}'_i)^T \Delta(\mathbf{x}'_i) - 2\Delta(\mathbf{x}'_i)^T \delta_i + \delta_i^T \delta_i) =$$

$$= \mathbf{x}'^T \underline{L^T M L} \mathbf{x}' - 2\mathbf{x}'^T \underline{L^T M} \delta + \text{const}$$



$n \times n$

cotan matrix

$$L^T M L = (M^{-1} L_w)^T M (M^{-1} L_w) = L_w M^{-1} M M^{-1} L_w =$$

$$= L_w M^{-1} L_w \longleftarrow \text{Symmetric sparse matrix!}$$

Minimizing the Laplacian Energy

- To find the minimum, gradient = 0 and substitute the modeling constraints

$$E(\mathbf{x}') = \mathbf{x}'^T L_w M^{-1} L_w \mathbf{x}' - 2\mathbf{x}'^T L_w \delta + \text{const}$$

$$\frac{\partial}{\partial \mathbf{x}'} E(\mathbf{x}') = 2L_w M^{-1} L_w \mathbf{x}' - 2L_w \delta$$

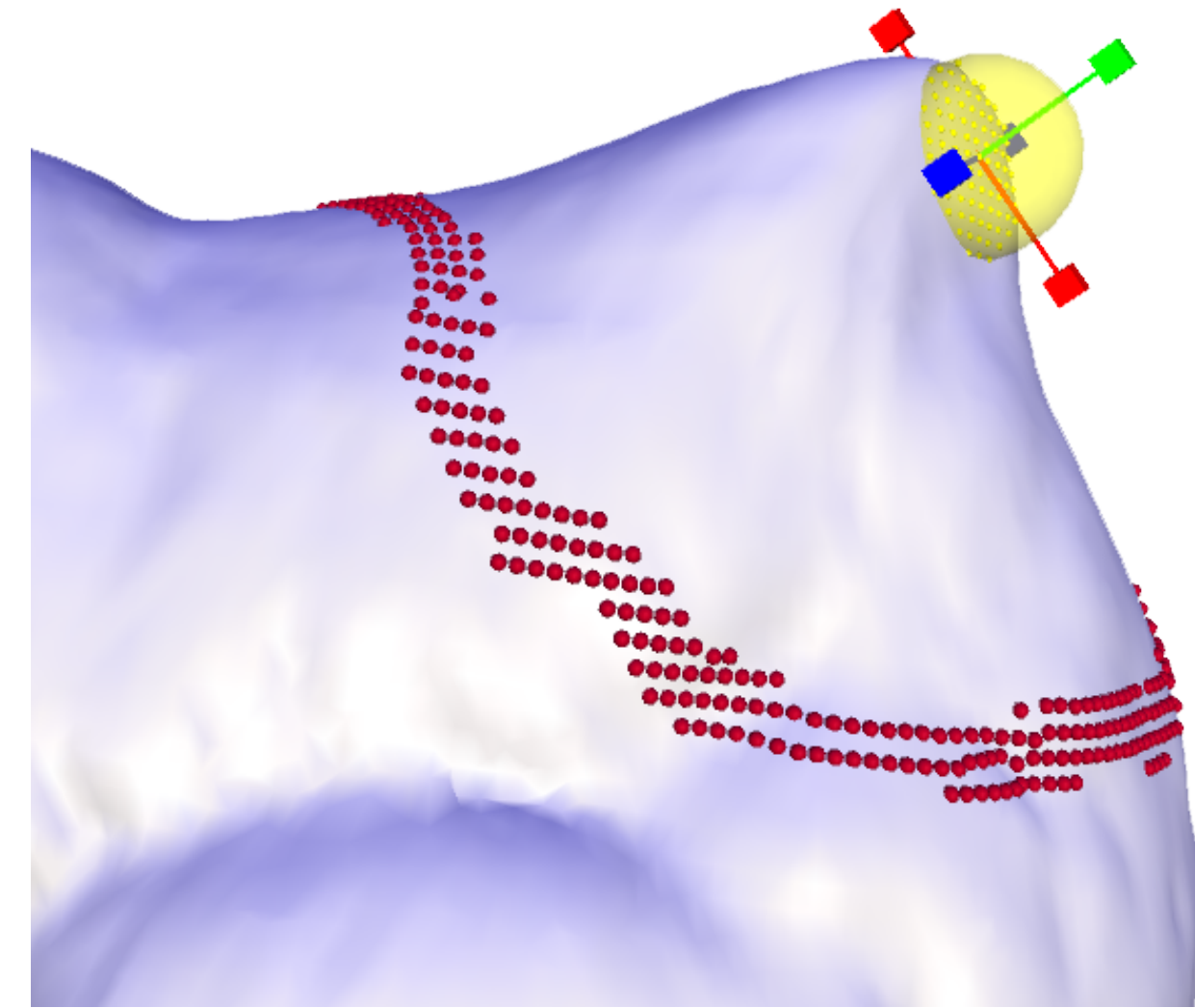
$$\mathbf{x}'_i = \mathbf{c}_i, \quad i \in \mathcal{C}$$

Minimizing the Laplacian Energy

$$Ax' = b$$

Matrix depends on the
initial mesh and the indices
of the constraints only.
Matrix is fixed!

Right-hand side contains
the coordinates of the
constraints (handles)

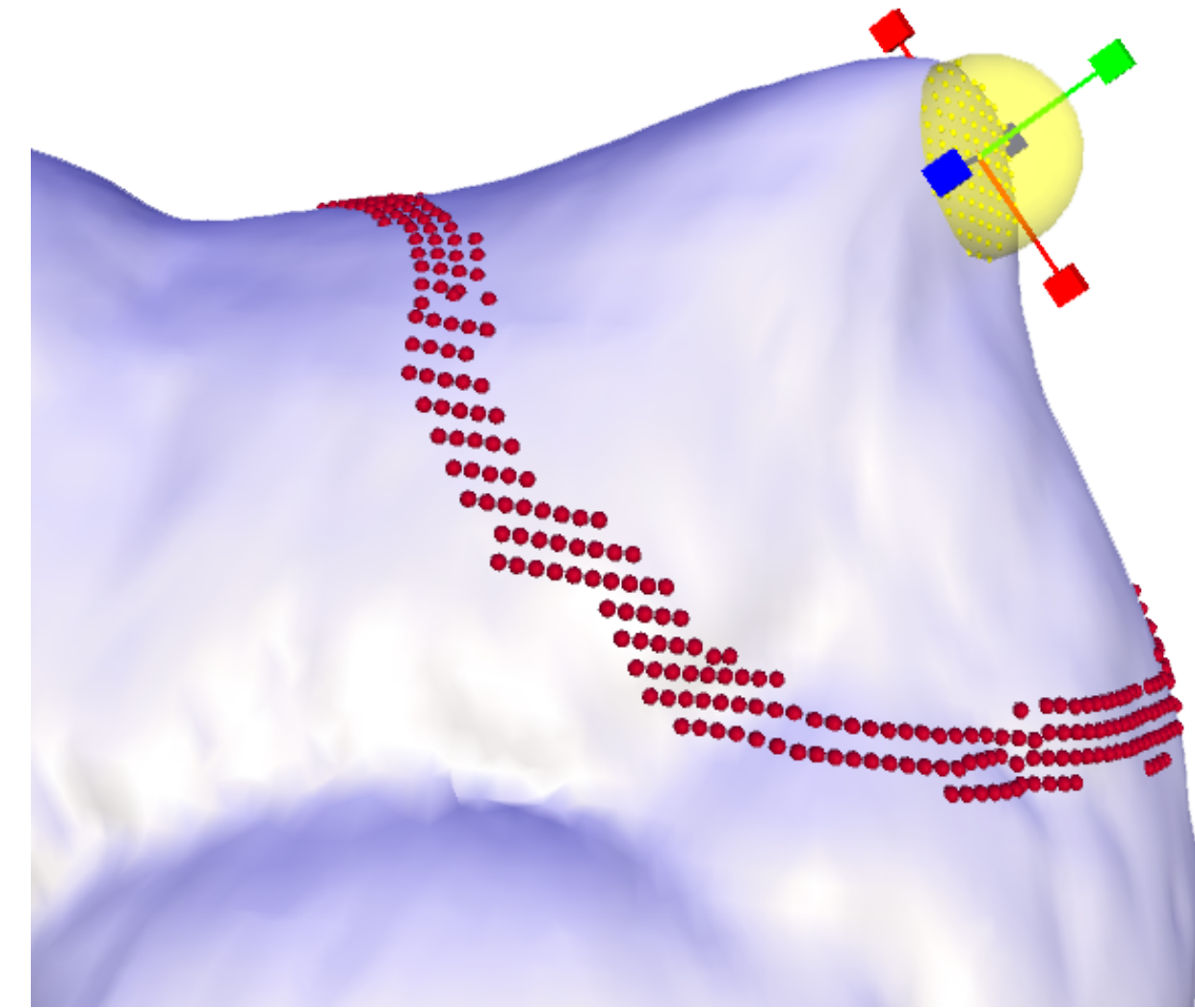
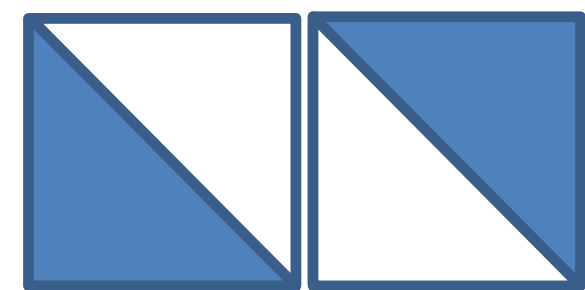


Minimizing the Laplacian Energy

$$Ax' = b$$

Sparse Cholesky
decomposition:

$$A = L_{\text{chol}} L_{\text{chol}}^T$$



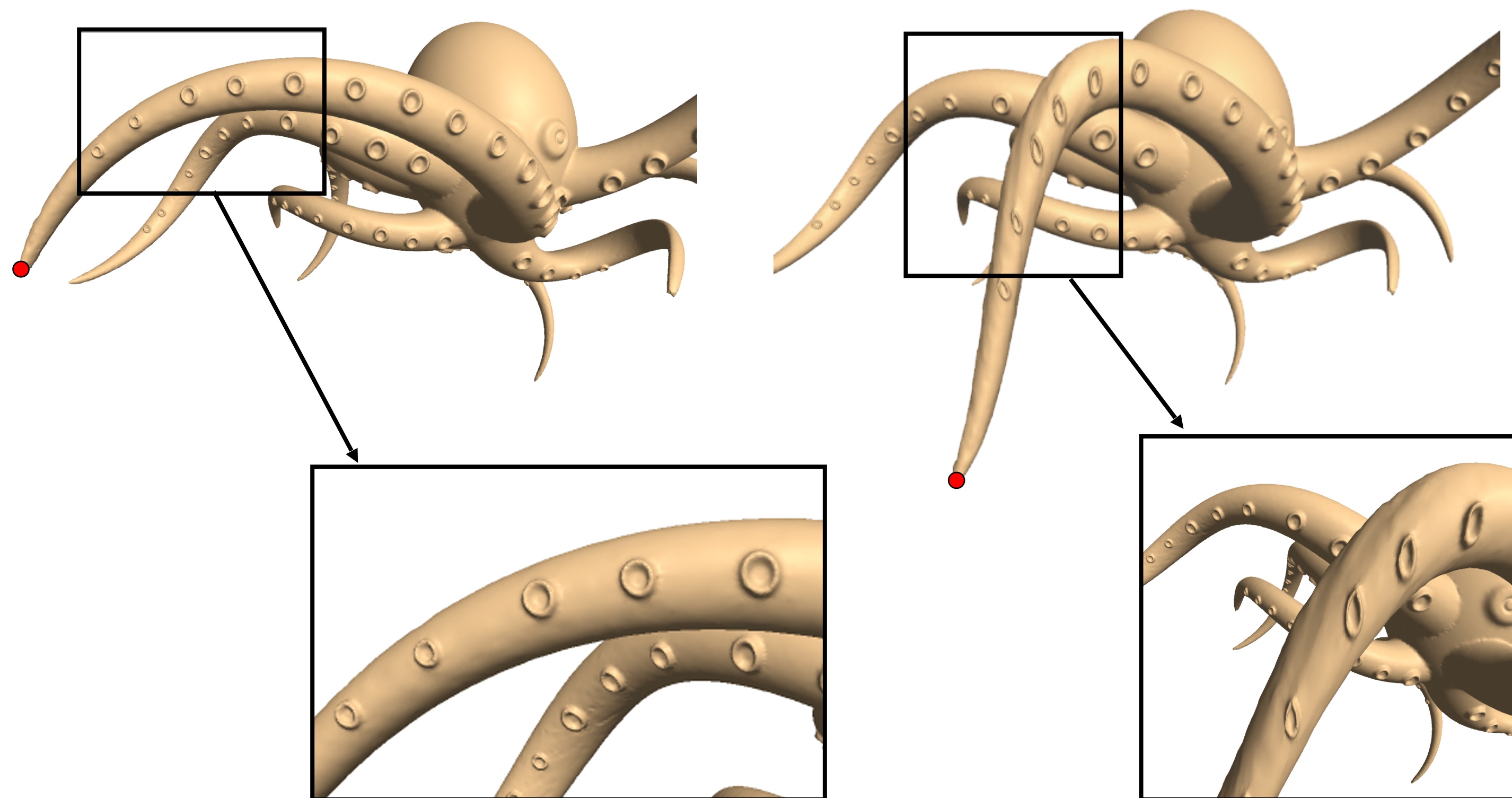
At run-time: just back-substitution!

$$L_{\text{chol}} \mathbf{y} = \mathbf{b}$$

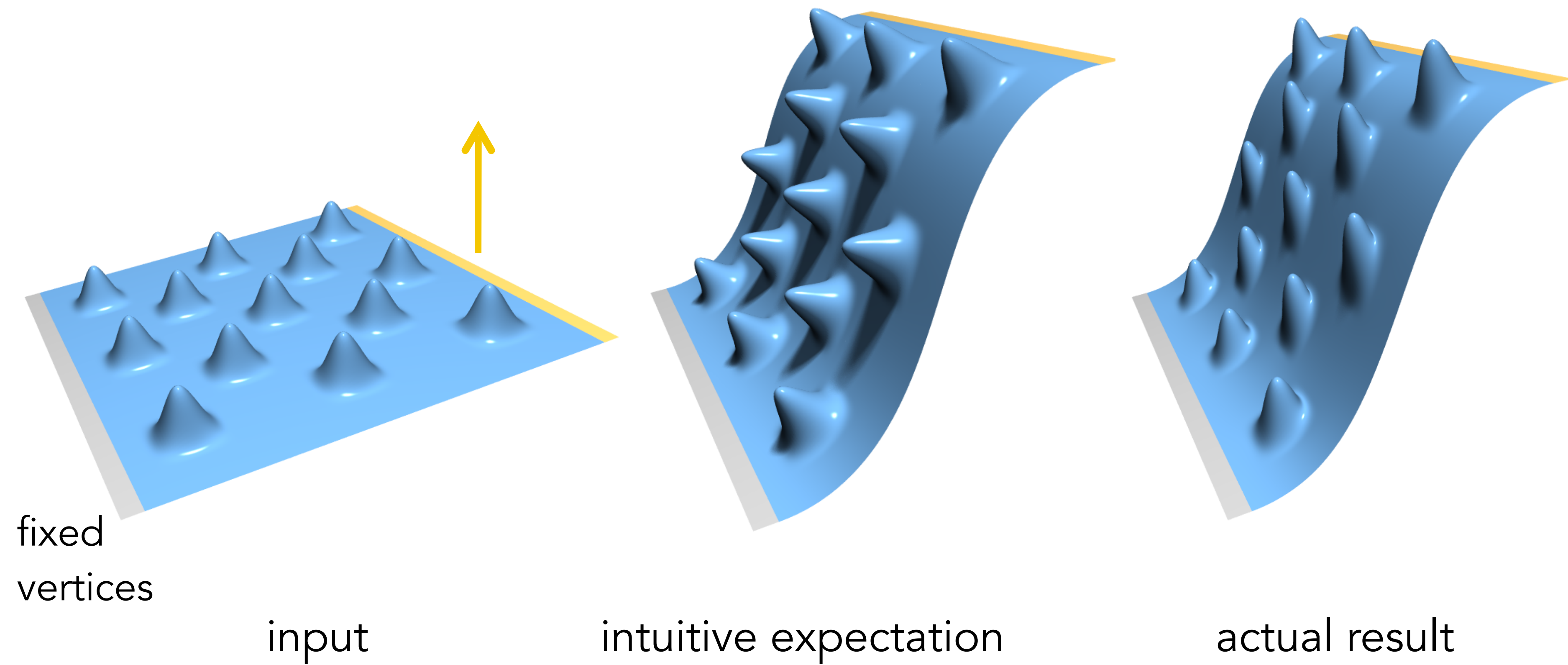
$$L_{\text{chol}}^T \mathbf{x}' = \mathbf{y}$$

Fundamental Problem: Invariance to Transformations

- The basic Laplacian operator is **translation**-invariant, but not **rotation**-invariant
- $E(\mathbf{x}')$ attempts to preserve the **original global** orientation of the details (the normal directions)

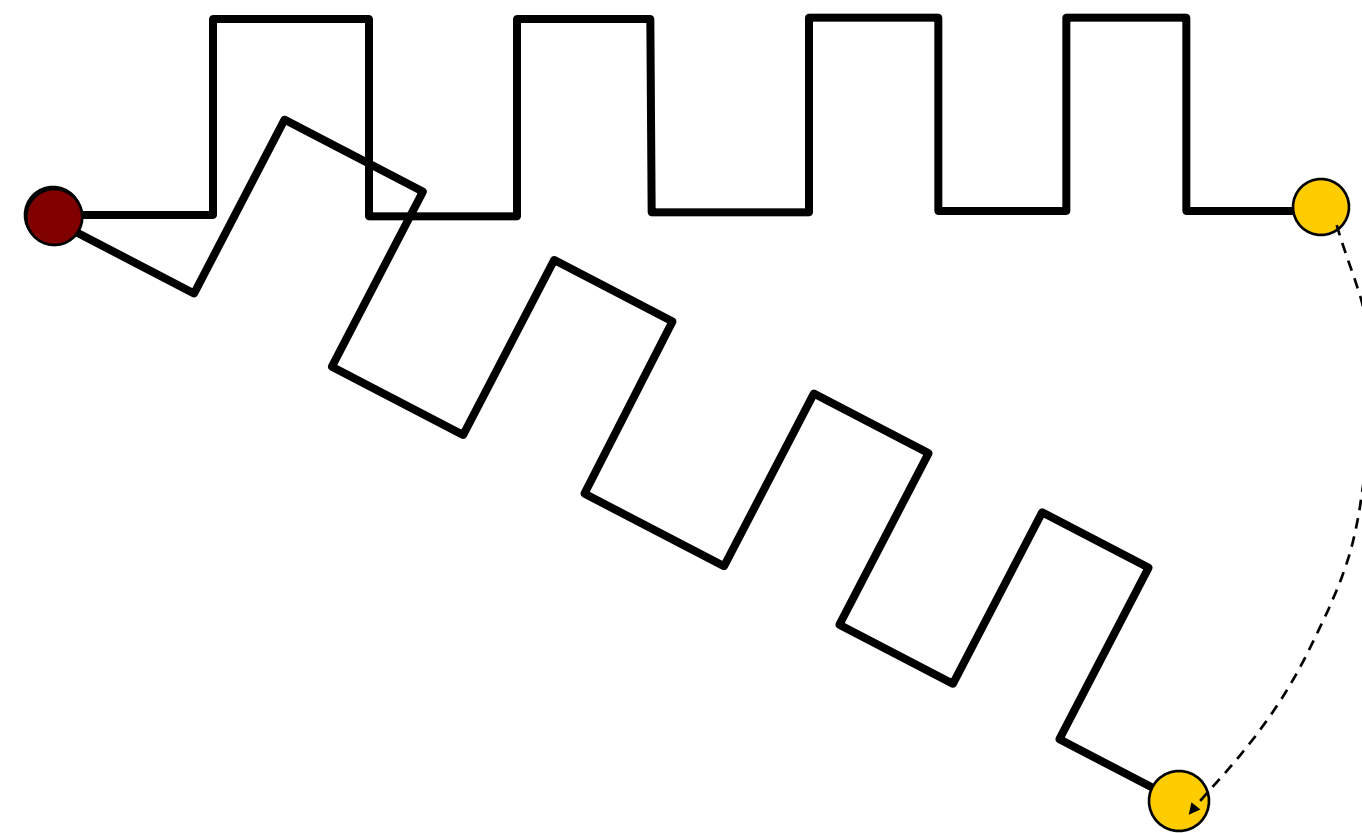


Fundamental Problem: Invariance to Transformations



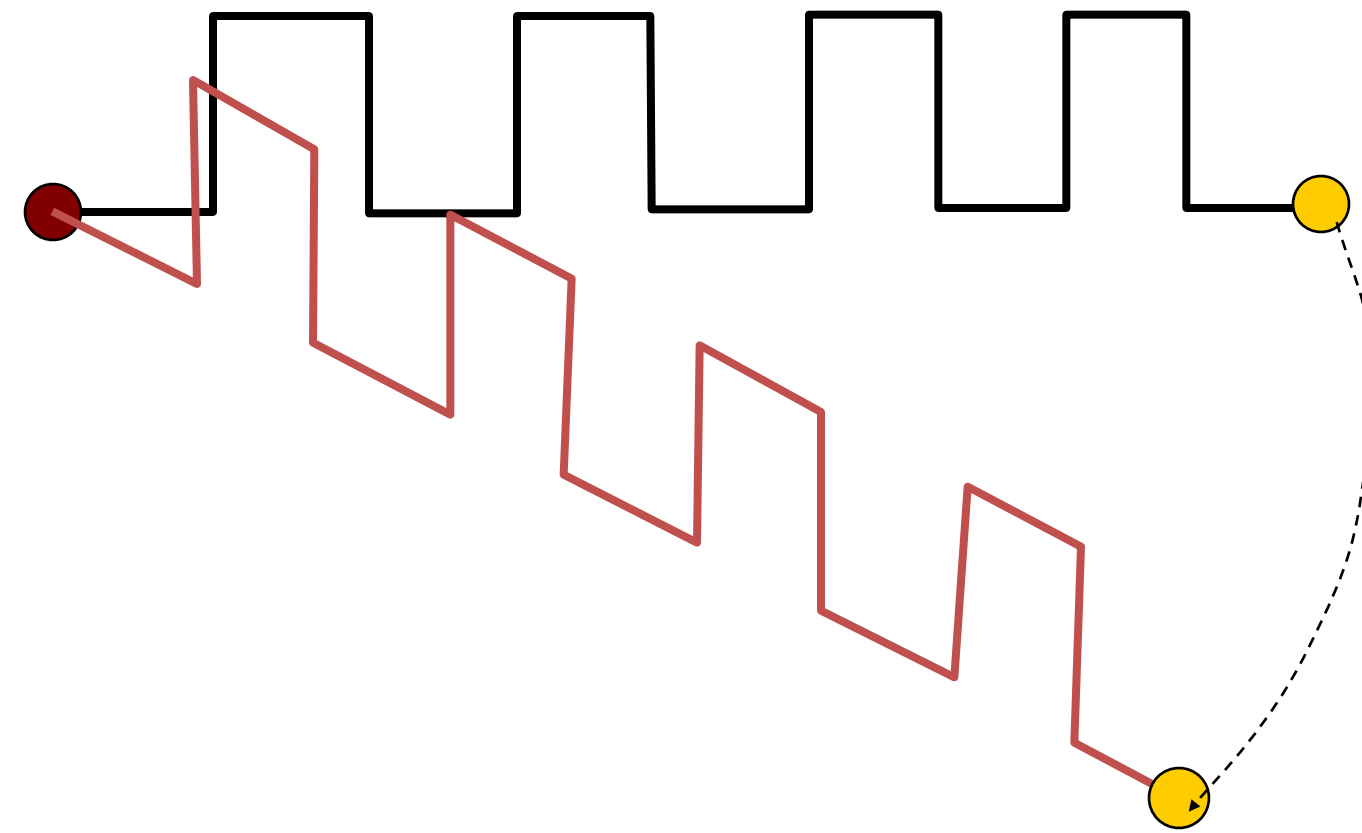
Fundamental Problem: Invariance to Transformations

- The basic Laplacian operator is **translation**-invariant, but not **rotation**-invariant
- $E(\mathbf{x}')$ attempts to preserve the **original global** orientation of the details (the normal directions)



Fundamental Problem: Invariance to Transformations

- The basic Laplacian operator is **translation**-invariant, but not **rotation**-invariant
- $E(\mathbf{x}')$ attempts to preserve the **original global** orientation of the details (the normal directions)



Energy Functional

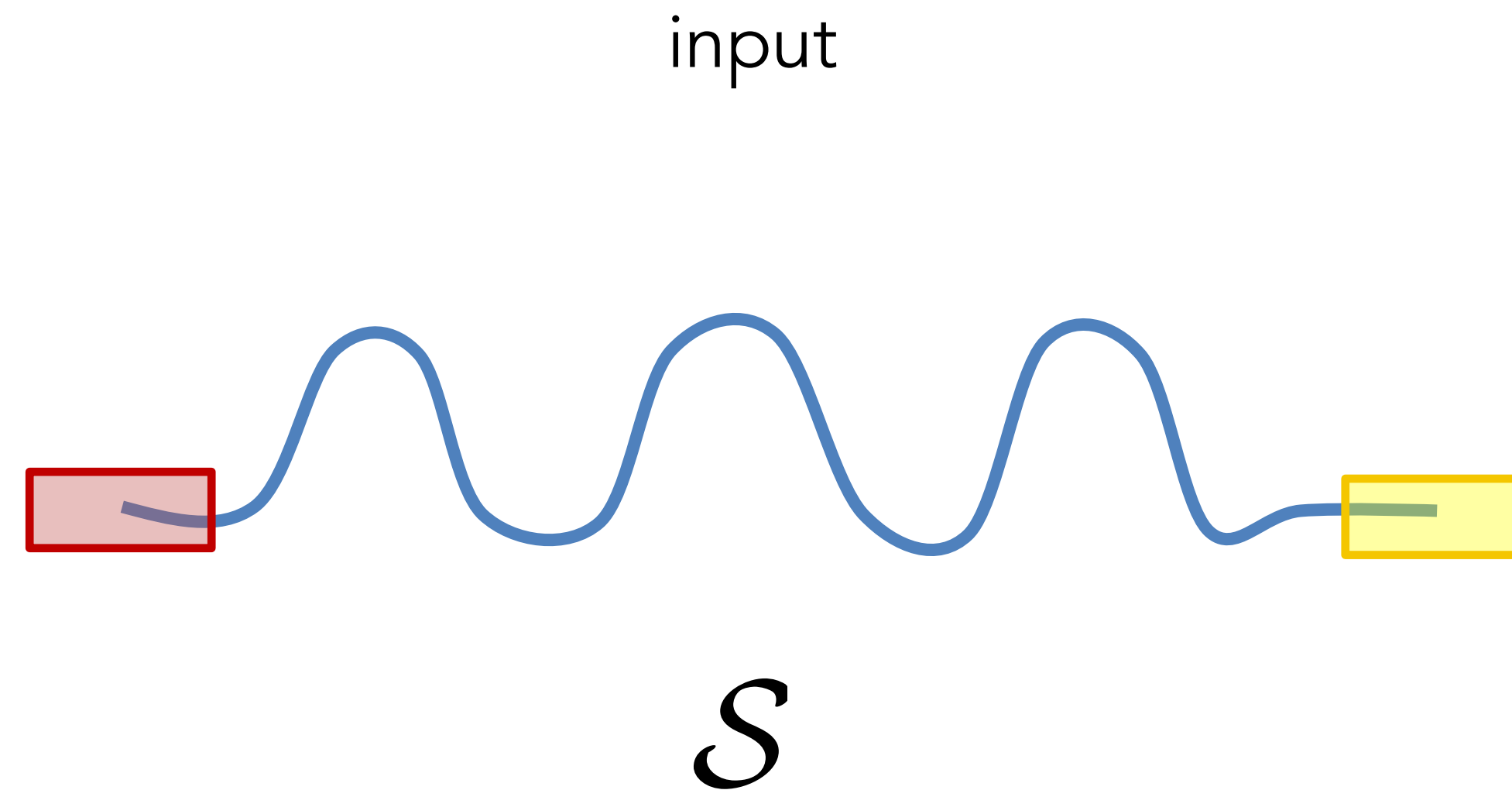
- We need a rigid-invariant energy...

$$E(\mathbf{x}') = \sum_{i=1}^n A_i \|\Delta(\mathbf{x}'_i) - \delta_i\|^2$$

Need to locally
rotate the *target*
m.c. normals

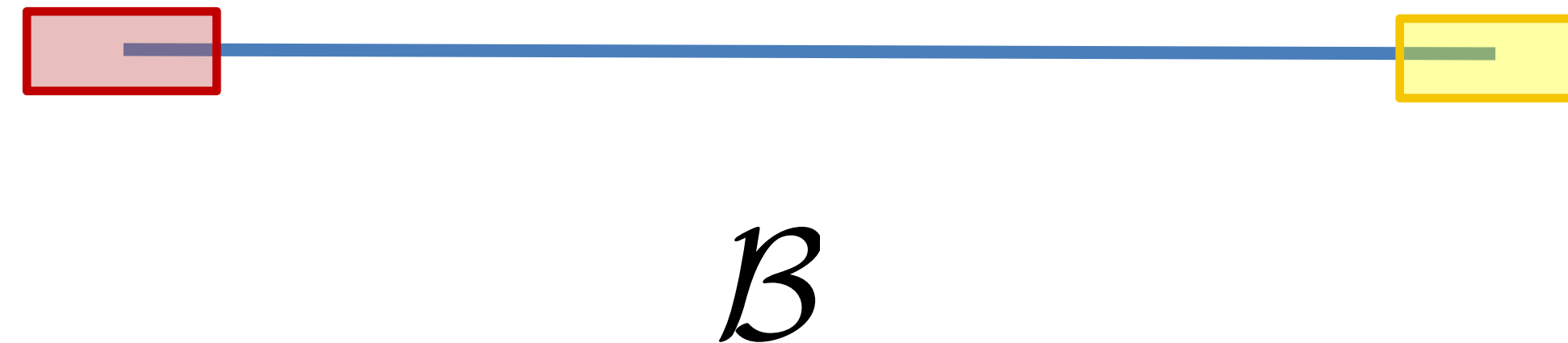
Approach 1: Multi-Resolution Mesh Editing

Multiresolution Approach



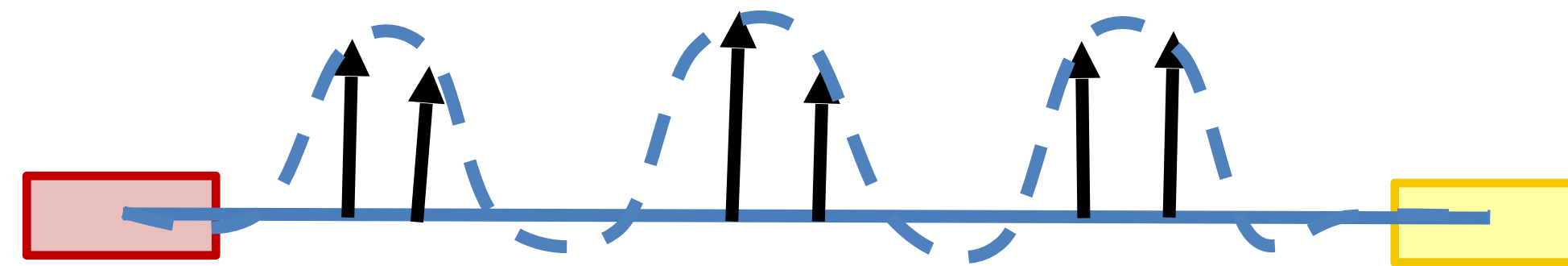
Multiresolution Approach

Smooth base surface



Multiresolution Approach

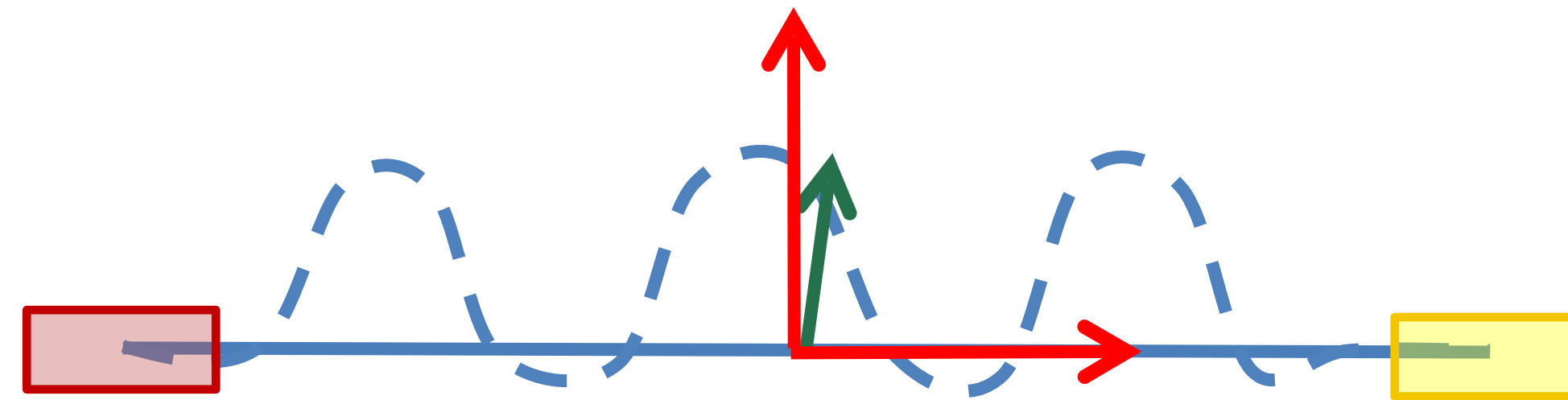
Details – displacement vectors



$$\mathcal{S} - \mathcal{B}$$

Multiresolution Approach

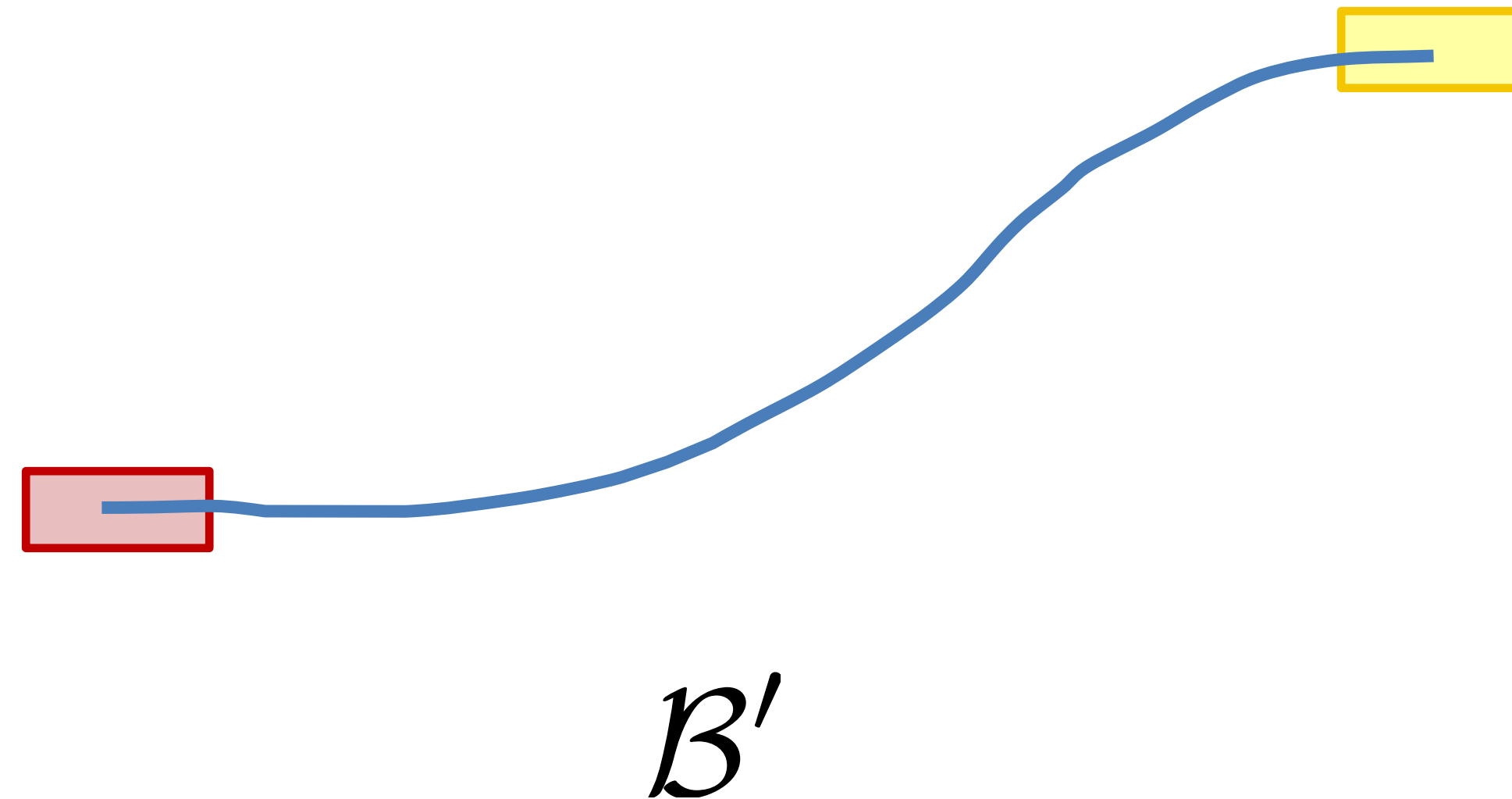
Encode details in the local frame of B



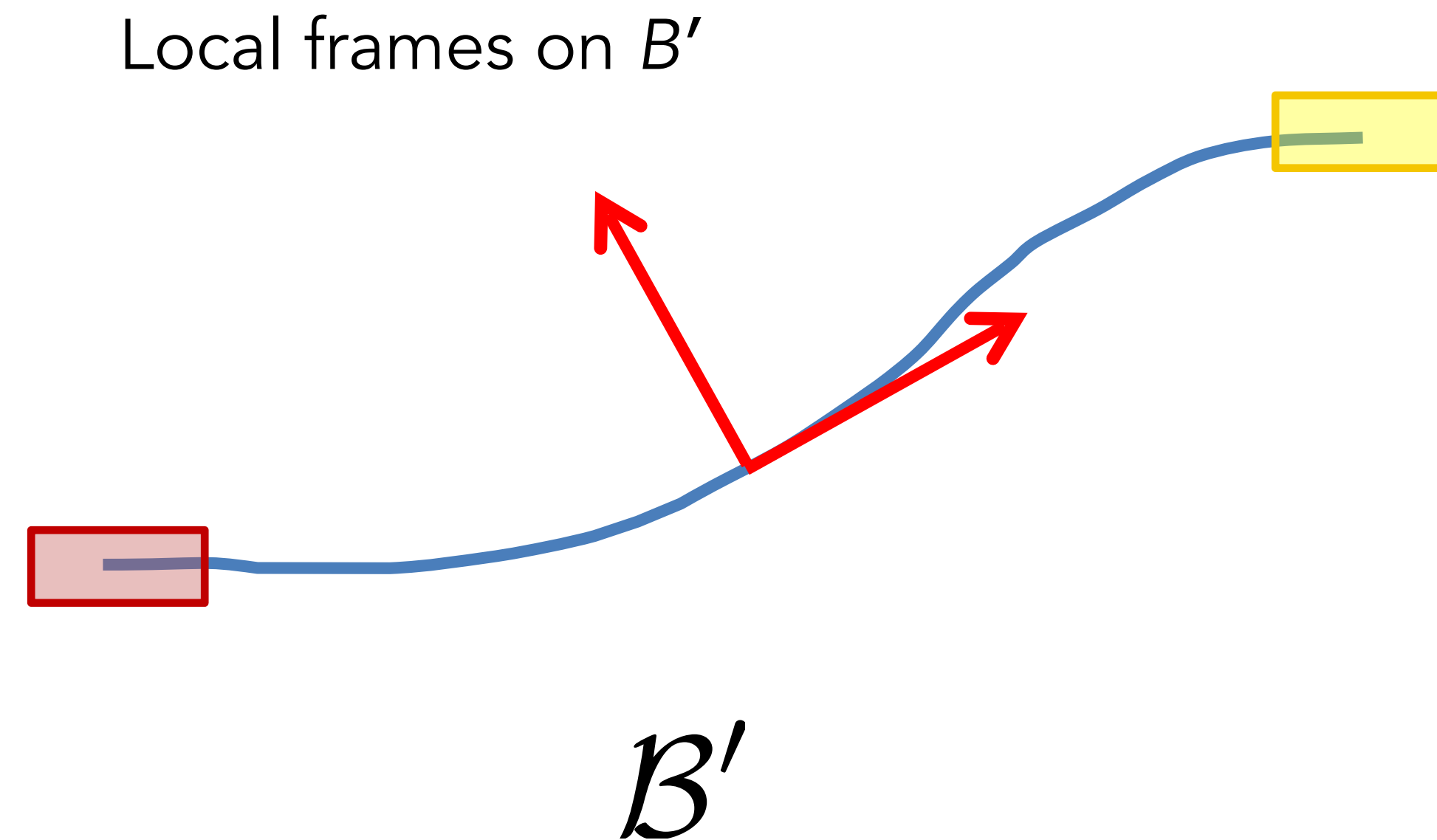
$$\mathbf{d}_i = a_1 \mathbf{t}_i + a_2 \mathbf{n}_i$$

Multiresolution Approach

Deform smooth base surface

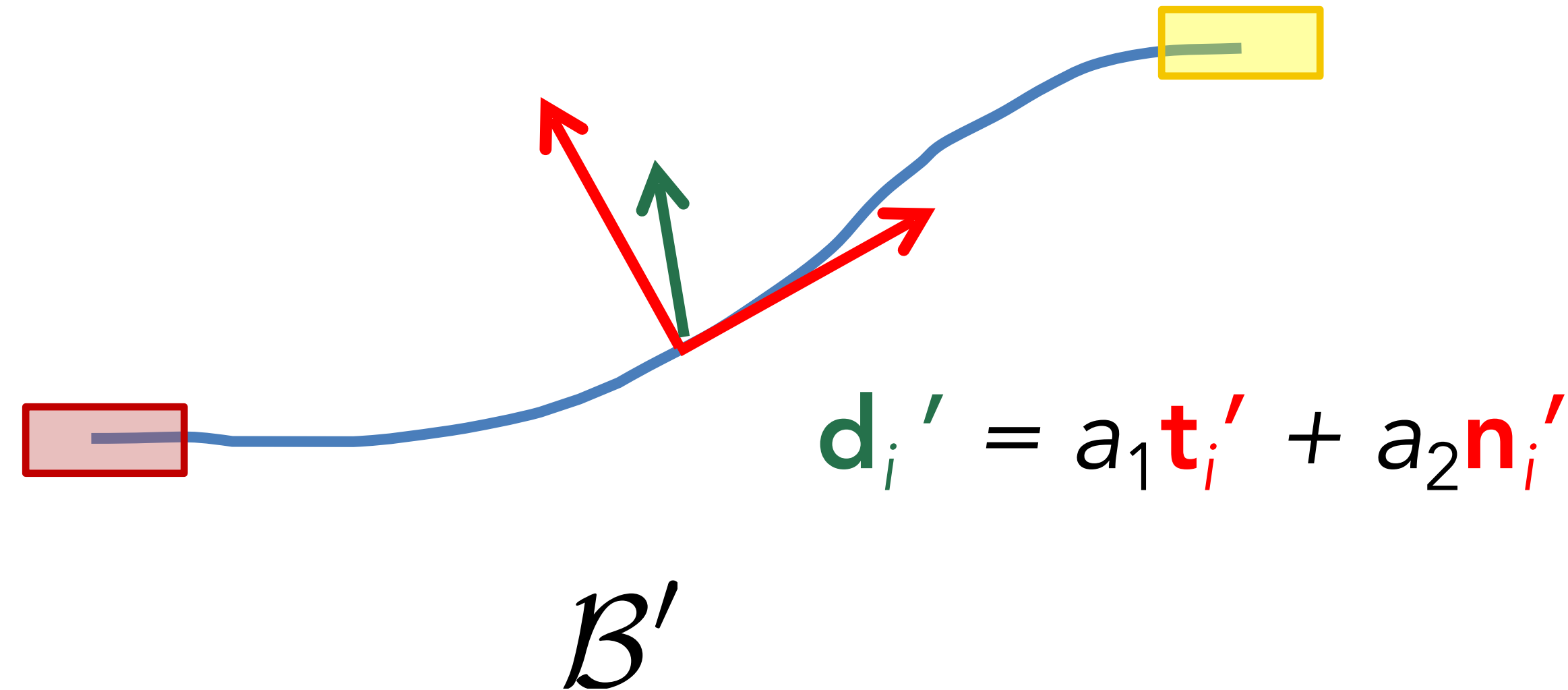


Multiresolution Approach



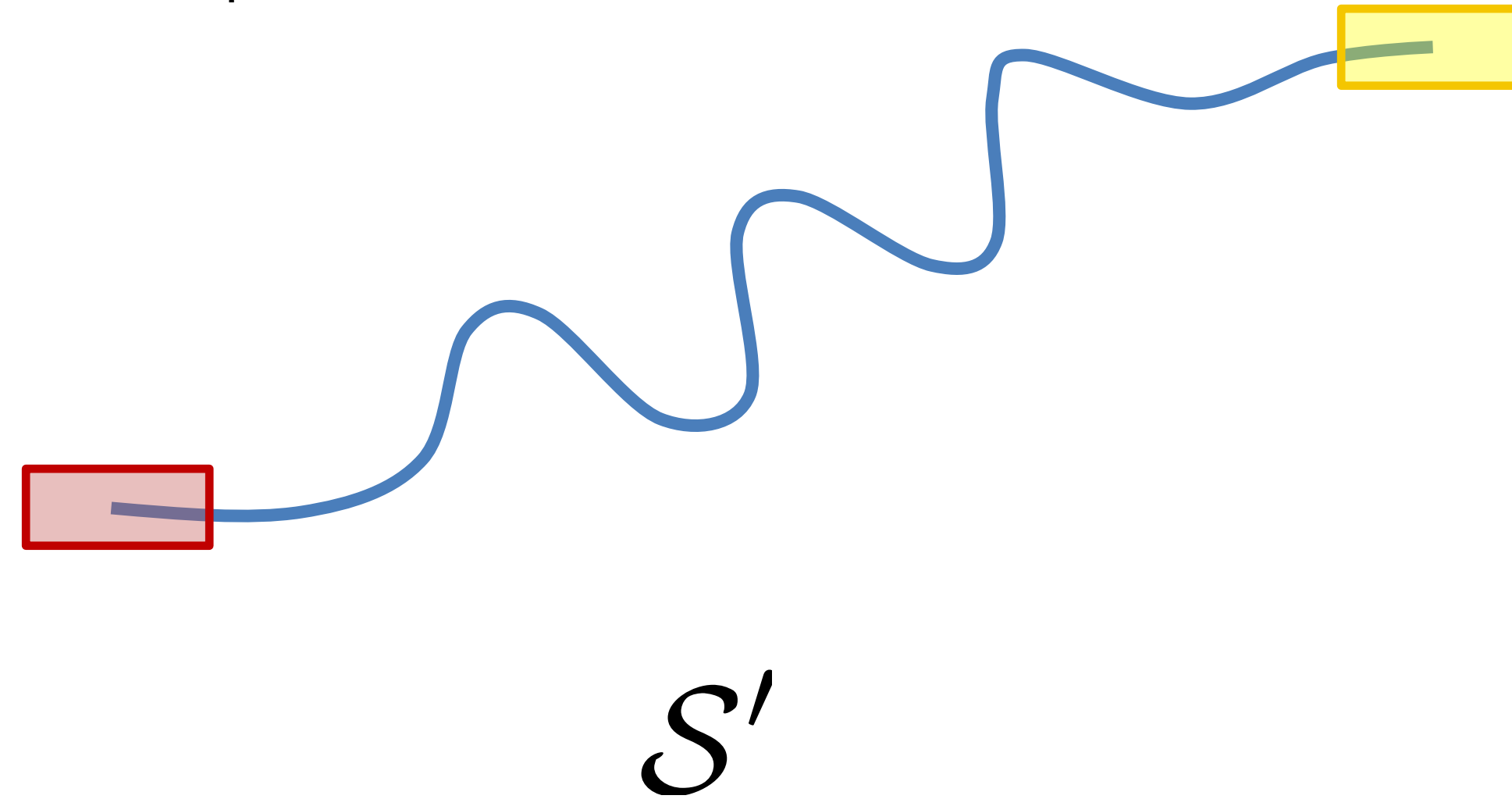
Multiresolution Approach

Add details back – in local frame!



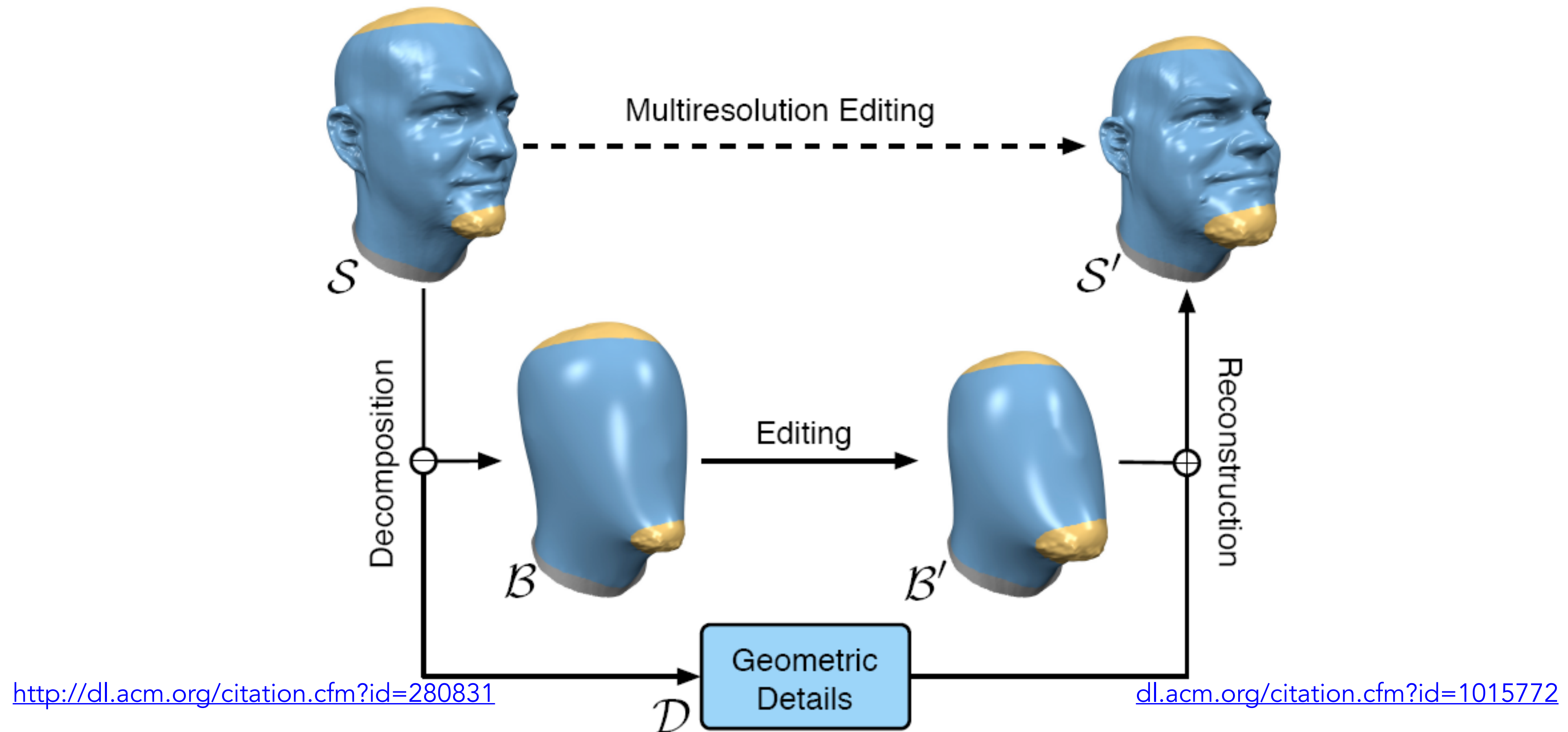
Multiresolution Approach

Displace the vertices to get the result



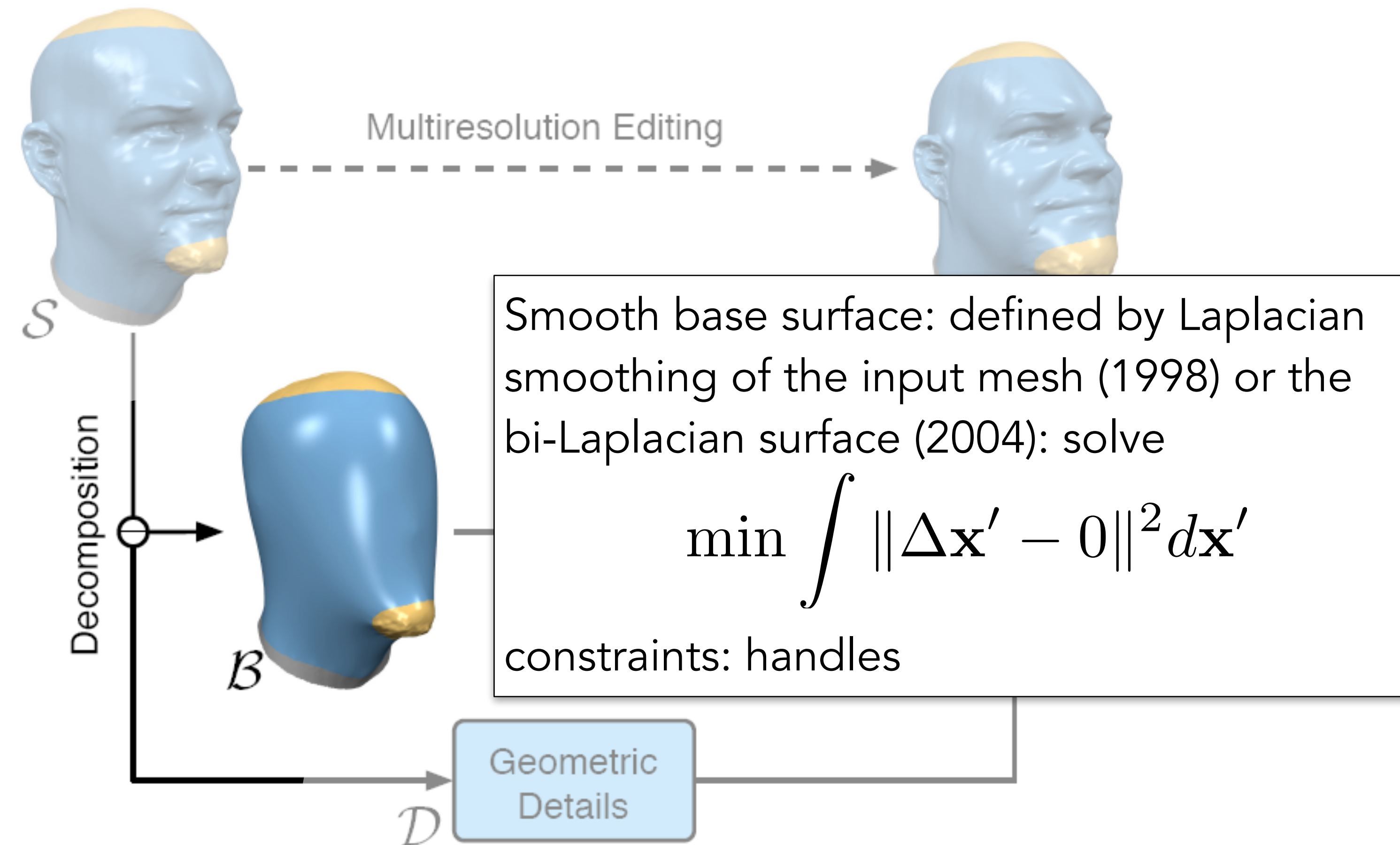
Multiresolution Approach

- Kobbelt et al. SIGGRAPH 98, Botsch and Kobbelt SIGGRAPH 2004



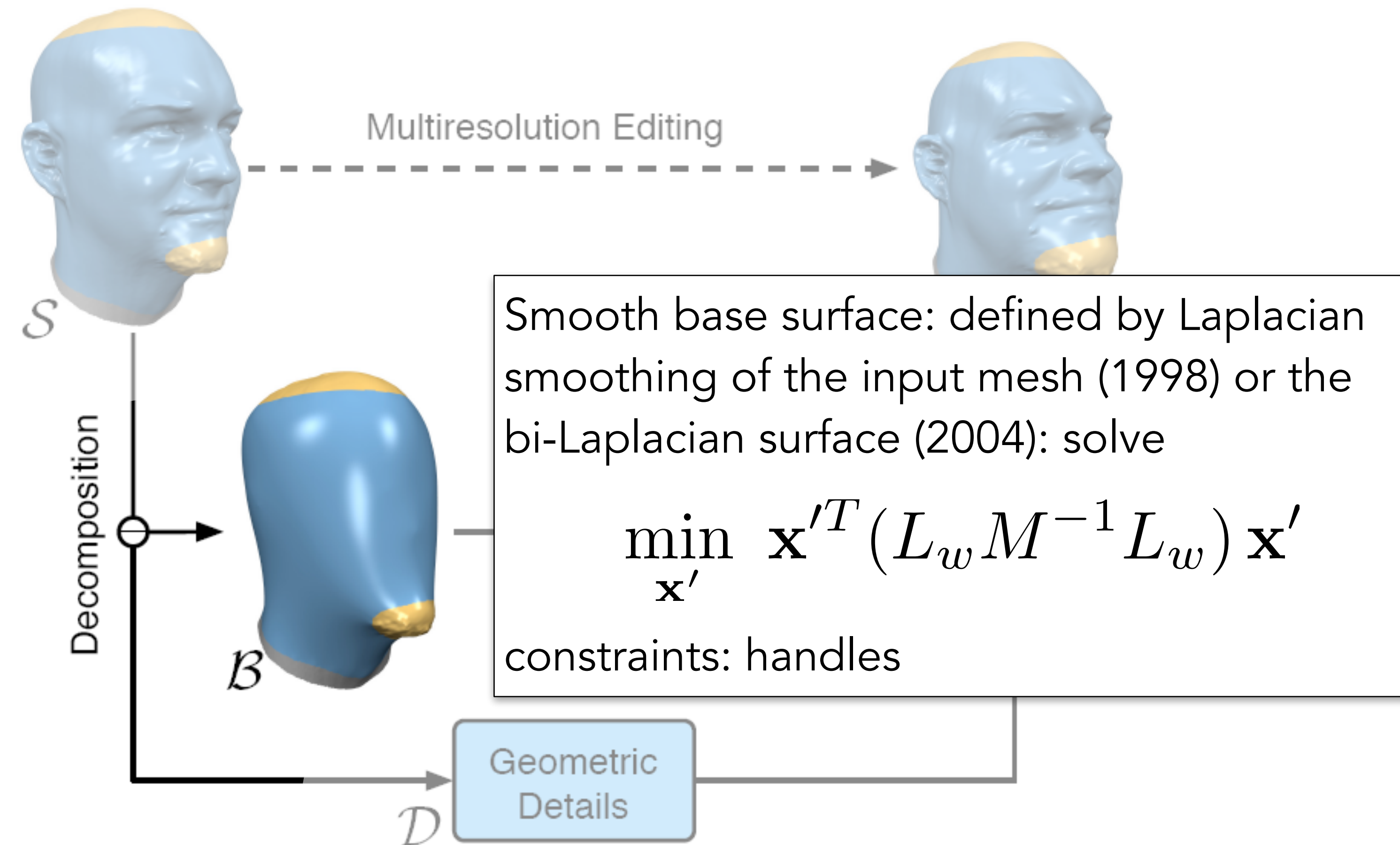
Multiresolution Approach

- Kobbelt et al. SIGGRAPH 98, Botsch and Kobbelt SIGGRAPH 2004



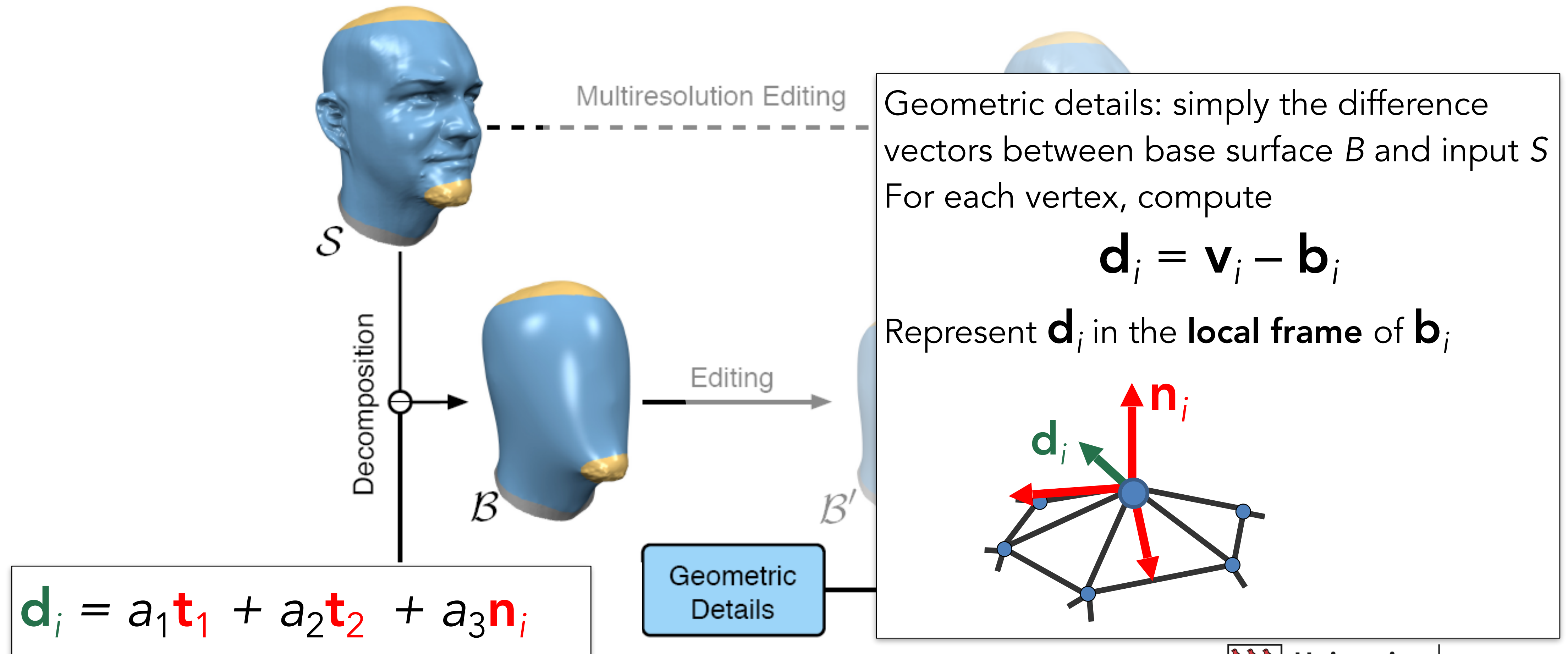
Multiresolution Approach

- Kobbelt et al. SIGGRAPH 98, Botsch and Kobbelt SIGGRAPH 2004



Multiresolution Approach

- Kobbelt et al. SIGGRAPH 98, Botsch and Kobbelt SIGGRAPH 2004



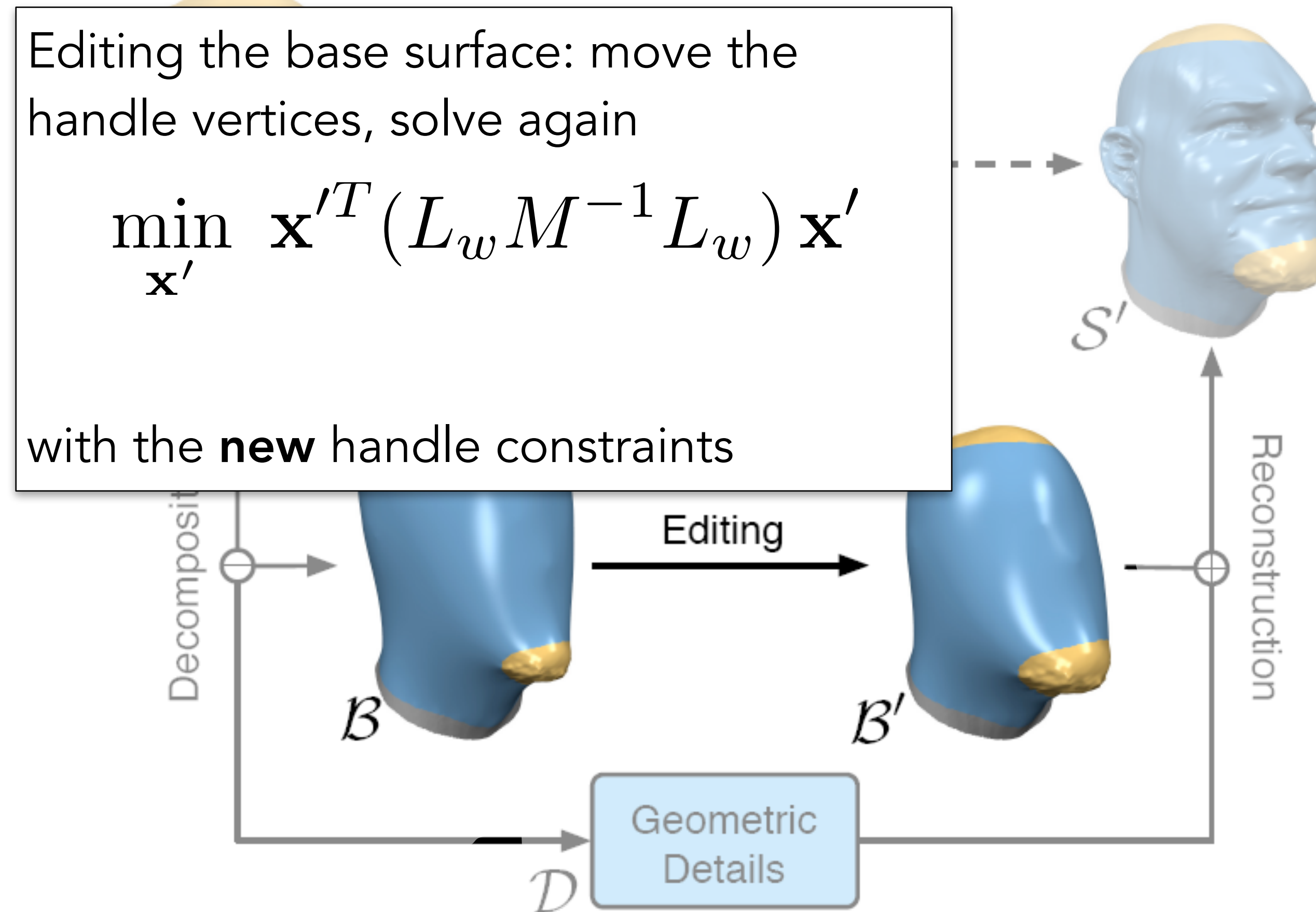
Multiresolution Approach

- Kobbelt et al. SIGGRAPH 98, Botsch and Kobbelt SIGGRAPH 2004

Editing the base surface: move the handle vertices, solve again

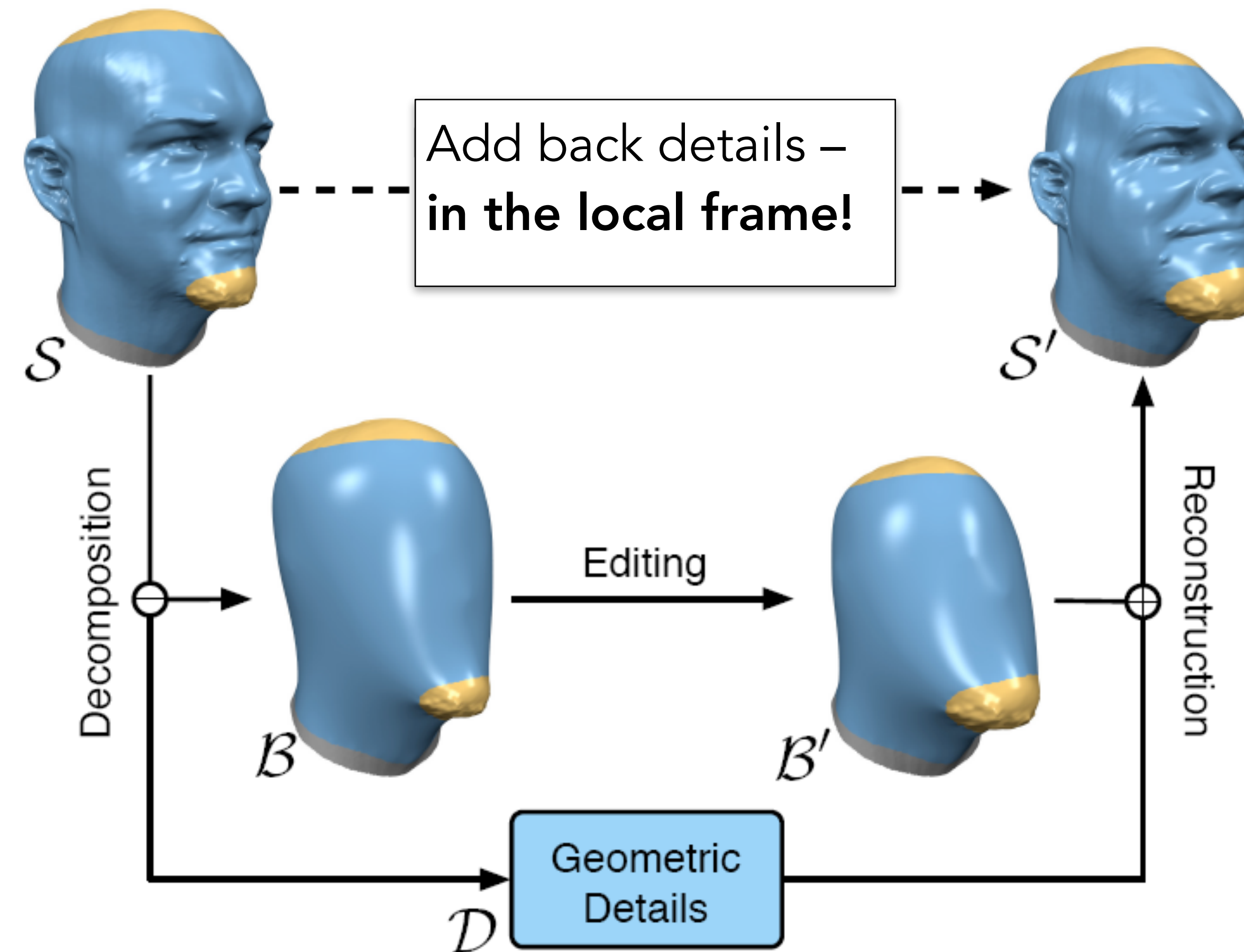
$$\min_{\mathbf{x}'} \mathbf{x}'^T (L_w M^{-1} L_w) \mathbf{x}'$$

with the **new** handle constraints



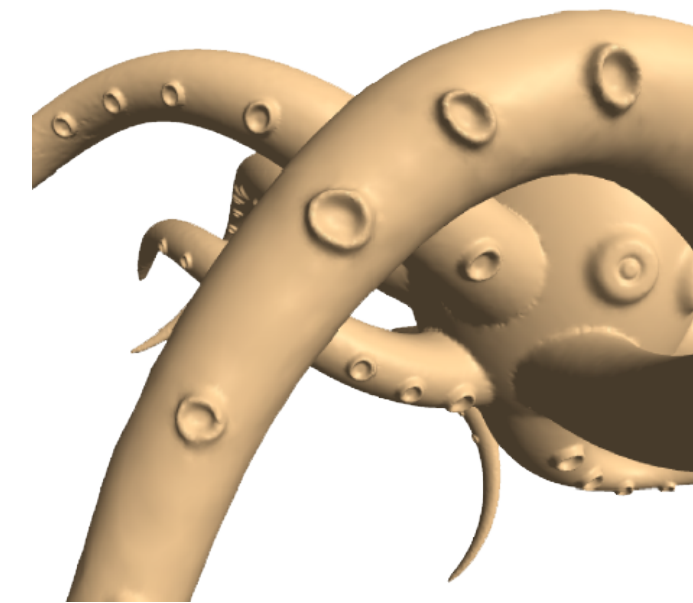
Multiresolution Approach

- Kobbelt et al. SIGGRAPH 98, Botsch and Kobbelt SIGGRAPH 2004

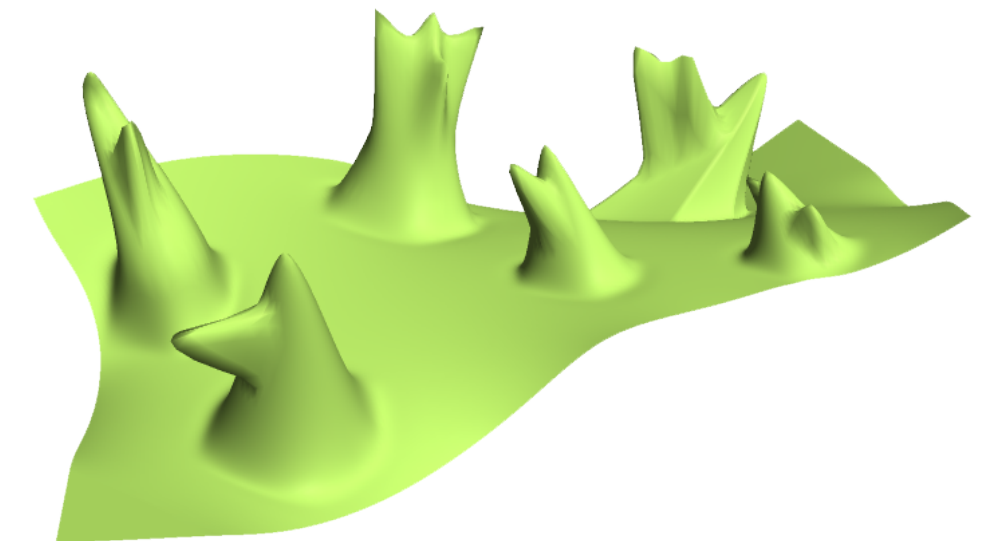


Discussion

- Pros:
 - Fast! Linear solve for the base surface deformation, and then add back displacements
 - Intuitive, easy to implement
- Cons:
 - works only for small height fields



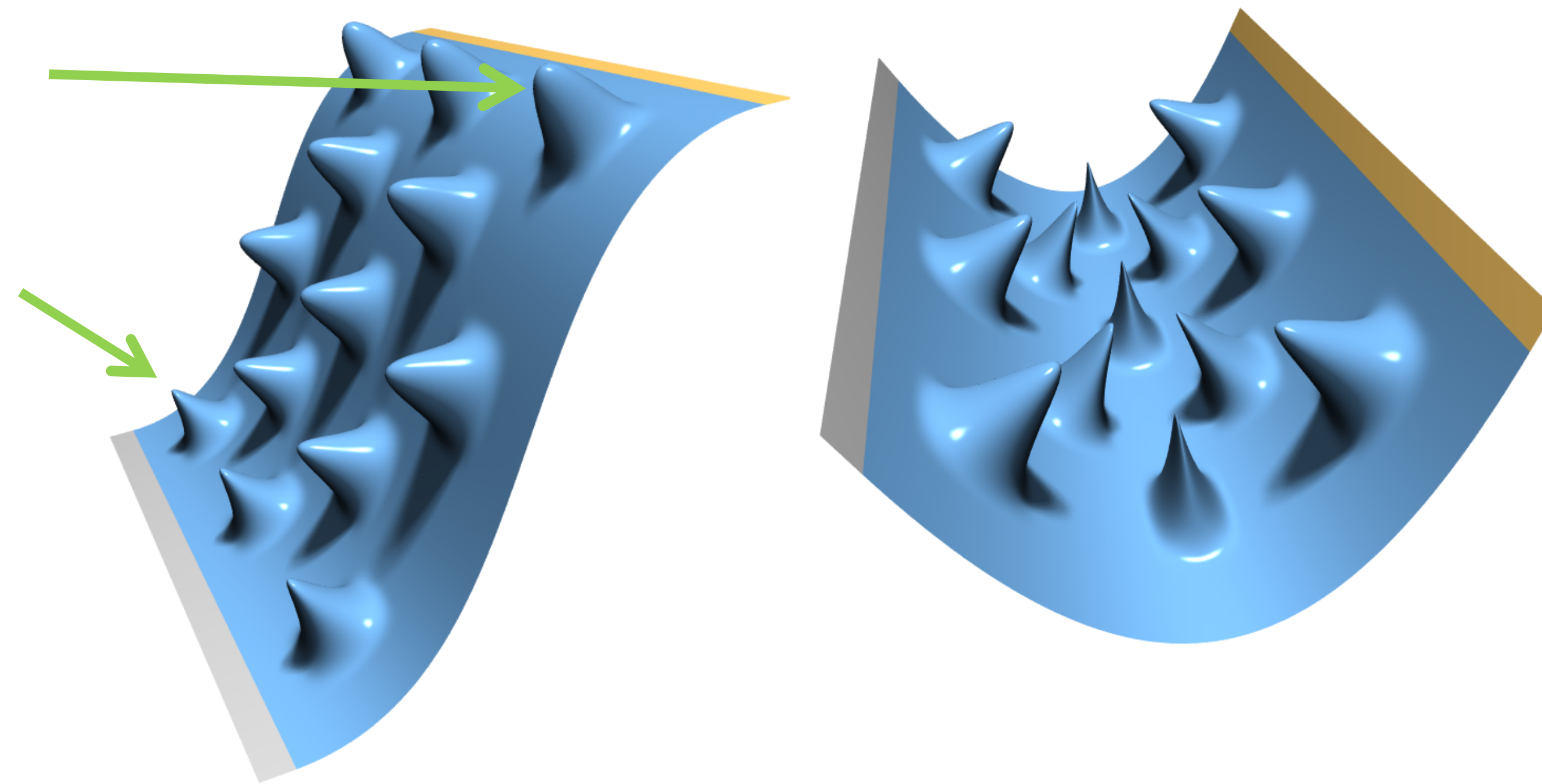
almost a height field



not a height field

Discussion

- If detail vectors are too big we get **overshooting** and **self-intersections**, especially in concave cases



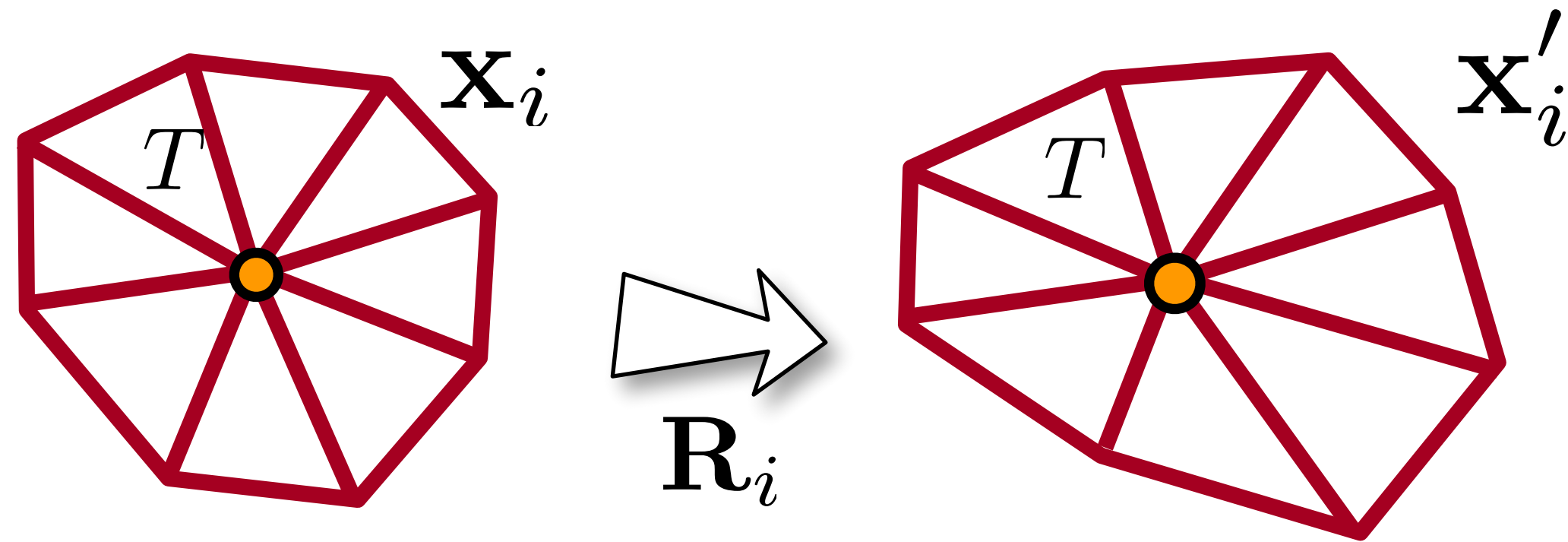
Approach 2: As-Rigid-As-Possible Mesh Editing

Demo

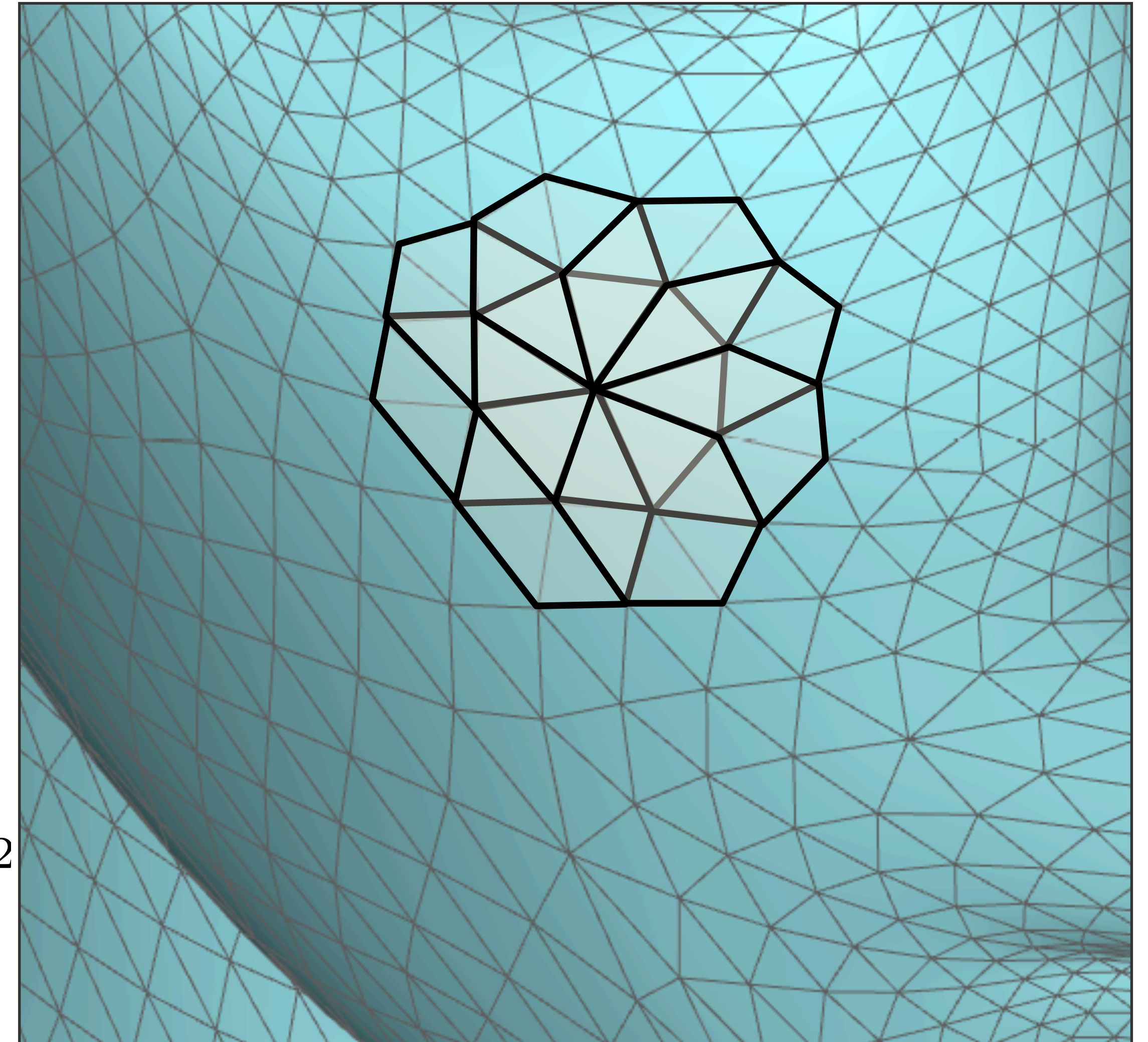
- <https://libigl.github.io/libigl-python-bindings/tut-chapter3/>

As-Rigid-As-Possible Deformation

- Preserve shape of cells covering the surface
- Ask each cell i to transform **rigidly** by best-fitting rotation $\mathbf{R}_i$

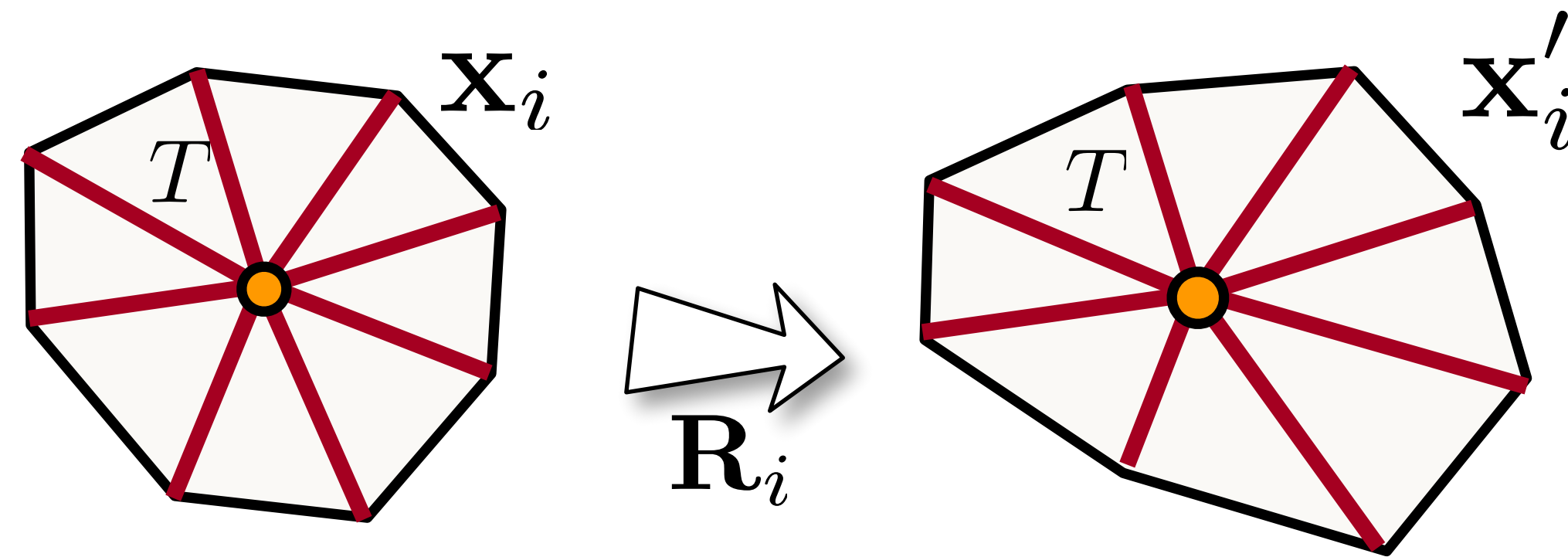


$$\min \sum_{T \in \text{Cell}_i} \sum_{(j,k) \in T} \|(\mathbf{x}'_j - \mathbf{x}'_k) - \mathbf{R}_i(\mathbf{x}_j - \mathbf{x}_k)\|^2$$



As-Rigid-As-Possible Deformation

- Optimal $\mathbf{R}_i$ is uniquely defined by $\mathbf{x}_i$ $\mathbf{x}'_i$

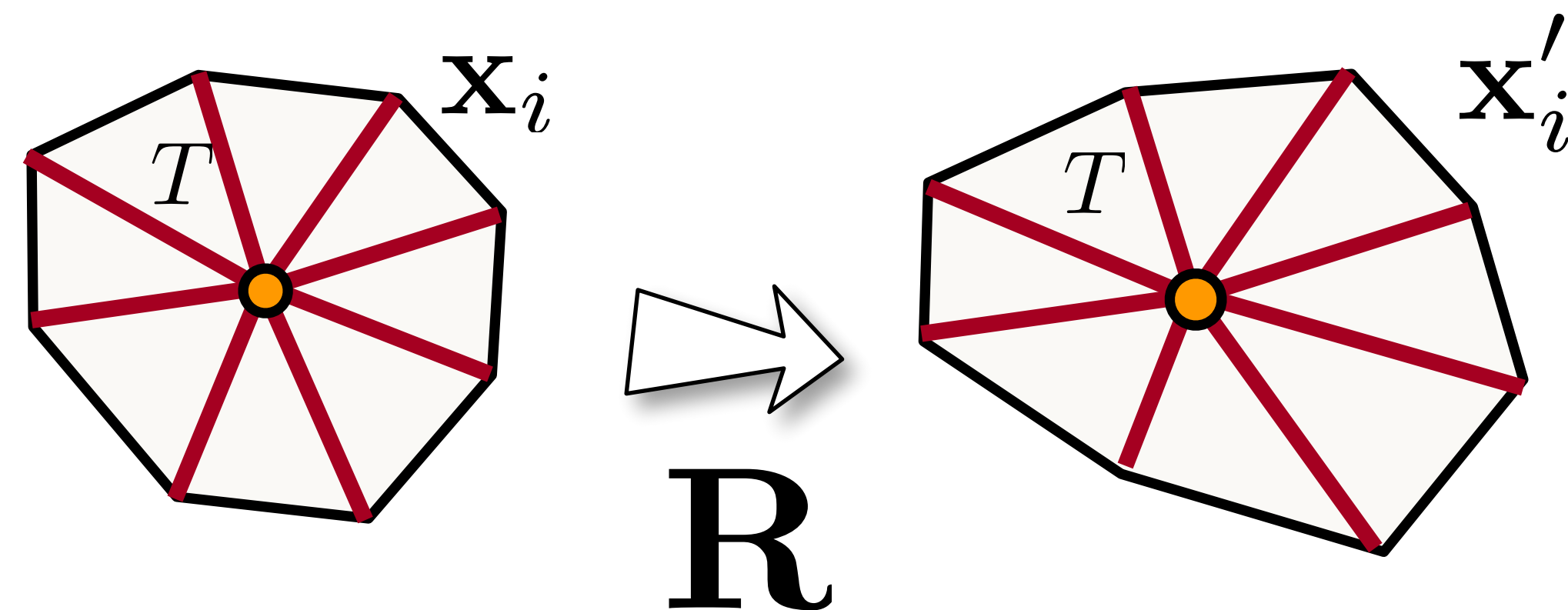


$$\min \sum_{T \in \text{Cell}_i} \sum_{(j,k) \in T} \|(\mathbf{x}'_j - \mathbf{x}'_k) - \mathbf{R}_i(\mathbf{x}_j - \mathbf{x}_k)\|^2$$

- so-called shape-matching problem, solved by a 3x3 SVD

$\mathbf{R}_i$ is a nonlinear function of $\mathbf{x}$

Optimal Rotation

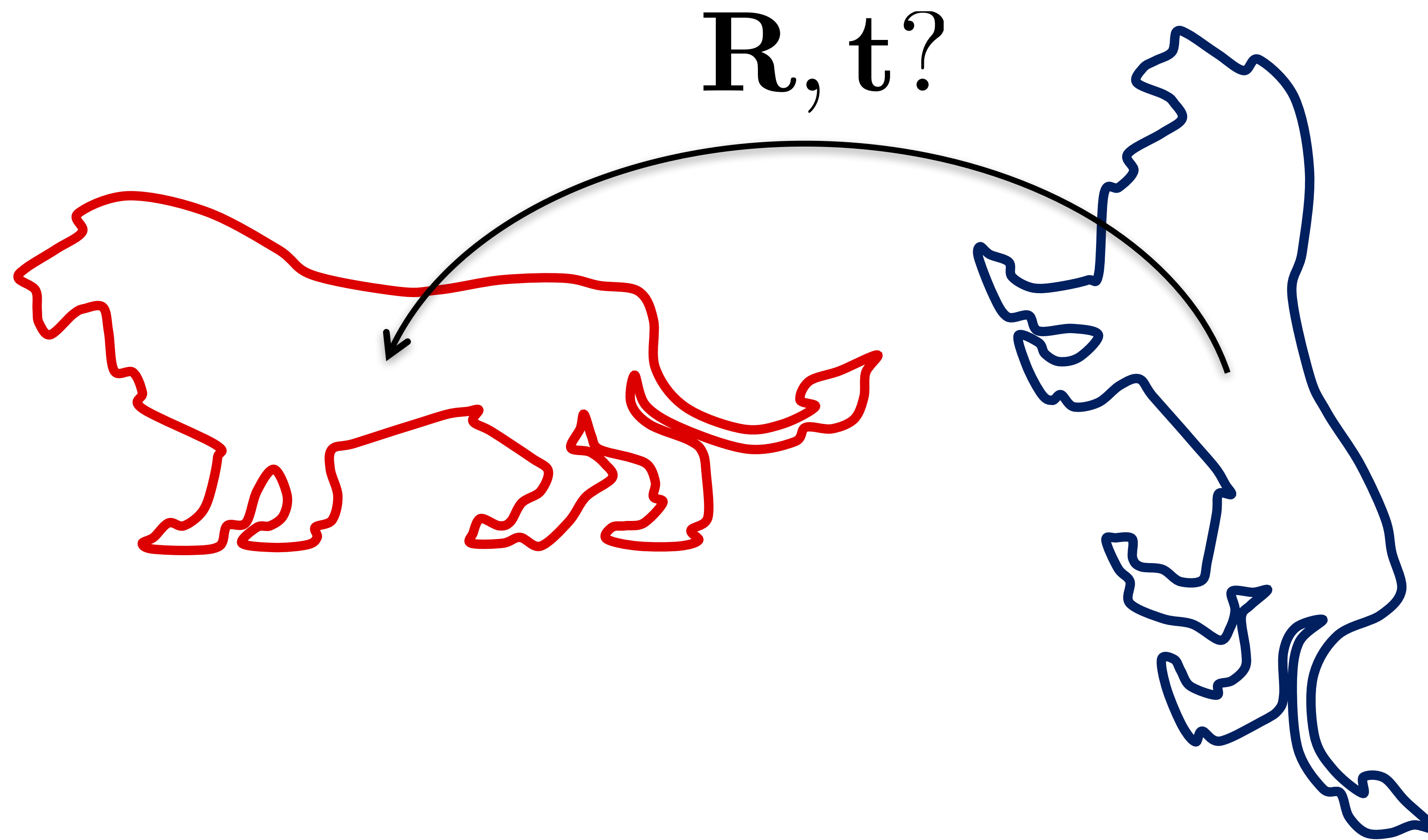


$$\min_{\mathbf{R} \in SO(3)} \sum_{T \in \text{Cell}_i} \sum_{(j,k) \in T} \|(\mathbf{x}'_j - \mathbf{x}'_k) - \mathbf{R}(\mathbf{x}_j - \mathbf{x}_k)\|^2$$

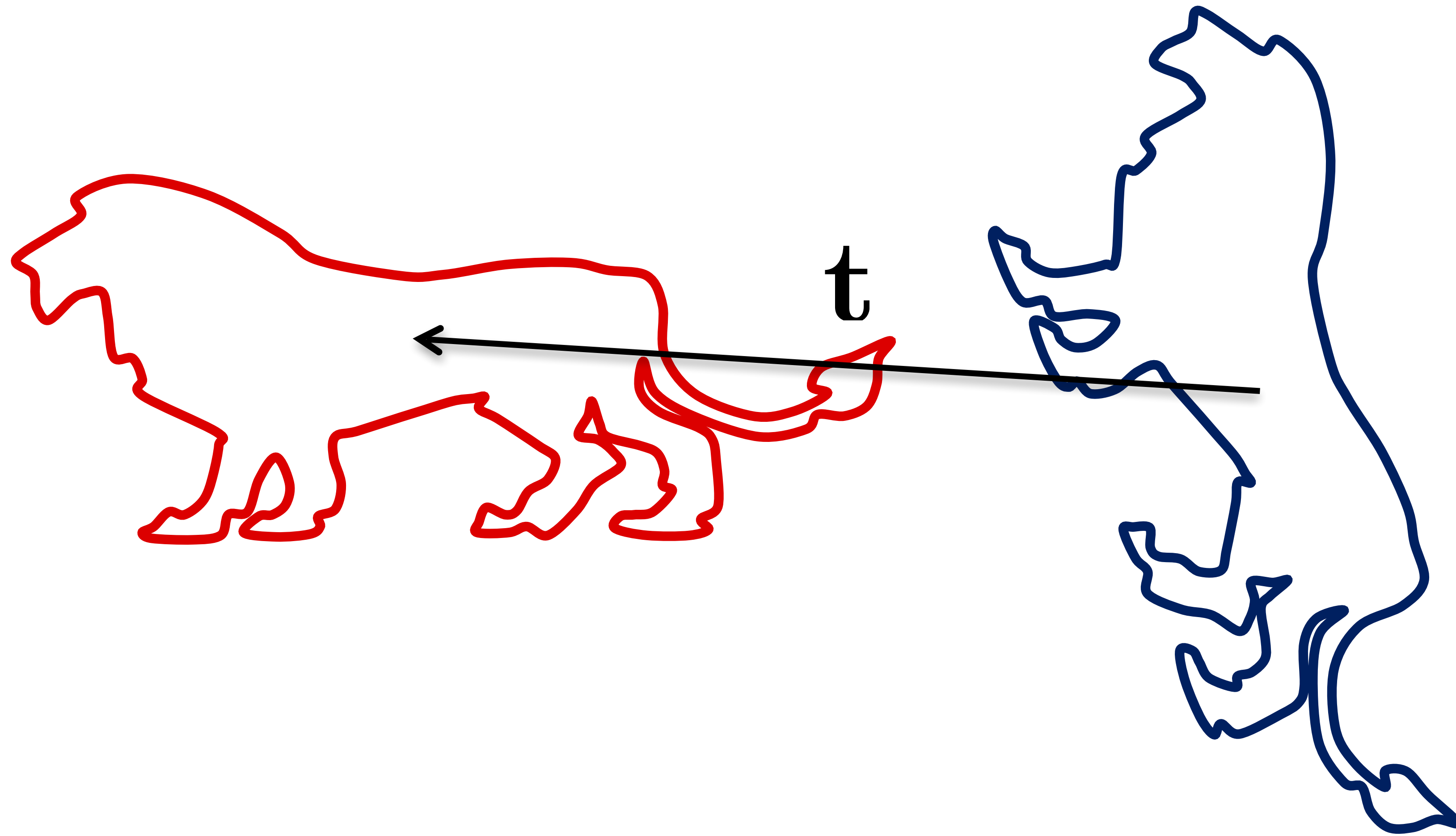
Rotation group



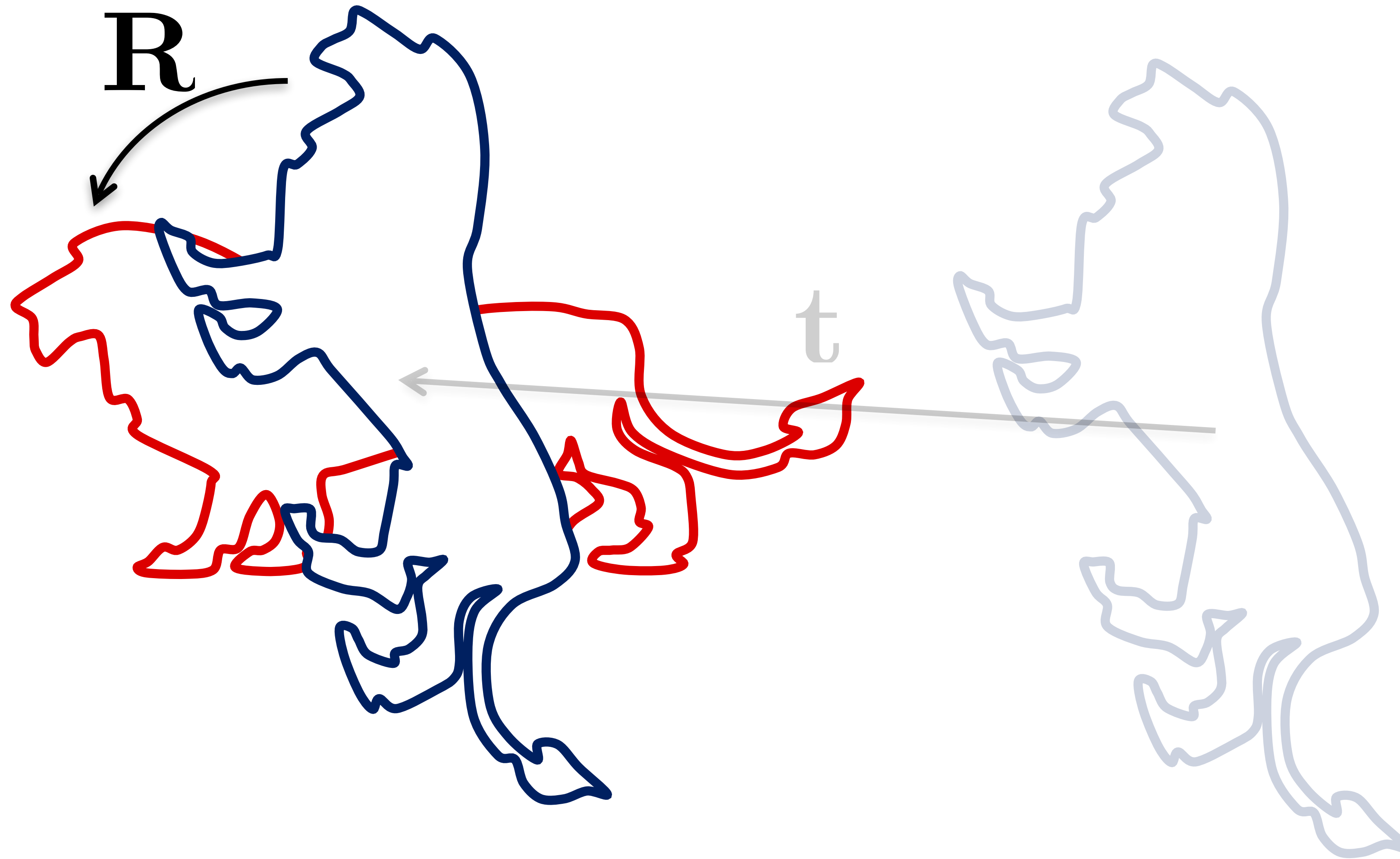
Shape Matching Problem



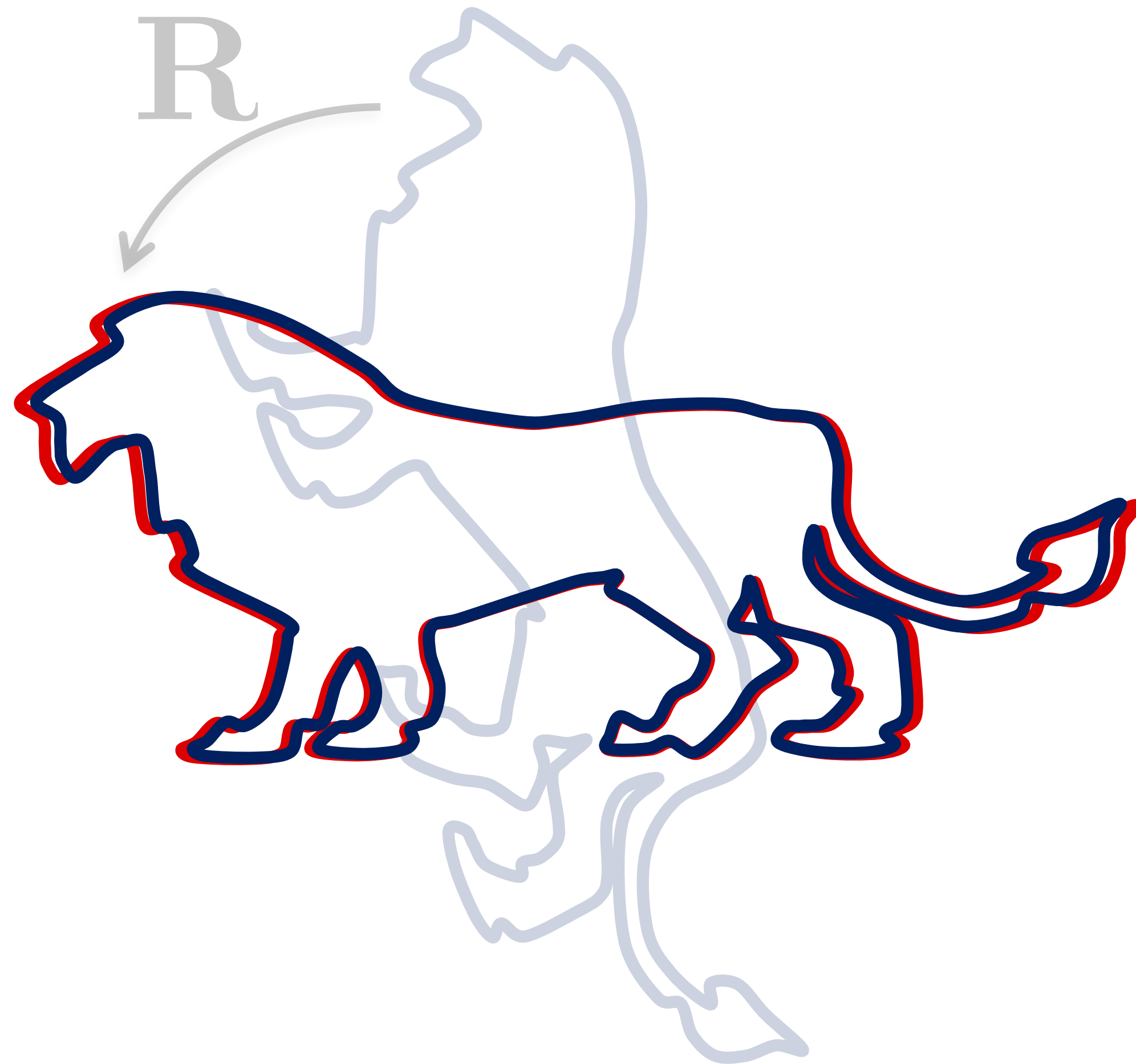
Shape Matching Problem



Shape Matching Problem



Shape Matching Problem



Shape Matching Problem

- Align two point sets

$$\mathcal{P} = \{\mathbf{p}_1, \dots, \mathbf{p}_n\} \text{ and } \mathcal{Q} = \{\mathbf{q}_1, \dots, \mathbf{q}_n\}$$

- Find a translation vector $\mathbf{t}$ and rotation matrix $\mathbf{R}$ so that

$$\sum_{i=1}^n \|(\mathbf{R}\mathbf{p}_i + \mathbf{t}) - \mathbf{q}_i\|^2 \text{ is minimized}$$

Shape Matching – Solution

- Solve for translation first (w.r.t. $\mathbf{R}$, $\mathbf{p}$, and $\mathbf{q}$)

$$\frac{\partial}{\partial \mathbf{t}} \sum_{i=1}^n \|(\mathbf{R}\mathbf{p}_i + \mathbf{t}) - \mathbf{q}_i\|^2 = \sum_{i=1}^n 2 ((\mathbf{R}\mathbf{p}_i + \mathbf{t}) - \mathbf{q}_i) \stackrel{!}{=} 0$$

$$\mathbf{R} \sum_{i=1}^n \mathbf{p}_i + \sum_{i=1}^n \mathbf{t} - \sum_{i=1}^n \mathbf{q}_i = 0$$

$$\mathbf{t} = \underbrace{\left(\frac{1}{n} \sum_{i=1}^n \mathbf{q}_i \right)}_{\bar{\mathbf{q}}} - \mathbf{R} \underbrace{\left(\frac{1}{n} \sum_{i=1}^n \mathbf{p}_i \right)}_{\bar{\mathbf{p}}}$$

Point sets $\{\mathbf{q}_i\}$ and $\{\mathbf{R}\mathbf{p}_i\}$
have the same center of
mass

Finding the Rotation **R**

- To find the optimal **R**, we bring the centroids of both point sets to the origin

$$\mathbf{v}_i = \mathbf{p}_i - \bar{\mathbf{p}}, \quad \mathbf{v}'_i = \mathbf{q}_i - \bar{\mathbf{q}}$$

- We want to find **R** that minimizes

$$\sum_{i=1}^n \|\mathbf{R}\mathbf{v}_i - \mathbf{v}'_i\|^2$$

Finding the Rotation **R**

$$\begin{aligned} \sum_{i=1}^n \|\mathbf{R}\mathbf{v}_i - \mathbf{v}'_i\|^2 &= \sum_{i=1}^n (\mathbf{R}\mathbf{v}_i - \mathbf{v}'_i)^T (\mathbf{R}\mathbf{v}_i - \mathbf{v}'_i) = \\ &= \sum_{i=1}^n \left(\underbrace{\mathbf{v}_i^T \mathbf{R}^T \mathbf{R} \mathbf{v}_i}_I - \mathbf{v}'_i{}^T \mathbf{R} \mathbf{v}_i - \mathbf{v}_i^T \mathbf{R}^T \mathbf{v}'_i + \underbrace{\mathbf{v}'_i{}^T \mathbf{v}'_i} \right) \end{aligned}$$

These terms do not depend on **R**,
so we can ignore them in the minimization

Finding the Rotation **R**

$$\operatorname{argmin}_{\mathbf{R} \in SO(3)} \sum_{i=1}^n \left(-\mathbf{v}'_i{}^T \mathbf{R} \mathbf{v}_i - \mathbf{v}_i{}^T \mathbf{R}^T \mathbf{v}'_i \right) = \operatorname{argmax}_{\mathbf{R} \in SO(3)} \sum_{i=1}^n \left(\mathbf{v}'_i{}^T \mathbf{R} \mathbf{v}_i + \underbrace{\mathbf{v}_i{}^T \mathbf{R}^T \mathbf{v}'_i}_{\text{green arrow}} \right) =$$

$$= \operatorname{argmax}_{\mathbf{R} \in SO(3)} \sum_{i=1}^n \mathbf{v}'_i{}^T \mathbf{R} \mathbf{v}_i$$

$$\mathbf{v}_i{}^T \mathbf{R}^T \mathbf{v}'_i = \left(\mathbf{v}_i{}^T \mathbf{R}^T \mathbf{v}'_i \right)^T = \mathbf{v}'_i{}^T \mathbf{R} \mathbf{v}_i$$

Finding the Rotation **R**

$$\sum_{i=1}^n \mathbf{v}_i'^T \mathbf{R} \mathbf{v}_i = \text{tr} \left(\mathbf{V}'^T \mathbf{R} \mathbf{V} \right)$$

$$\begin{array}{|c|} \hline \mathbf{v}_1'^T \\ \mathbf{v}_2'^T \\ \vdots \\ \mathbf{v}_n'^T \\ \hline \end{array} \begin{array}{|c|} \hline \mathbf{R} \\ \hline \end{array} \underbrace{\begin{array}{|c|} \hline \mathbf{v}_1 \quad \mathbf{v}_2 \quad \cdots \quad \mathbf{v}_n \\ \hline \end{array}}_{\mathbf{V}} = \begin{array}{|c|} \hline \mathbf{v}_1'^T \\ \mathbf{v}_2'^T \\ \vdots \\ \mathbf{v}_n'^T \\ \hline \end{array} \begin{array}{|c|} \hline \mathbf{R} \mathbf{v}_1 \quad \mathbf{R} \mathbf{v}_2 \quad \cdots \quad \mathbf{R} \mathbf{v}_n \\ \hline \end{array}$$

$\mathbf{V}'^T$



Finding the Rotation **R**

$$\sum_{i=1}^n \mathbf{v}'_i{}^T \mathbf{R} \mathbf{v}_i = \text{tr} \left(\mathbf{V}'^T \mathbf{R} \mathbf{V} \right)$$

$$\begin{bmatrix} \mathbf{v}'_1{}^T \\ \mathbf{v}'_2{}^T \\ \vdots \\ \mathbf{v}'_n{}^T \end{bmatrix} \begin{bmatrix} \mathbf{R} \mathbf{v}_1 & \mathbf{R} \mathbf{v}_2 & \cdots & \mathbf{R} \mathbf{v}_n \end{bmatrix} = \begin{bmatrix} \mathbf{v}'_1{}^T \mathbf{R} \mathbf{v}_1 & & & \\ & \mathbf{v}'_2{}^T \mathbf{R} \mathbf{v}_2 & & \\ & & \ddots & \\ & & & \mathbf{v}'_n{}^T \mathbf{R} \mathbf{v}_n \end{bmatrix}$$



Finding the Rotation **R**

Take a look at the Matrix Cookbook!

- Find **R** that maximizes

$$\text{tr} \left(\mathbf{V}'^T \mathbf{R} \mathbf{V} \right) = \text{tr} \left(\mathbf{R} \mathbf{V} \mathbf{V}'^T \right)$$

- SVD: $\mathbf{V} \mathbf{V}'^T = \mathbf{U} \mathbf{\Sigma} \tilde{\mathbf{U}}^T$

$$\text{tr} \left(\mathbf{R} \mathbf{V} \mathbf{V}'^T \right) = \text{tr} \left(\underbrace{\mathbf{R} \mathbf{U}} \underbrace{\mathbf{\Sigma} \tilde{\mathbf{U}}^T} \right) = \text{tr} \left(\mathbf{\Sigma} \underbrace{\tilde{\mathbf{U}}^T \mathbf{R} \mathbf{U}}_{\substack{\text{orthonormal} \\ \text{matrix}}} \right)$$

Finding the Rotation **R**

- We want to maximize
$$\text{tr}(\Sigma \mathbf{M})$$

M: orthonormal matrix
all entries ≤ 1

σ_1	
	σ_2
	σ_3

m_{11}	$\dots$
$\vdots$	m_{22}
	$\dots$
	m_{33}

$$\text{tr}(\Sigma \mathbf{M}) = \sum_{i=1}^3 \sigma_i m_{ii} \leq \sum_{i=1}^3 \sigma_i$$



Finding the Rotation **R**

$$\text{tr}(\Sigma \mathbf{M}) = \sum_{i=1}^3 \sigma_i m_{ii} \leq \sum_{i=1}^3 \sigma_i$$

- Our best shot is $m_{ii} = 1$, i.e. to make **M** = **I**

$$\mathbf{M} = \tilde{\mathbf{U}}^T \mathbf{R} \mathbf{U} \stackrel{!}{=} \mathbf{I}$$

$$\mathbf{R} \mathbf{U} = \tilde{\mathbf{U}}$$

$$\boxed{\mathbf{R} = \tilde{\mathbf{U}} \mathbf{U}^T}$$

Summary of Rigid Alignment

- Translate the input points to the centroids

$$\mathbf{v}_i = \mathbf{p}_i - \bar{\mathbf{p}}, \quad \mathbf{v}'_i = \mathbf{q}_i - \bar{\mathbf{q}}$$

- Compute the "covariance matrix" $\mathbf{V}\mathbf{V}'^T$

- Compute its SVD: $\mathbf{V}\mathbf{V}'^T = \mathbf{U}\mathbf{\Sigma}\tilde{\mathbf{U}}^T$

- The optimal orthonormal $\mathbf{R}$ is $\mathbf{R} = \tilde{\mathbf{U}}\mathbf{U}^T$

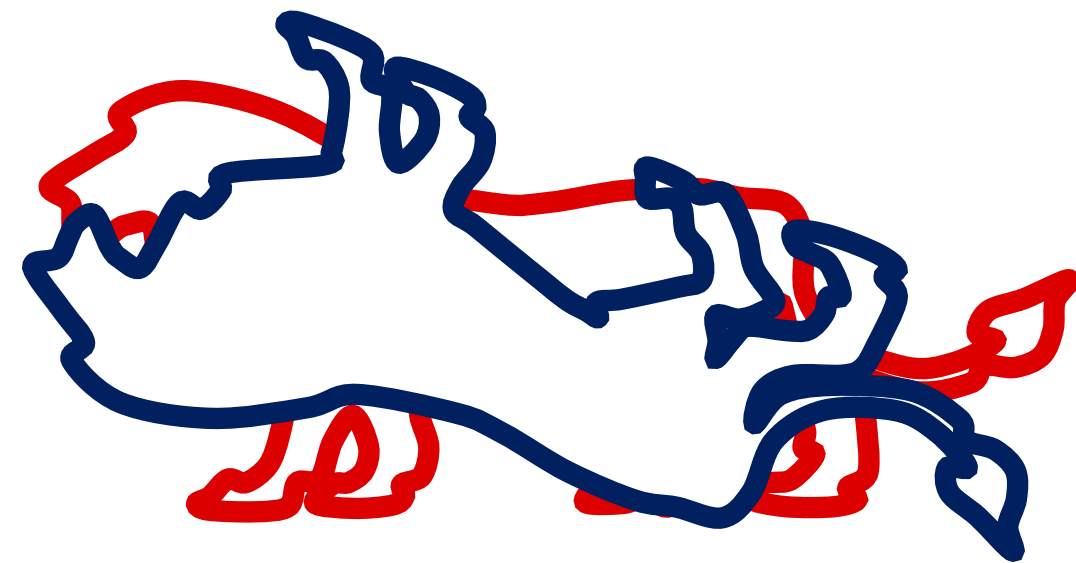
Sign Correction

- It is possible that $\det(\tilde{\mathbf{U}}\mathbf{U}^T) = -1$: sometimes reflection is the best orthonormal transform



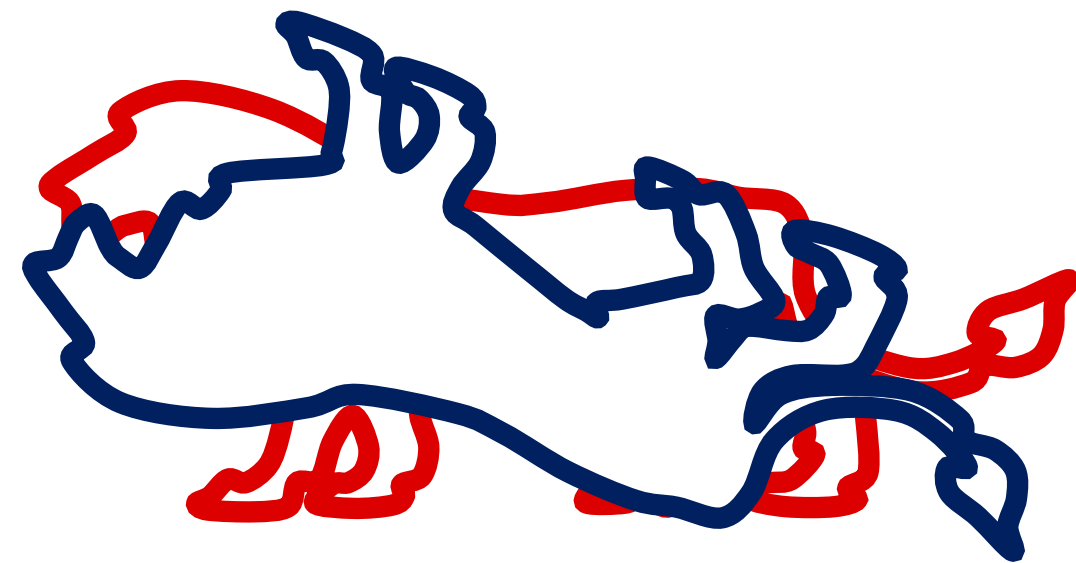
Sign Correction

- It is possible that $\det(\tilde{\mathbf{U}}\mathbf{U}^T) = -1$: sometimes reflection is the best orthonormal transform



Sign Correction

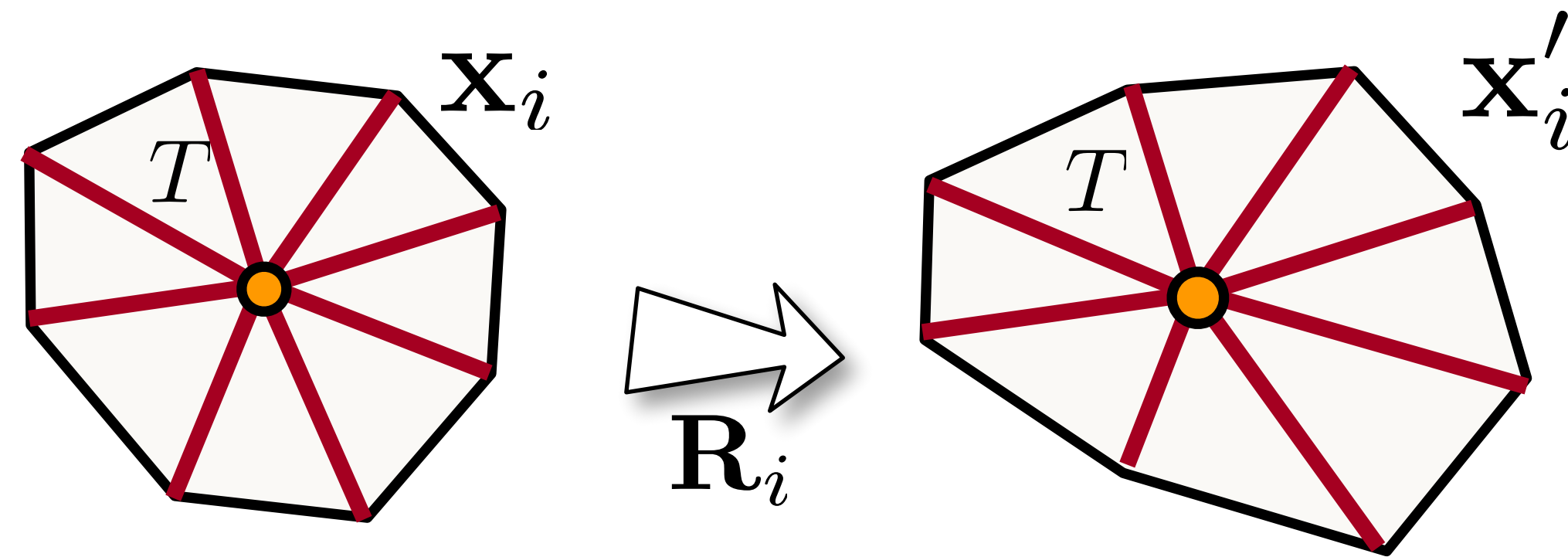
- To restrict ourselves to rotations only:
take the last column of $\mathbf{U}$ (corresponding to the smallest singular value) and invert its sign.



- Why? See http://igl.ethz.ch/projects/ARAP/svd_rot.pdf

As-Rigid-As-Possible Deformation

- Optimal $\mathbf{R}_i$ is uniquely defined by $\mathbf{x}_i$ $\mathbf{x}'_i$



$$\min \sum_{T \in \text{Cell}_i} \sum_{(j,k) \in T} \|(\mathbf{x}'_j - \mathbf{x}'_k) - \mathbf{R}_i(\mathbf{x}_j - \mathbf{x}_k)\|^2$$

- so-called shape-matching problem, solved by a 3x3 SVD

$\mathbf{R}_i$ is a nonlinear function of $\mathbf{x}$

As-Rigid-As-Possible Deformation

- Total ARAP energy: sum up for all the cells i

$$\sum_i \sum_{T \in \text{Cell}_i} \sum_{(j,k) \in T} \|(\mathbf{x}'_j - \mathbf{x}'_k) - \mathbf{R}_i(\mathbf{x}_j - \mathbf{x}_k)\|^2$$

- Treat $\mathbf{x}$ and $\mathbf{R}$ as separate sets of variables
- Simple **local-global** iterative optimization process
 - Decreases the energy at each step

As-Rigid-As-Possible Deformation

- Total ARAP energy: sum up for all the cells i

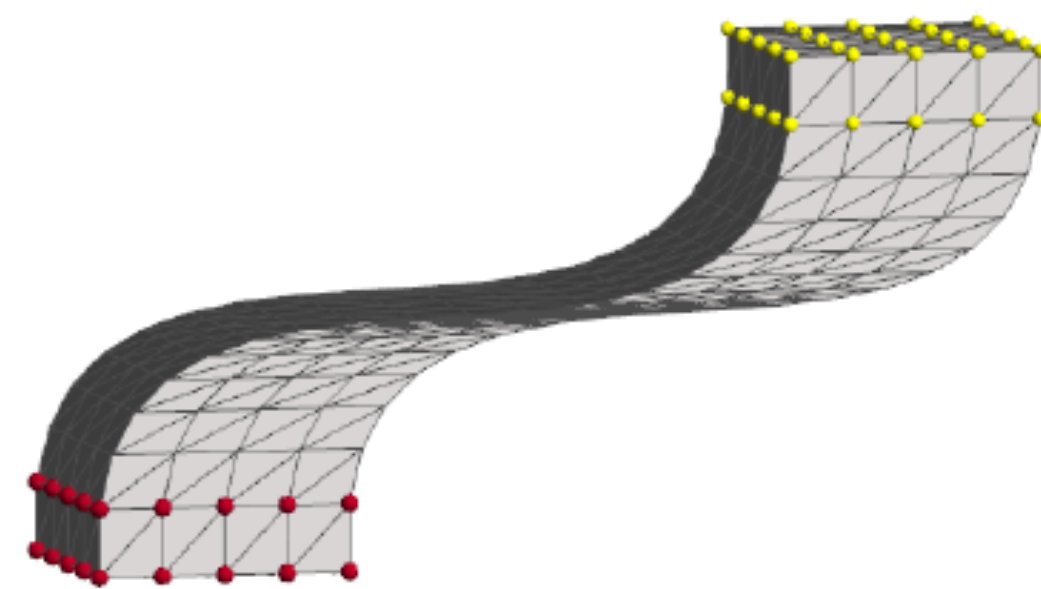
$$\sum_i \sum_{T \in \text{Cell}_i} \sum_{(j,k) \in T} \|(\mathbf{x}'_j - \mathbf{x}'_k) - \mathbf{R}_i(\mathbf{x}_j - \mathbf{x}_k)\|^2$$

- 
- Local step: keep $\mathbf{x}'$ fixed, find optimal $\mathbf{R}_i$ per cell i
 - Global step: keep $\mathbf{R}_i$ fixed, solve for $\mathbf{x}' \rightarrow \mathbf{L}\mathbf{x}' = \mathbf{b}$
quadratic minimization problem
 - The matrix $\mathbf{L}$ stays fixed, can pre-factorize

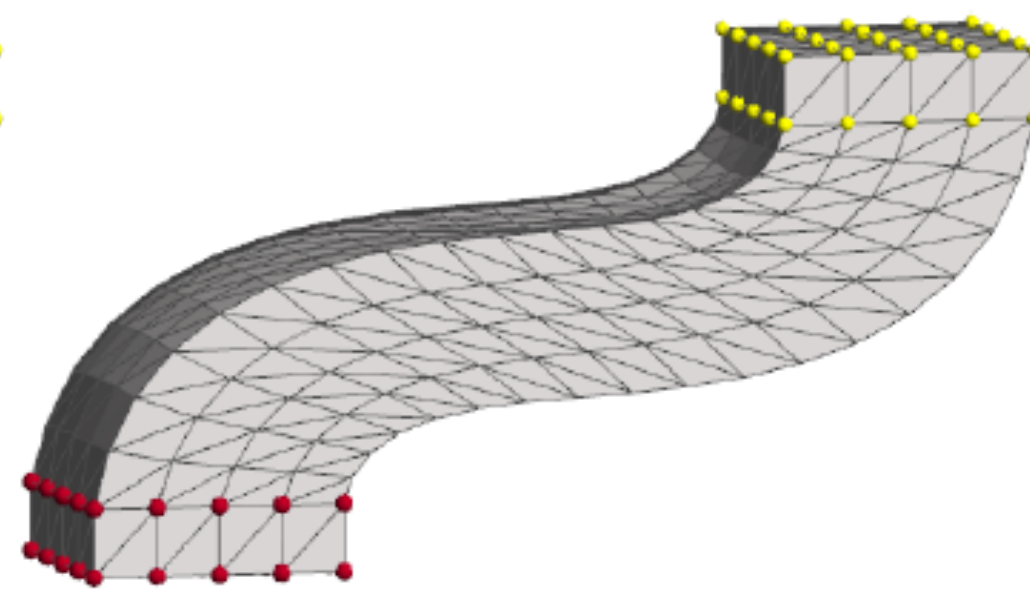


Initial Guess

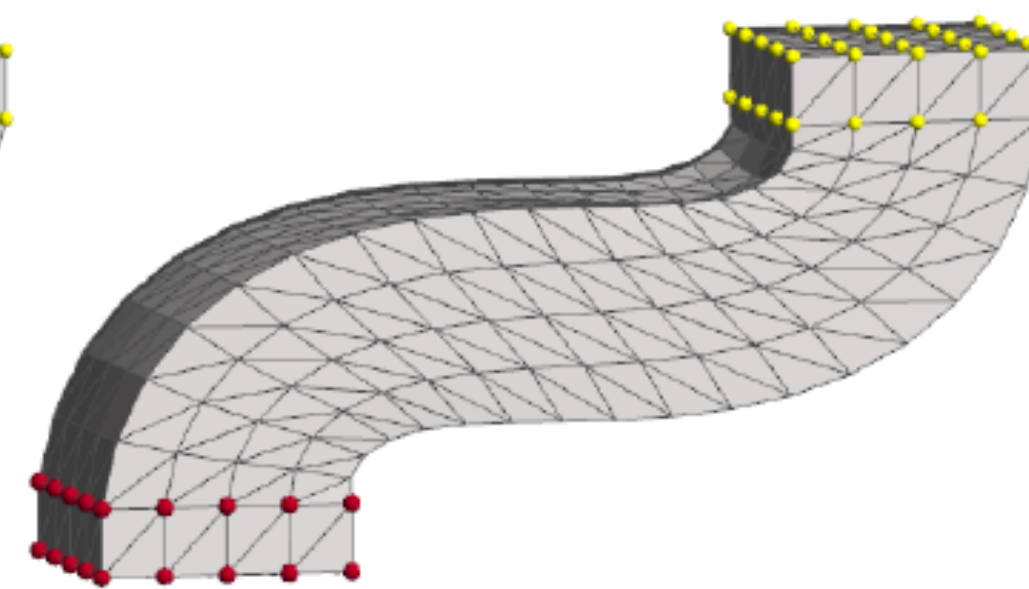
- Can use naïve Laplacian editing



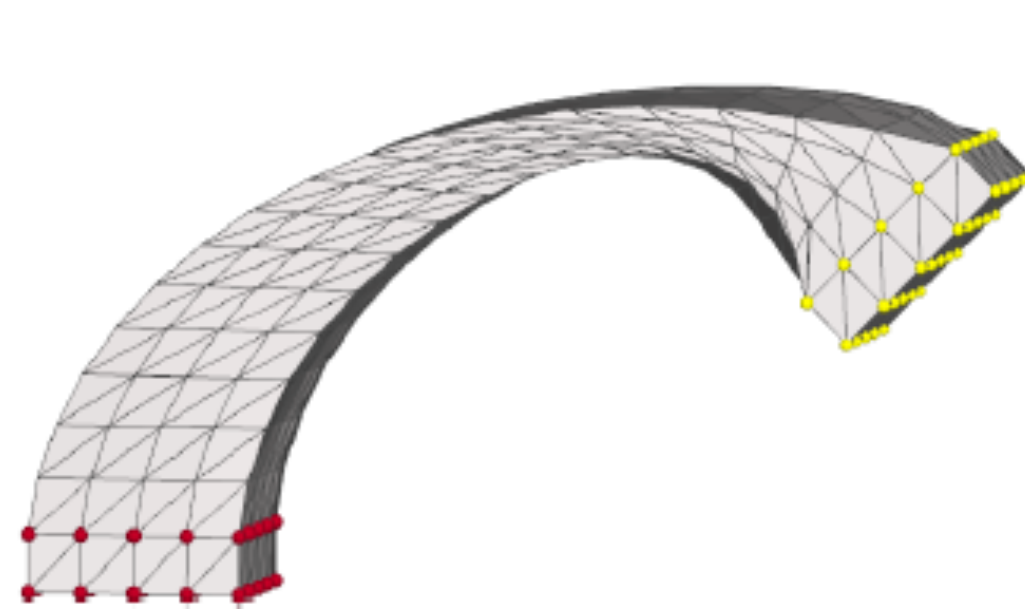
initial guess



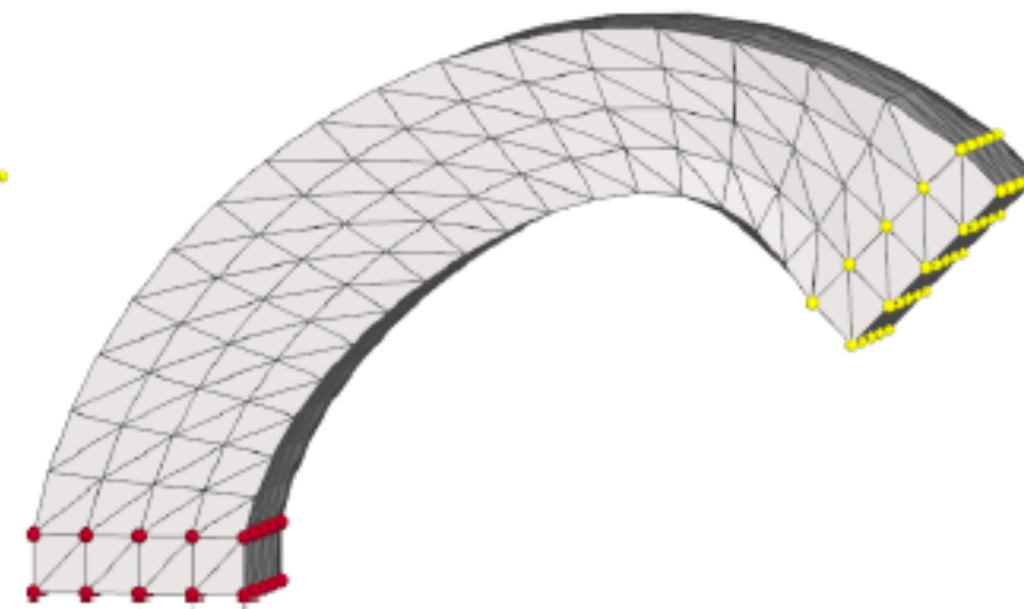
1 iteration



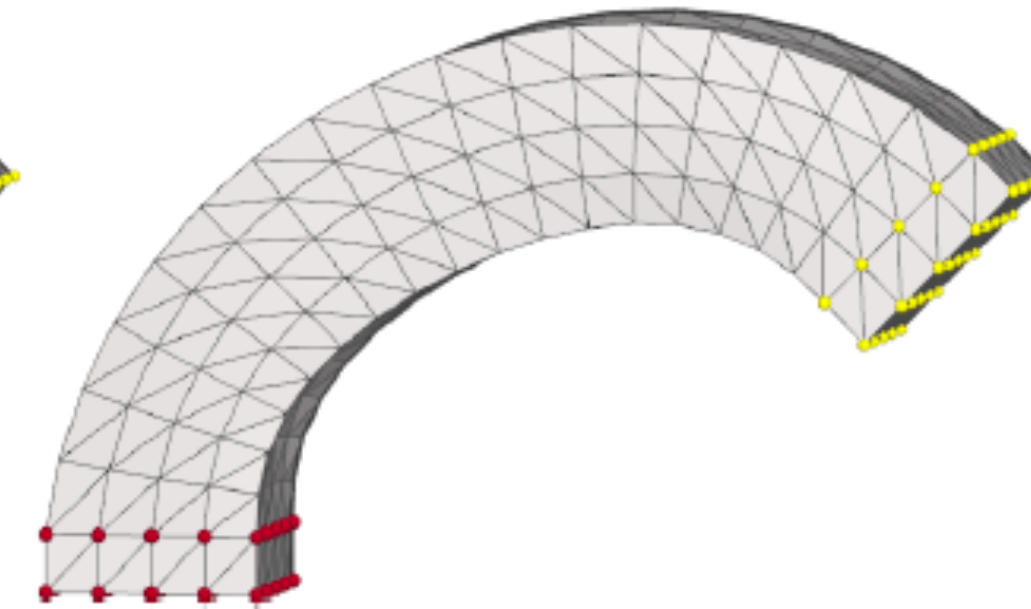
2 iterations



initial guess



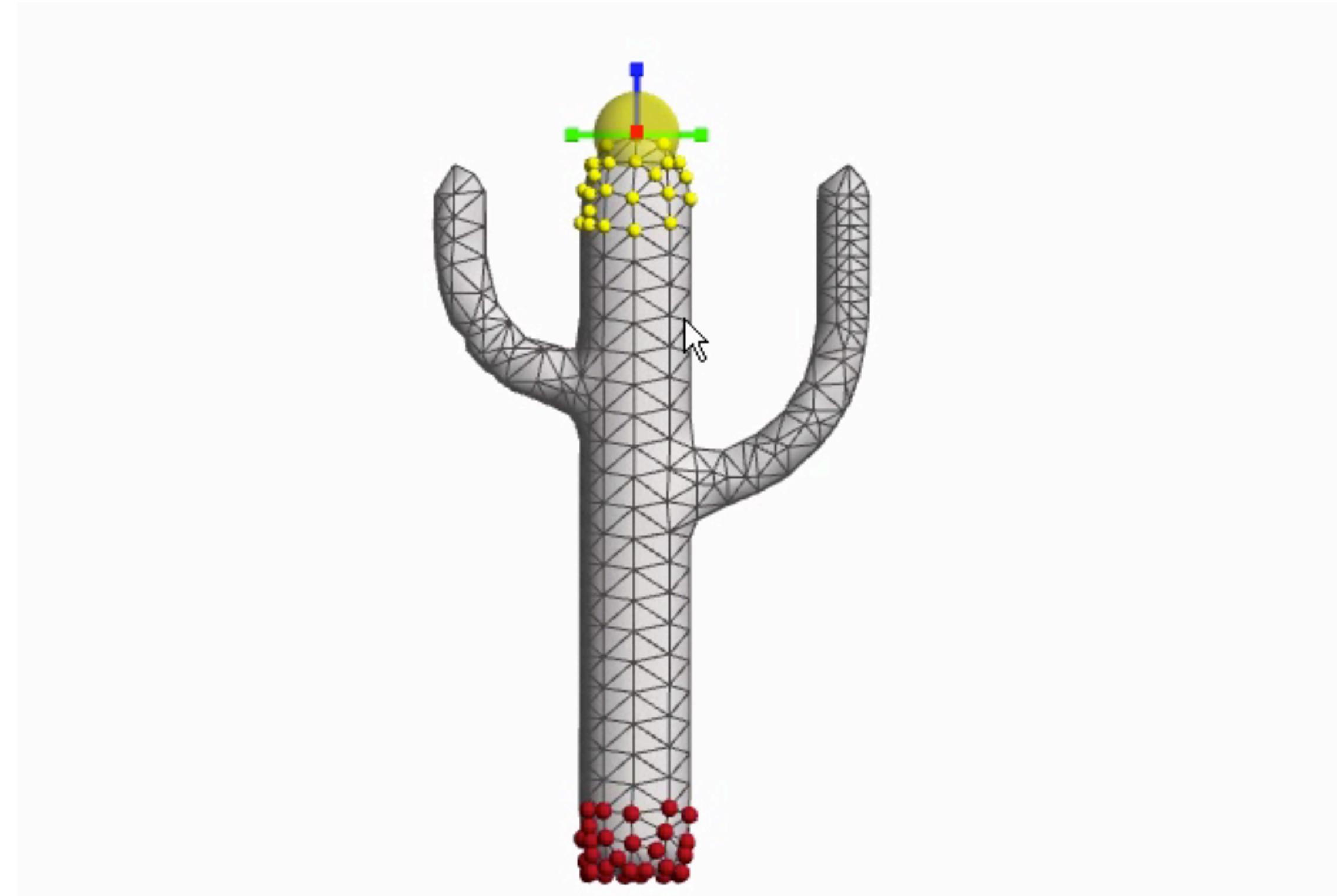
1 iterations



4 iterations

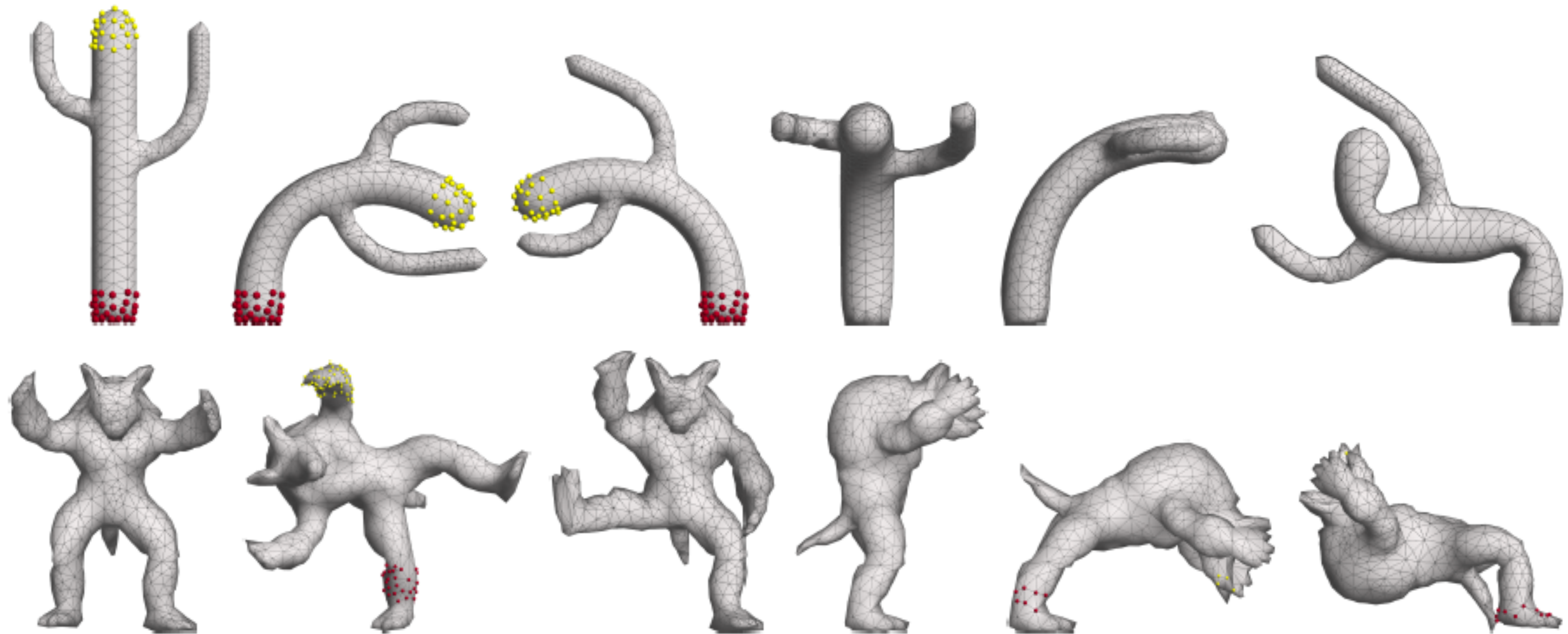
Initial Guess

- Can also use the previous frame
- Replace all handle vertex positions by the currently prescribed ones
- Fast convergence

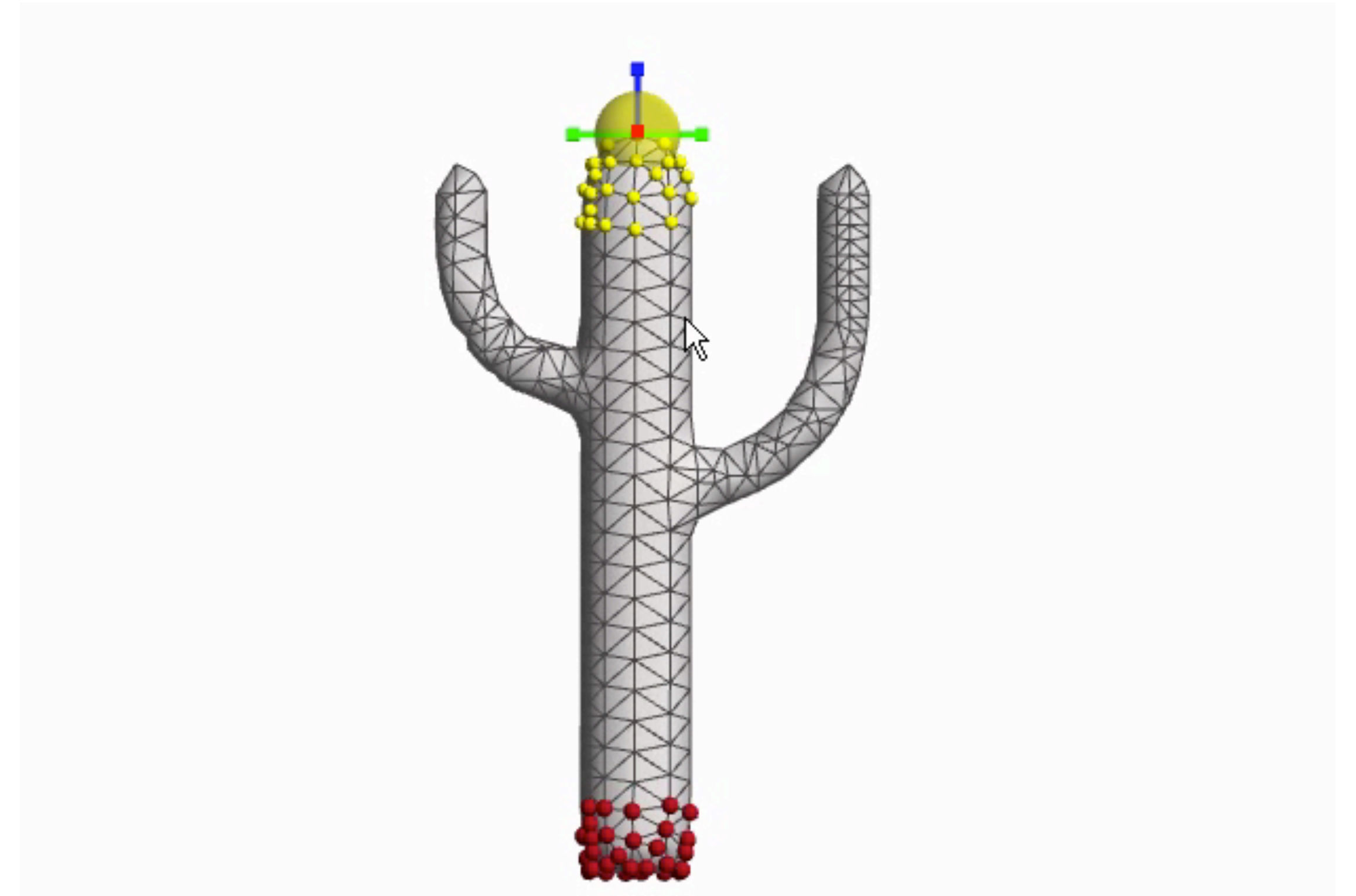
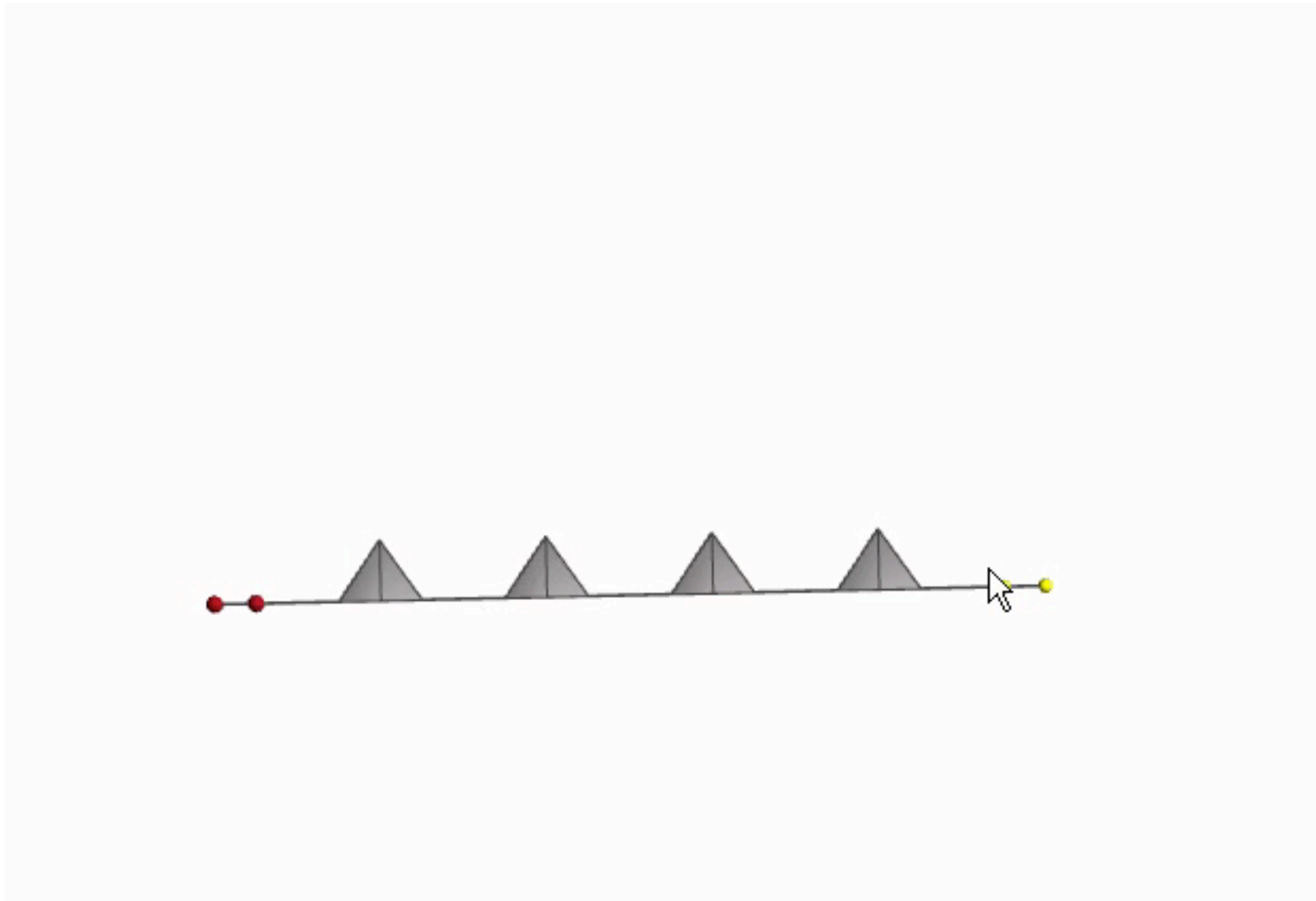


Large Rotations

- Use previous frame as the initial guess



Examples



Discussion

- Nonlinear deformation that models a elastic shell
- Simple to implement, no parameters to tune except number of iterations
- Each step is guaranteed to not increase the energy
- Each iteration is relatively cheap, no matrix re-factorization necessary

Discussion

- Works fine on small meshes
- On larger meshes: slow convergence
 - Each iteration is more expensive
 - Need more iterations because the conditioning of the system becomes worse as the matrix grows
- Material stiffness depends on the cell size
 - lots of wrinkles for fine meshes when using 1-rings as cells

Acceleration using Subspace Techniques

- Subspace created by influence weight functions for each handle
- Drastically reduces the number of degrees of freedom in the optimization

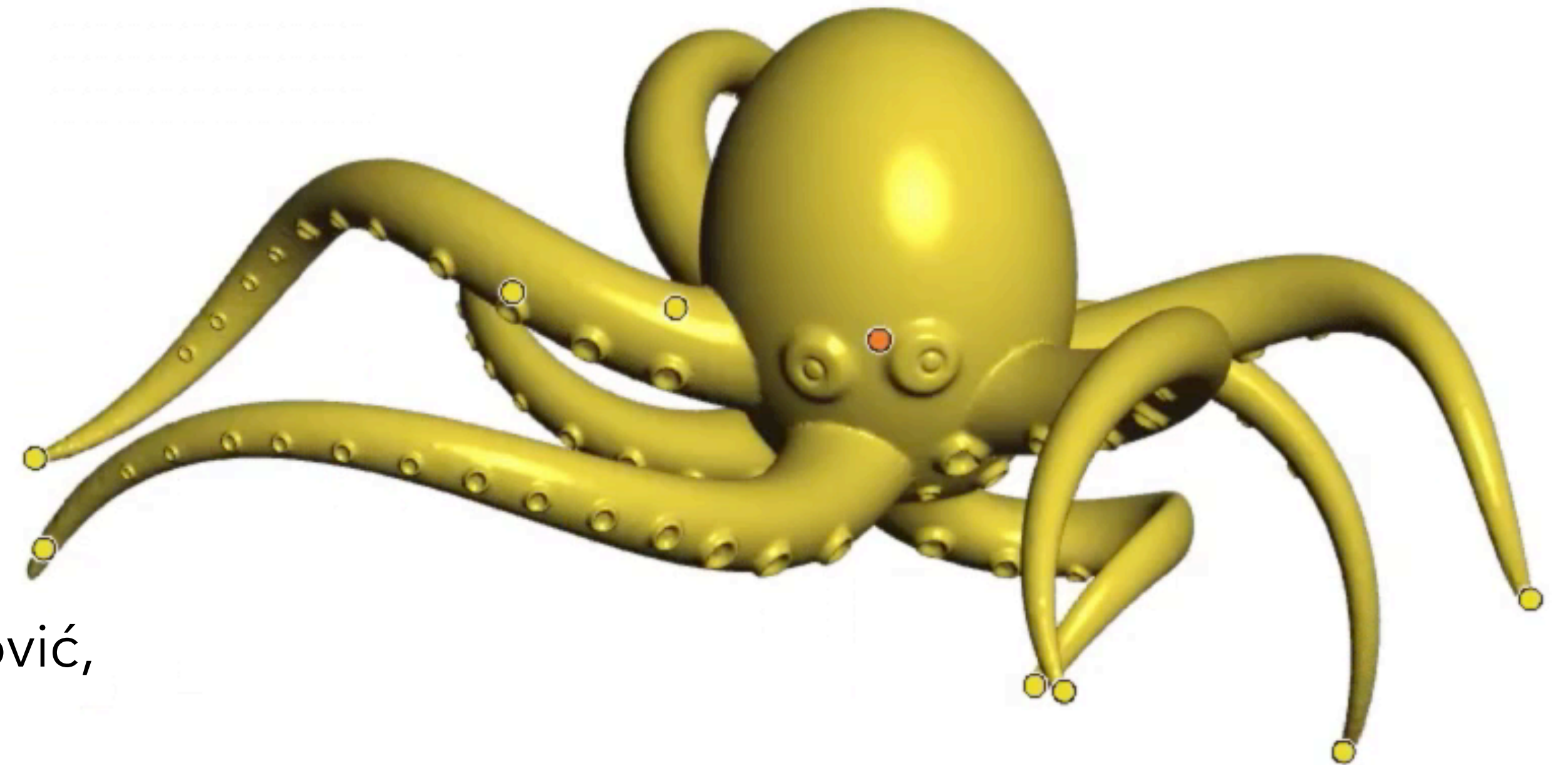


Alec Jacobson, Ilya Baran, Ladislav Kavan, Jovan Popović, and Olga Sorkine. ["Fast Automatic Skinning Transformations," 2012.](#)



Acceleration using Subspace Techniques

- Subspace created by influence weight functions for each handle
- Drastically reduces the number of degrees of freedom in the optimization

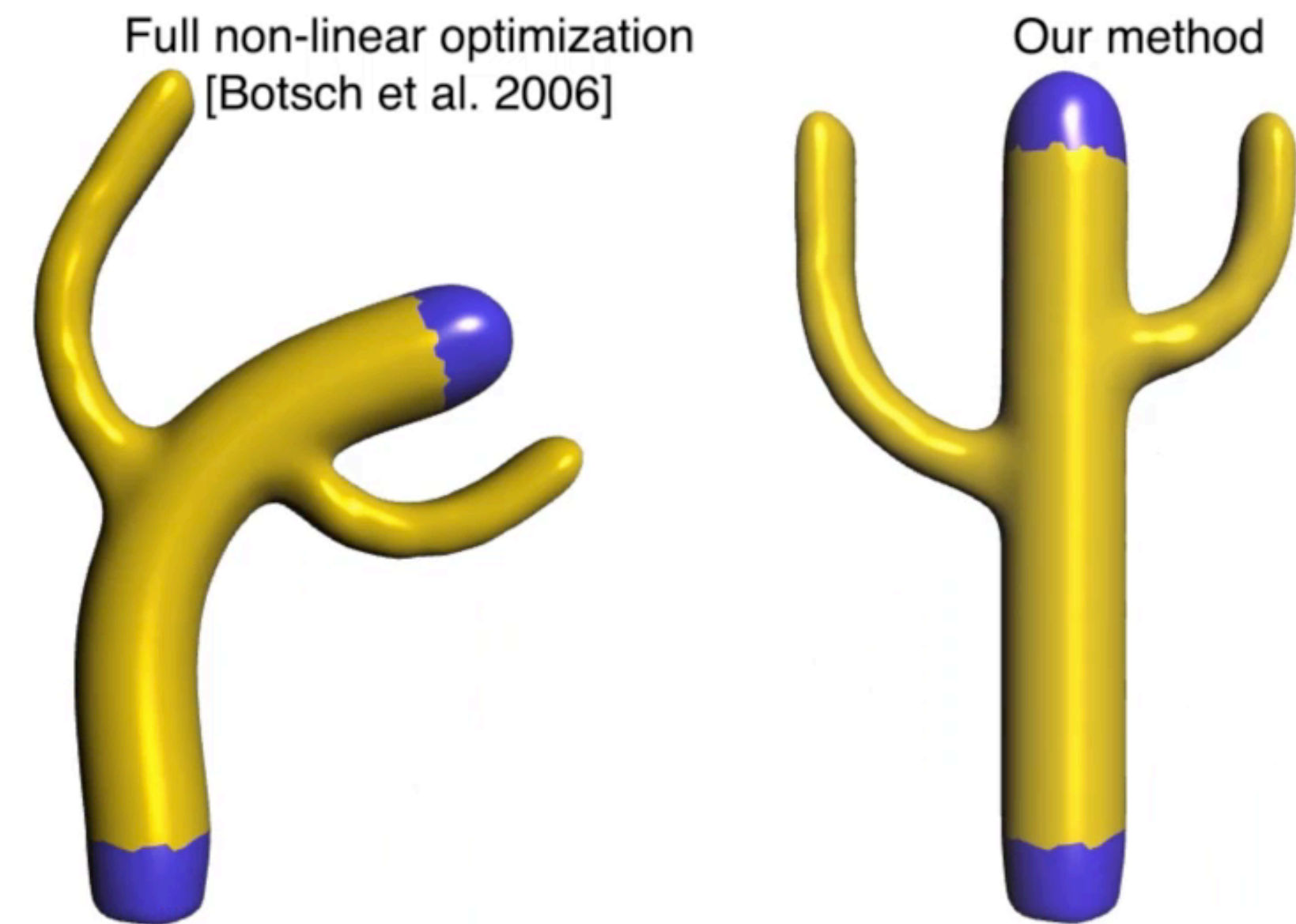


Alec Jacobson, Ilya Baran, Ladislav Kavan, Jovan Popović, and Olga Sorkine. ["Fast Automatic Skinning Transformations," 2012.](#)

Acceleration using Subspace Techniques

- Subspace created by influence weight functions for each handle
- Drastically reduces the number of degrees of freedom in the optimization

Alec Jacobson, Ilya Baran, Ladislav Kavan, Jovan Popović, and Olga Sorkine. ["Fast Automatic Skinning Transformations," 2012.](#)



Summary on Surface-Based Deformation

Summary on Surface-Based Deformation

- Find a mesh that optimizes some objective functional and satisfies modeling constraints

$$\mathbf{x}_{\text{def}} = \underset{\mathbf{x}'}{\operatorname{argmin}} E(\mathbf{x}') \quad s.t. \quad \mathbf{x}'_i = \mathbf{c}_i$$

Linear Methods

- Triangle gradient methods (2004-2005)

<http://dl.acm.org/citation.cfm?id=1015774>

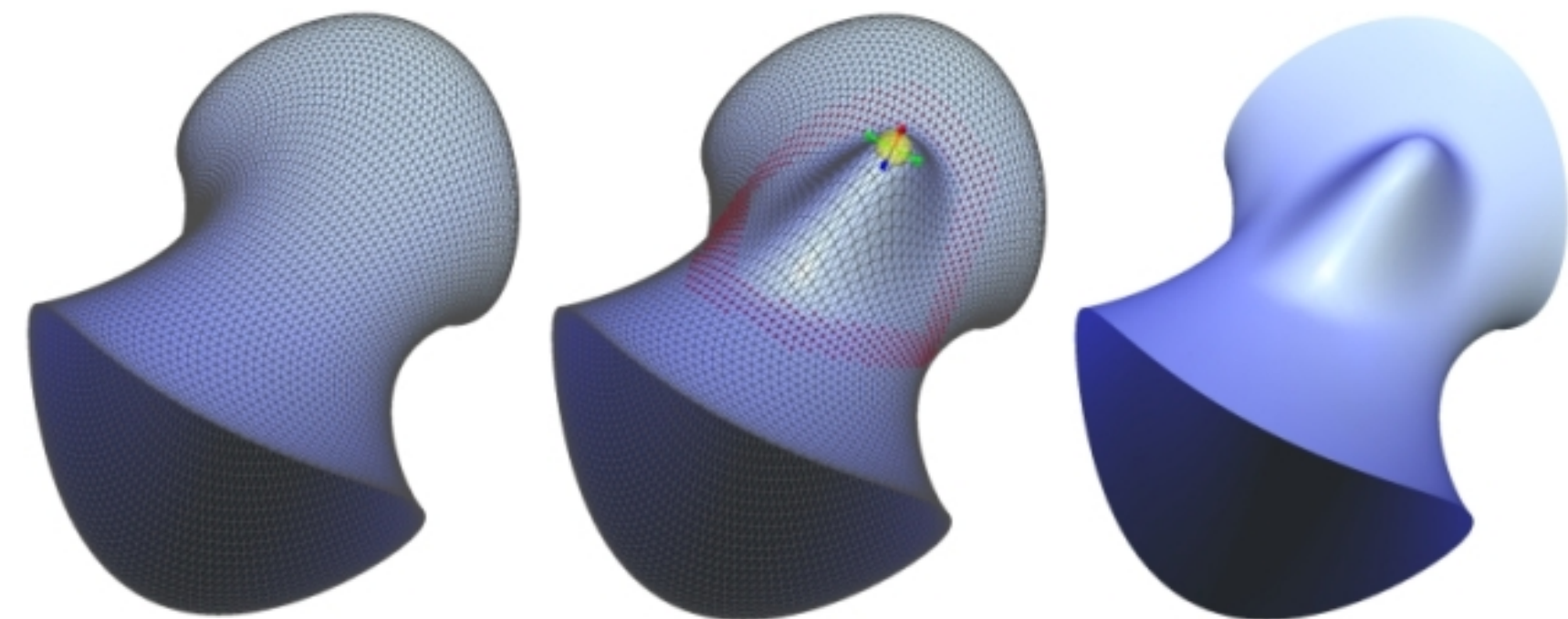
<http://www-ui.is.s.u-tokyo.ac.jp/~takeo/research/rigid/>



- Laplacian surface editing (2004-2005)

<http://dl.acm.org/citation.cfm?id=1015772>

<http://igl.ethz.ch/projects/Laplacian-mesh-processing/Laplacian-mesh-editing/>



Nonlinear Methods

- As Rigid as Possible surface modeling

<http://igl.ethz.ch/projects/ARAP/>

<http://dl.acm.org/citation.cfm?id=1778775>

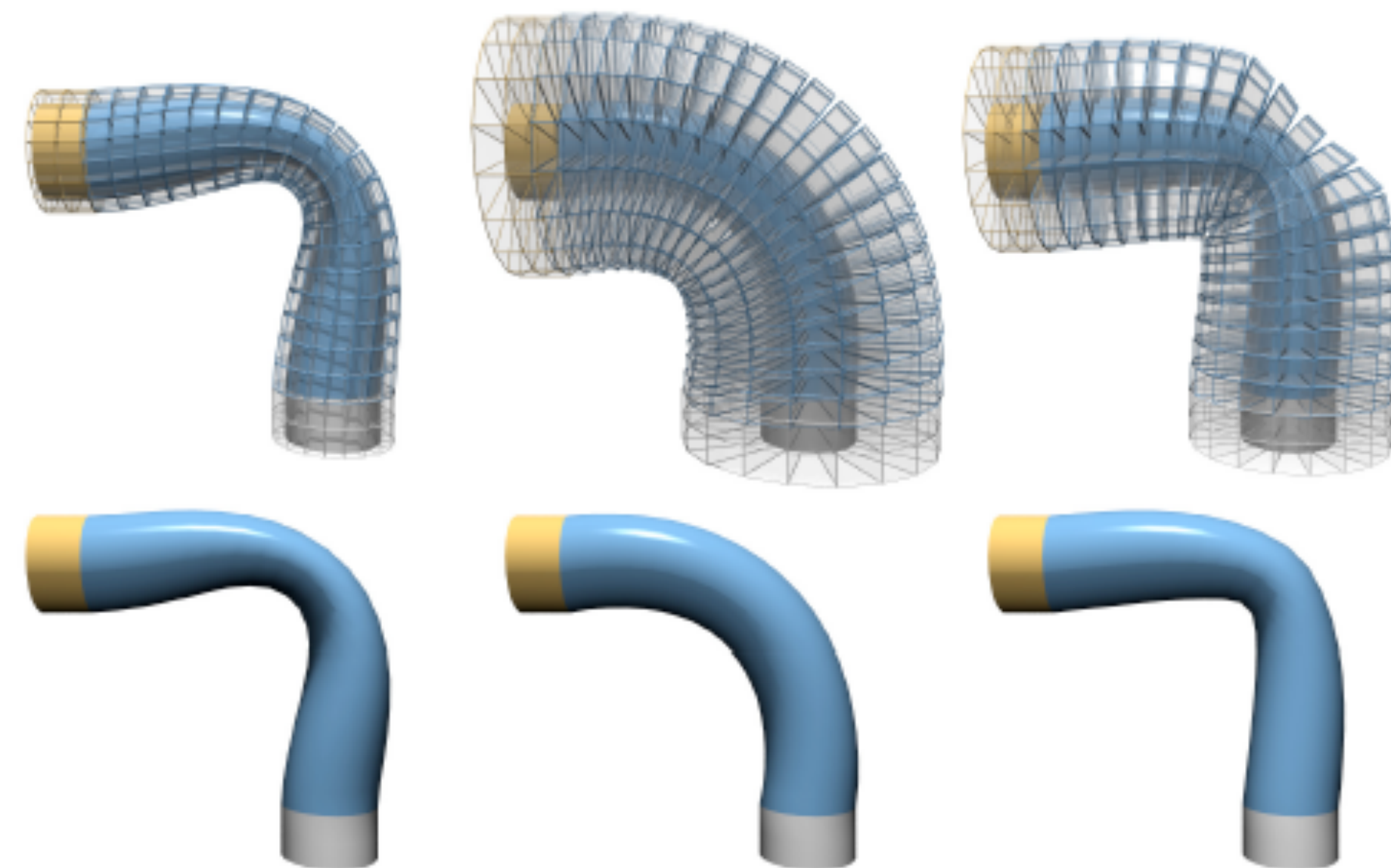
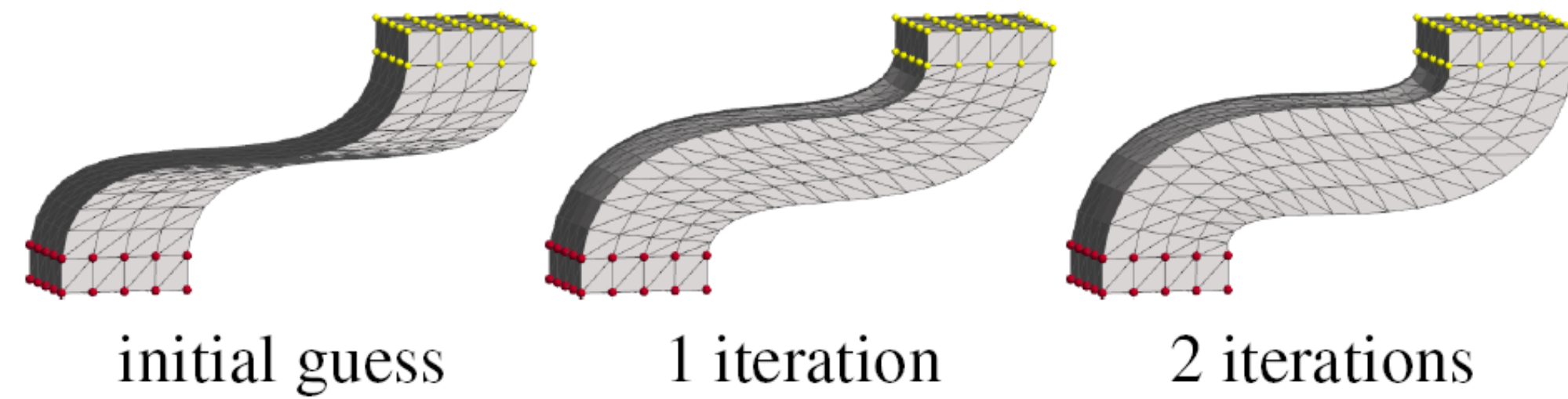
<http://igl.ethz.ch/projects/fast/>

- PriMo

<http://dl.acm.org/citation.cfm?id=1281959>

- Mesh Puppetry

<http://dl.acm.org/citation.cfm?id=1275808.1276479>



Surface-based Deformations

- Objective functional expressed in the mesh elements (vertices)
- Complexity depends on the mesh resolution
- Linear methods:
 - Solve a global linear system on the mesh
 - Usually suffer from some artifacts
- Nonlinear methods
 - Fewer artifacts but slower, and harder to implement

References

- Polygon Mesh Processing, Chapter 9